

AN ABSTRACT OF THE DISSERTATION OF

Samuel T. Woodrich for the degree of Doctor of Philosophy in Environmental Sciences presented on July 16, 2026.

Title: The Ecology of Grazing Lawns

Abstract approved:

William J. Ripple

Herbivores can instill significant changes to both local habitat characteristics and to the functions of ecosystems through herbivory and through the indirect ecological effects associated with herbivory pressure. My dissertation examines and expands upon a specific consumer-resource interaction in ecology caused by frequent and concentrated grazing by native herbivores in landscapes worldwide, deemed grazing lawns. I conducted the first systematic review of grazing lawn ecology and performed a field study that describes the characteristics, functions, and impacts of Roosevelt elk (*Cervus canadensis roosevelti*) and black-tailed deer (*Odocoileus hemionus columbianus*) maintained grazing lawns in the central Oregon Coast range. My systematic review of journal articles related to grazing lawn ecology revealed predation, and predator effects are frequently not considered in grazing lawn research, even when predators of grazing lawn maintaining herbivores are present within the study sites. I also identified various inconsistencies and deviations in the language used to describe the characteristics and functions of grazing lawns and suggest a modern definition that mitigates these discrepancies for future research on the subject.

In my field study chapter, I identified and described the characteristics of grazing lawns in the temperate coastal rainforest ecosystem of the Oregon coast, a system previously not associated with grazing lawn presence. Grazing lawns in the Tenmile Creek and Cummins Creek watersheds were associated with increased cover of alien plants in the herbaceous layer, suggesting that repeated herbivory gives alien vegetation a competitive advantage over native species. Following the exclusion of herbivory in grazing lawns, alien plants were no longer present, indicating the cessation of herbivory can shift plant structure and composition towards conditions of the native forest that surround these lawns. Finally, I provide commentary on the effects frequent and intense herbivory pressure has on degraded habitats of the Tenmile Creek watershed and offer restoration prescriptions that will passively improve the ecological conditions based on the results of my grazing lawn ecology research.

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The Ecology of Grazing Lawns

by
Samuel T. Woodrich

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APPROVED:

Major Professor, representing Environmental Sciences

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Dean of Graduate Education

I understand that my thesis will become part of the permanent collection of Oregon State University libraries. My signature below authorizes release of my thesis to any reader upon request.

Samuel T. Woodrich, Author

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“There’s a bird in the sky. Look at him fly. Why?”
-Mac Miller

1. CHAPTER 1: Introduction

Herbivory is a powerful influence that shapes the characteristics and functions of ecosystems worldwide. To study herbivory effects, ecologists often examine systems that have had herbivore density or behavior changes over time, or through the comparison of similar systems that differ in food web or trophic structure (Ripple et al. 2016). To quantify herbivory pressure within these systems, herbivory frequency, herbivore population or herd size, and herbivore distributions or migrations are researched within systems. Furthermore, when the dynamics of food webs are altered, either by the loss or addition of species within trophic levels, researchers can measure ecological impacts and effects caused by these changes. This has led to substantial discoveries involving the influence herbivores have on plant diversity and species assemblages (Olf and Ritchie 1998, Borer et al. 2014, Ye et al. 2025), hydrological conditions (Beschta and Ripple 2008, 2019), landscape heterogeneity (Porensky and Veblen 2012, Cromsigt and te Beest 2014), ecosystem services (Metcalf et al. 2014, Moreira and Abdala-Roberts 2023), and soil conditions (Vaieretti et al. 2021, Moon et al. 2025) among many other impacts to the ecological characteristics of systems. The ecological drivers responsible for these changes consist of, but are not limited to, predator extirpation or reintroduction (Beschta and Ripple 2016), invasive species introduction (Gruner 2004), disease (Holdo et al. 2009), changes in land management (Newsome et al. 2001), climate change (Ripple et al. 2024) and extreme weather events (Barton et al. 2023). Finally, the effects of herbivory can vary across spatial and temporal scales (Brown and Allen 1989), often limiting our comprehension of

herbivory related impacts to the areas they are observed in at a singular moment in time. While these limitations may appear daunting, researchers worldwide continue to examine the effects of herbivores on system-wide scales through the experimental manipulation of herbivory pressure and through the natural study of systems that have experienced changes in their food web structure.

A possibility for mitigating the uncertainties of herbivory effects across spatial and temporal scales involves the examination of herbivory effects on the patch dynamics of a system. For instance, herbivores can facilitate positive consumer-resource interactions to maximize nutritional intake in patches. If herbivory rates remain unchecked across patches, plant richness, reproduction, and morphology may be affected at system-wide scales (Schreiner et al. 1996). One example of patch level changes in response to grazing pressure is the creation and maintenance of grazing lawns. Grazing lawns persist through a positive consumer-resource loop maintained by herbivores, in which grazing increases the concentration of nutrients in plant parts, resulting in a feedback that promotes greater herbivory pressure (McNaughton 1984). Grazing lawn patch presence can instill several ecological knock-on effects that influence the conditions of a system in its entirety. For instance, grazing lawn maintenance can facilitate a competitive advantage in grazing tolerant plant species (Zwerts et al. 2015), reduce woody plant recruitment (Staver et al. 2009), establish alien plant species (Trager et al. 2004, Kellett and Alisauskas 2022), increase nutrient availability in soils (Johnson and Matchett 2001, Coetsee et al. 2011), and increase visibility for predators (Fleazar et al. 2019). As such, ecologists use patch-scale studies and experiments to comprehend the factors, interactions, and influences associated

with sustained herbivory and infer how patch dynamics can influence the functions and characteristics of a system in its entirety.

To gain a greater understanding of the ecological functions and effects herbivory has on patch dynamics, I conducted the first ever systematic review of empirical studies on grazing lawn ecology (Chapter 2). I then applied my understanding of grazing lawn ecology in a field study that characterized the grazing lawns of the temperate coastal rainforest ecosystem and evaluated the effects of herbivory pressure mitigation on grazing lawns of the system (Chapter 3). Finally, I used the results of my systematic review and field study to make recommendations for more effective environmental restoration practices that facilitate passive restoration within the heavily herbivorized habitats of the temperate coastal rainforest ecosystem (Chapter 4). In the remainder of my introduction chapter, I briefly review the history of grazing lawn ecology, justify the need for a systematic review of the subject, introduce the ecological characteristics and management history of the temperate coastal rainforest ecosystem, and discuss the processes responsible for the formation and maintenance of grazing lawns within the temperate coastal rainforest ecosystem.

1.1 Grazing Lawn Ecology

Grazing lawns, first characterized as nutrient dense habitats where vegetation is short in stature and highly concentrated, were discovered in the African savanna. Since they were first described by McNaughton (1984), grazing lawn research has been conducted in numerous ecoregions worldwide, expanding ecologists

understanding of the processes responsible for their maintenance (du Toit and Olff 2014), promoting discussion of the mechanisms that create them (Archibald 2008, Mouissie et al. 2008, Singh et al. 2024), and increasing the number of herbivores known to maintain them. As the number of scientific publications related to grazing lawn ecology increases, the defining characteristics and processes associated with grazing lawns has changed over time. Such changes include the ecoregions where grazing lawns persist, the herbivores that facilitate them, the dominant vegetation types associated with grazing lawns, and the indirect effects associated with lawn maintenance. Subsequently, grazing lawn research is fragmented, lacks interconnectivity in function and characteristic across the ecoregions they persist in, and lacks comprehensive analysis of the methods, impacts, and ecological effects associated with herbivore-maintained lawn patches on a global scale. While reviews of grazing lawn ecology have been published previously, they lack a systematic approach in their data collection and analysis methods (du Toit and Olff 2014), pertain to specific interactions in grazing lawns (Howison et al. 2017, Kaspari 2020), are focused on specific herbivore species (Voysey et al. 2023), or are limited in their geographical extent (Hempson et al. 2015). As such, a systematic review of grazing lawn ecology that identifies trends in the subject previously unnoticed, depicts the scientific advancements made over time, describes the weaknesses and blind spots pertaining to the subject, and makes recommendations for conducting future grazing lawn research is worthwhile.

Formulating research questions suited for a systematic review of grazing lawns is complex. Given the large variety in study design, data analysis, and

technology used across scientific publications, exploratory data analysis of simple, universal components between grazing lawn publications provides meaningful preliminary trends for future research to be based upon, benefiting the subject in its entirety. To identify global trends in the subject, I collected, standardized, and summarized the species name, taxonomic group, and frequency of every animal associated with grazing lawns publications world-wide. Additionally, I identified the geographical location of every study site where grazing lawn research has been conducted and categorized them by ecoregion and biome. Lastly, I identified every grazing lawn definition provided and summarized the most frequently used vocabulary used to describe the ecological characteristics and processes associated with grazing lawns.

Using these data, I found most of the grazing lawn research consisted of just herbivore-producer relationships and failed to include predators or predation effects. I suggest predation is an influence on where grazing lawns persist world-wide but is not frequently considered in grazing lawn research. Furthermore, I found grazing lawn study sites to persist most commonly in protected areas world-wide, indicating habitat and biodiversity preservation may facilitate grazing lawns, or that grazing lawn research is favored in areas closest to baseline conditions of protected systems. Additionally, grazing lawns are maintained by a wide variety of animals, ranging in size from mega herbivores to rodents, and in taxonomic groups including insects, birds, and mammals. Finally, I provide a clear definition for grazing lawns based on the language and characteristics used in other grazing lawn definitions previously published in peer reviewed journal articles of the subject. By summarizing the

ecological characteristics and functions of grazing lawns, my systematic review serves as a foundation for subsequent research of the subject and contributes to grazing lawn ecology by illustrating trends across the research conducted by hundreds of scientists over the past four decades.

1.2 The Temperate Coastal Rainforest Ecosystem

The temperate coastal rainforest ecosystem of the Oregon coast has experienced vast ecological changes in recent history. Alterations in land use and ownership, native herbivore and predator populations, and plant species assemblages have resulted in landscape heterogeneity throughout the system. Furthermore, historical fire disturbances, timber harvesting, and livestock ranching has altered the successional states and trajectories of habitats and fragmented the ecosystem. Along the central Oregon Coast, the Tenmile Creek and Cummins Creek watersheds exhibit spatial heterogeneity caused by the ecological influences and disturbances that have occurred historically. While some habitats in these watersheds resemble those traditionally associated with the temperate coastal rainforest ecosystem, other habitats, particularly areas historically used for livestock ranching are degraded.

1.3 Tenmile Creek Watershed History

Various properties in the Tenmile Creek watershed were purchased following the Forest Homestead Act of 1906. These properties were purchased for farming and ranching, though the latter proved to be more viable due to the microclimate and soil conditions of the watershed. Properties near the creek were most desirable for

livestock ranching while other parcels in wetland habitats or at increased elevations along the valley of the Tenmile Creek watershed were not purchased (Radtke 2017). The trees of the properties purchased were often removed to make room for livestock grazing and to generate additional income through timber sales. Livestock ranching decreased in popularity in the following century, resulting in the establishment of private residencies. Forestry operations, a more profitable land practice during the mid-20th century, were established further inland in the Tenmile Creek watershed or on the southern valley closest to the Pacific Ocean (Sleeper 2003) (Figure 1.1). Today, livestock ranching is absent in Tenmile Creek, and forestry operations are spatially sparse, temporally infrequent, and are located further inland in the watershed. Additionally, land unpurchased during the Forest Homestead Act is mostly owned by the federal government and managed by Siuslaw National Forest of the United States Forest Service (USFS). Some portions of USFS managed land in Tenmile Creek exhibit late successional and old growth assemblages of Sitka spruce (*Picea sitchensis*), western hemlock (*Tsuga heterophylla*), and western redcedar (*Thuja plicata*) trees, along with assemblages of other late successional understory plant species such as cascara (*Rhamnus purshiana*), red elderberry (*Sambucus racemosa*), red huckleberry (*Vaccinium parvifolium*), evergreen huckleberry (*Vaccinium ovatum*), and sword fern (*Polystichum munitum*). These sections of the Tenmile Creek watershed are frequently used as a baseline ecosystem for restoration efforts in degraded habitats of the same system. Other parcels purchased during the Forest Homestead Act are now owned and managed by various environmental

agencies interested in restoring the ecological conditions of their properties to enhance biodiversity, promote system resilience, and sequester atmospheric carbon.

1.4 Cummins Creek Watershed History

In 1984, the Cummins Creek Wilderness Area was established in effort to protect the remaining old growth, and late successional Sitka spruce stands historically prevalent along the Oregon coast. The wilderness incorporates most of the Cummins Creek watershed within its boundary, including the main waterbodies and tributaries of Bob Creek, Little Cummins Creek, and Cummins Creek. Prior to the establishment of the wilderness area, timber harvests within the Cummins Creek watershed were spatially minimal and mostly outside of the present-day boundary of the Cummins Creek Wilderness Area (Figure 1.1). Today, the Cummins Creek Wilderness Area is regarded as the only old-growth Sitka spruce forest in the Oregon Wilderness system (United States Forest Service 2026). Since the creation of the wilderness area, timber harvests have been outlawed and land management practices, such as trail maintenance, are infrequent and minimally impactful to the function of the ecosystem. In addition to the tree and shrub assemblages depicted in the Tenmile Creek watershed, portions of the Cummins Creek watershed high in elevation consist of natural Douglas fir (*Pseudotsuga menziesii*) stands with salal (*Gaultheria shallon*) and Oregon grape (*Berberis aquifolium*) in the understory. Restoration prescriptions have not been implemented within the wilderness area of the Cummins Creek watershed.

1.5 Native Fauna

Roosevelt elk (*Cervus canadensis roosevelti*) and black-tailed deer (*Odocoileus hemionus columbianus*) are the dominant ungulate herbivores of the Cummins Creek and Tenmile Creek watersheds. Settlers of the Tenmile Creek watershed claim elk were not present in the early 1900s due to the profitability of hind and front quarter harvests granted by elk hunting. A statewide ban on Roosevelt elk hunting was enforced in 1900 to encourage elk population recovery, but later repealed in the Tenmile Creek watershed from 1937-1951 (Radtke 2017). Populations of Roosevelt elk in the Alsea unit 18 hunting area, an area that incorporates both watersheds, increased dramatically following a 1947-71 reintroduction program by Oregon Department of Fish and Wildlife (ODFW) (Harper 1971). Today, the population density of elk in the Alsea unit 18 hunting area is estimated to be between 4,800 – 5,800 elk for the region (0.95 - 1.1 Roosevelt elk per km²) (Oregon Department of Fish and Wildlife 2023) and elk hunting in the Alsea unit is permitted but regulated by a hunting tag and dedicated hunting period system enforced by ODFW.

The most recent population estimate for black-tailed deer in Oregon was 320,000 for the state (1.26 black-tailed deer per km²) in 2004 (Kuzyk and Heffelfinger 2015) but there are no estimates in Alsea unit 18. Settlers of the Tenmile Creek watershed claim black-tailed deer were relatively abundant in the 1950s (Radtke 2017). Today, black-tailed deer hunting is regulated by ODFW through a hunting tag and dedicated hunting period system.

Several predators are present in the temperate coastal rainforest ecosystem, such as cougars (*Puma concolor*), black bears (*Ursus americanus*), bobcats (*Lynx rufus*), and coyotes (*Canis latrans*). Grey wolves (*Canis Lupus*) were present within both watersheds before their extirpation from Oregon in the early 1900s. The last documented encounter of wolves in the region is rumored to have occurred in 1927 when an early settler trapped 7 wolves in 1927 along the ridge connecting the Tenmile Creek and Cummins Creek watersheds (Radtke 2017). Wolves have not been reintroduced or observed in the temperate coastal rainforest ecosystem of Oregon since their extirpation (Oregon Department of Fish and Wildlife 2026b).

Cougars have an estimated population density of 1,200 across the coast and northern cascade range (0.03 cougar per km²) (Broman et al. 2026). From 1913 to 1961, Oregon established a bounty for predators, including cougars. In response, homesteaders in the Tenmile Creek watershed frequently used dogs to hunt cougars in the 1940s (Radtke 2017). Historic cougar population records for Oregon are limited. However, ODFW estimates a minimum of 6,718 cougars were taken from Oregon in response to the bounty program and subsequent management practices by the state (Oregon Department of Fish and Wildlife 1987) indicating a substantial portion of the native population may have been removed.

Black bears have a high presence in the Alsea hunting area (Oregon Department of Fish and Wildlife 2012), and have a population density estimate of 25,000 - 30,000 for the state (0.10 – 0.12 black bear per km²) in 2026 (Oregon Department of Fish and Wildlife 2026a). Historic records of black bear hunting in both watersheds are lacking. However, early settlers of the Tenmile Creek watershed

experienced bear predation of livestock frequently (Radtke 2017), and may have protected their livestock by hunting bears.

Population estimates for bobcats in Oregon are scarce, though the species is thought to be stable or increasing in the American west (Roberts and Crimmins 2010). Historic records of bobcat hunting or trapping in either watershed are lacking. Oregon coyote populations are thought to be stable, although a formalized population density estimate for the state has not been published. A coyote population density estimate of 300,000 individuals (1.18 coyote per km²) for the state was suggested by Shannon Hurn, the Deputy Director for Fish and Wildlife Programs of ODFW in 2019 (Hurn 2019). Coyote predation of livestock is hypothesized to have persuaded settlers in the Tenmile Creek watershed to cease livestock ranching (Radtke 2017), indicating coyote preyed upon local livestock during the 1900s.

1.6 Disturbance Events and Alien Plants

Fires in the 19th century burned throughout the Cummins and Tenmile Creek watersheds (Figure 1.2). Crown fires in the Cummins Creek and Tenmile watersheds have not occurred since the 19th century. Other disturbances common within the system include the leveling of the upper canopy by blowdown events from strong gales and heat wave events such as the 2021 Pacific Northwest Heat Dome event (Heeter et al. 2023, Li et al. 2024). Substantial environmental degradation or changes within the entirety of each watershed have not been documented. However, active restoration efforts within the most degraded habitats of the Tenmile Creek watershed have been implemented or plan to be implemented on the private properties owned by

various environmental groups (Sleeper 2003, Engelmeyer 2003, Suring and Lance 2021).

Alien and invasive plant species dominate the terrestrial habitats where historic timber harvests and/or livestock ranching have occurred in the Tenmile Creek watershed. These plant species consist of, but are not limited to, reed canary grass (*Phalaris arundinacea*), velvet grass (*Holcus lanatus*), sweet vernal grass (*Anthoxanthum odoratum*), orchard grass (*Dactylis glomerata*), foxglove (*Digitalis purpurea*), common dandelion (*Taraxacum officinale*), common bird's-foot trefoil (*Lotus corniculatus*), double flowered creeping buttercup (*Ranunculus repens*), tansy ragwort (*Senecio jacobaea*), and buckhorn plantain (*Plantago lanceolata*). Some alien plants are palatable to native herbivores (e.g. common dandelion, double flowered creeping buttercup) (Woodrich, personal observation) while others are not (e.g. foxglove) (Corrigall et al. 1978). Some grasses can compensate for high herbivory pressure with increased growth (McNaughton 1983), such as alien orchard grass (Hicks and Reader 1995). Additionally, alien grasses such as velvet grass can gain a competitive advantage over other herbaceous plants when grazing pressure is high, especially if nutrients are readily available (Tahmasebi Kohyani et al. 2009). The competitive and compensatory advantages alien plants have over native ones has contributed to their established dominance in areas where historic land management practices occurred.

1.7 Grazing Lawns Effects on The Temperate Coastal Rainforest Ecosystem

Grazing lawns in the temperate coastal rainforest ecosystem have not been described previously in ecological literature, and are not a recognized plant association type of the northern Oregon coast range (McCain 2002, USDA Forest Service Pacific Northwest Region 2005). However, frequent herbivory pressure by native ungulates, decreased predator populations and predation risk in the system, and historical land management practices in these watersheds may have contributed to the creation of grazing lawn patches.

The Forest Homestead Act promoted the conversion of forested habitat to pastures in the Tenmile Creek watershed for livestock ranching, resulting in habitats composed of alien herbaceous vegetation. After the cessation of livestock ranching, the herbivory pressure caused by non-native herbivores was substituted by the recovering population of native herbivores in the system. The trophic consumptive and non-consumptive effects typically caused by native predators were minimal due to the simultaneous human induced decrease in predator populations of the system. The subsequent recovery in native ungulate populations increased herbivory pressure within the system, further facilitating alien plant dominance in the areas historically used for ranching. Alien plants can spread through native herbivore seed dispersal to the other areas herbivores frequently use within a system (Calvino-Cancela 2011). I suggest additional grazing lawn patches are created by the movement of alien plant seeds from established lawn patches to other areas frequently utilized by native ungulates, leading to the formation of additional grazing lawns. The historical, environmental, and food web context of this system satisfies the conditions necessary

for grazing lawns to form, while the herbivore population and limited predation risk facilitates the maintenance of grazing lawns (Figure 1.3).

Patch-level transitions from forested to grazing lawn habitat could instill numerous changes to the dynamics of the ecosystem and landscapes of the temperate coastal rainforest ecosystem. For instance, the spread of alien plants could decrease biodiversity (Langmaier and Lapin 2020). Sustained herbivory pressure could result in simplified vegetation structures of forest habitats (Reed et al. 2022) and reduced above ground carbon storage (Tanentzap and Coomes 2012). Increased herbivore populations can alter nutrient cycling and primary productivity (Stephan et al. 2017) or litter decomposition rates (Chollet et al. 2021). As such, a greater understanding of the processes that maintain grazing lawn patches within this system would illuminate the potential impacts of sustained herbivory in the system's entirety. Furthermore, the manipulation of herbivory pressure in grazing lawn patches of this system provides insight regarding how the ecological forces that prohibit passive regeneration of native plants in degraded habitats of the system can be mitigated, leading to the implementation of more effective restoration practices for improving landscape and ecosystem function and quality.

1.8 Conclusion

A systematic review of grazing lawn ecology has been lacking in the current literature of the subject. Likewise, the historical land management practices, increased herbivore populations, and decreased predator presence in the temperate coastal rainforest ecosystem has led to the formation and maintenance of grazing

lawns within my study area. In the next chapter of my dissertation, I present the first systematic review of grazing lawn ecology. In chapter three, I identify and describe the characteristics and ecological functions associated with grazing lawns along the central Oregon Coast and examine the effect herbivory pressure has had on plant composition and structure in grazing lawns. Finally, in my concluding chapter, I apply the synthesized trends from my systematic review and results from my field experiments to make recommendations on how to improve the ecological conditions of degraded habitats of restoration sites similar in ecological characteristics and functions to grazing lawns in the temperate coastal rainforest ecosystem.

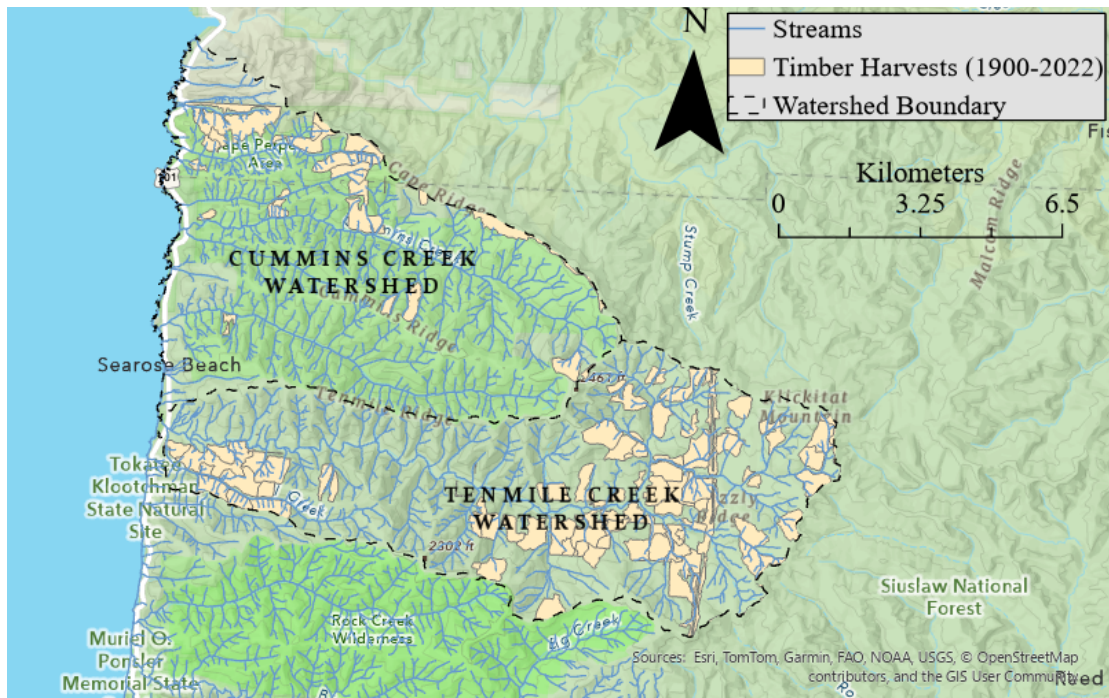


Figure 1.1: Map of the commercial forestry operations between 1900 and 2022 in the Cummins and Tenmile Creek watersheds. Forestry operations that took place in within the Cummins Creek Wilderness area, indicated in lime green within the Cummins Creek Watershed, occurred prior to the establishment of the wilderness area in 1984. Timber harvest location data was collected by Siuslaw National Forest of the United States Forest Service and provided by Doug Glavich of the Northwest Oregon Area Ecology Program.

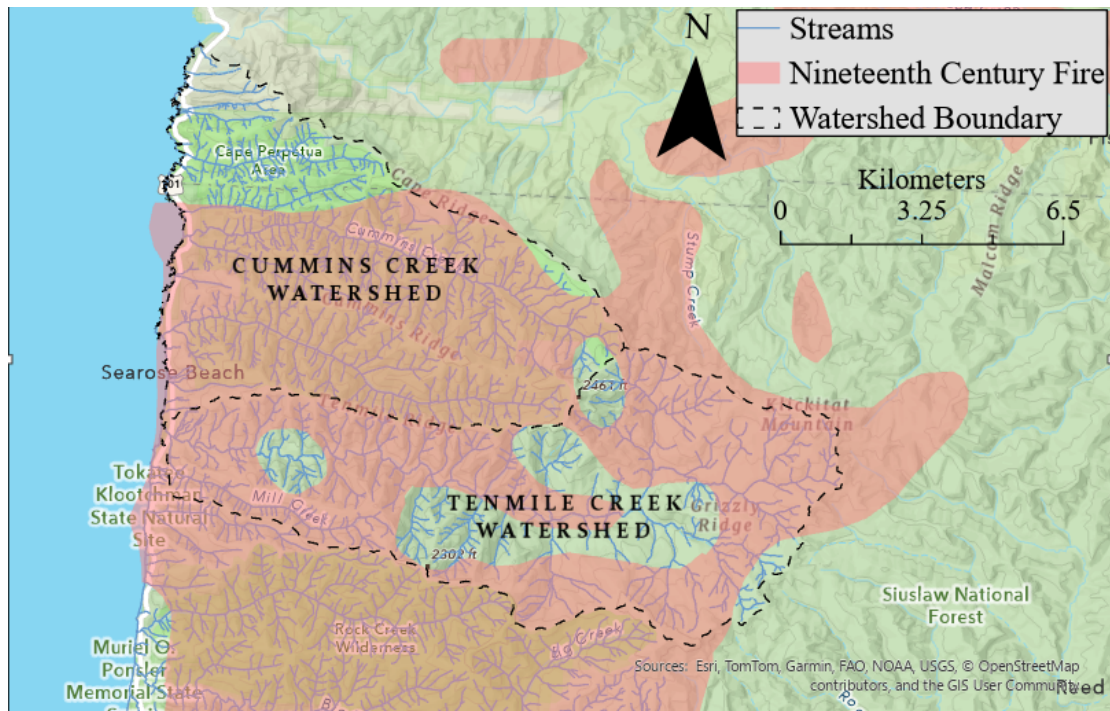


Figure 1.2: Map of the historic fire disturbance in the Cummins and Tenmile Creek watersheds. The nineteenth century fire boundary data was collected by Siuslaw National Forest of the United States Forest Service and provided by Doug Glavich of the Northwest Oregon Area Ecology Program.

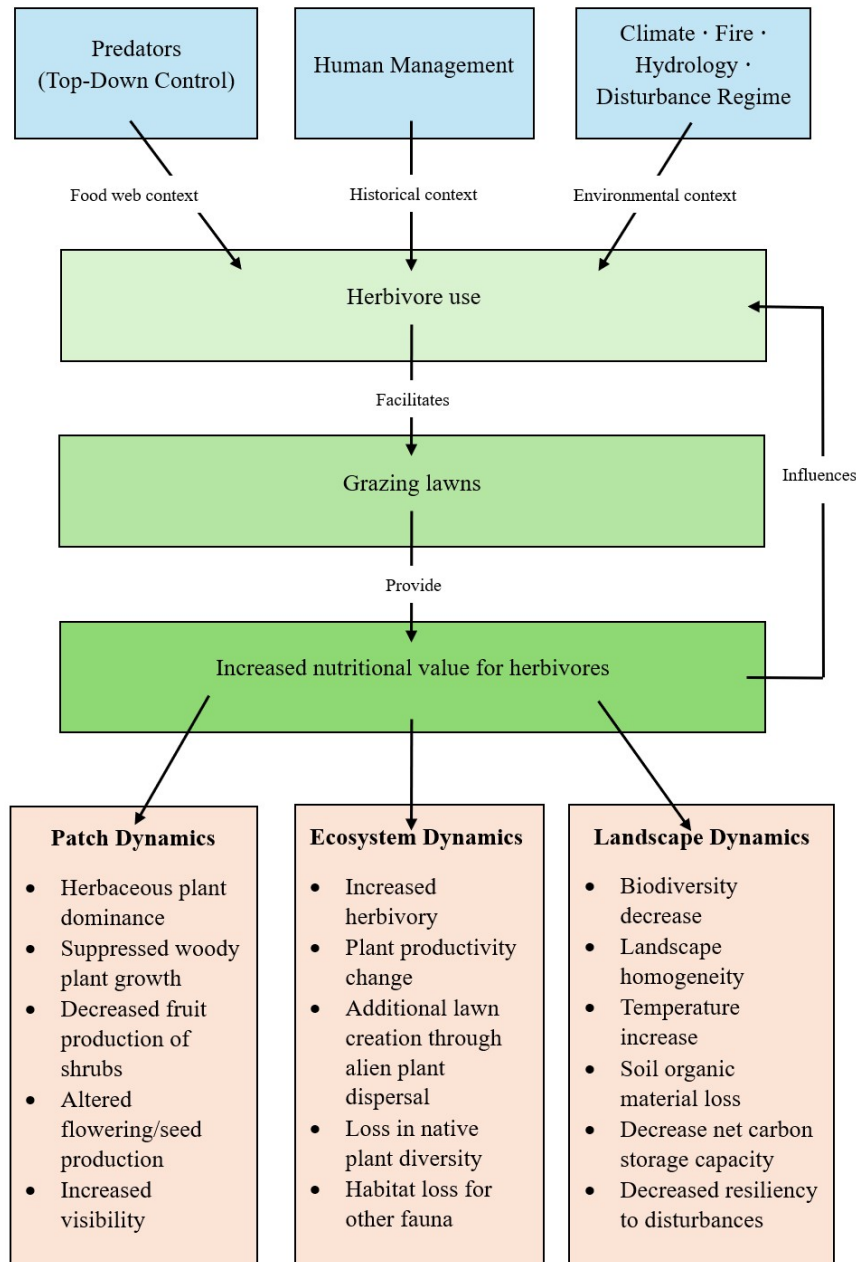


Figure 1.3: Conceptual framework diagram for grazing lawn maintenance in the temperate rainforest. This conceptual framework diagram illustrates the feedback-driven ecological systems that facilitate the maintenance of grazing lawns in the temperate coastal rainforest ecosystem. At the top of the diagram in blue are environmental, historical, and food web influences that provide context for the effects native herbivores have on system dynamics. In the middle of the diagram in green are the processes and feedback that occur in the system that facilitate grazing lawn patch creation and maintenance. At the bottom of the diagram in orange are the potential characteristics and effects associated with grazing lawn presence in the system, organized by the spatial scales they influence.

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2. CHAPTER 2: A Systematic Review of Grazing Lawn Ecology

2.1 Abstract

Grazing lawns, defined as “a patch with increased nutritional value and short statured vegetation maintained by grazing herbivores” have been described in several ecoregions worldwide. Although several reviews of grazing lawn ecology exist, a systematic synthesis of where grazing lawns occur, which animals maintain them, and how they are defined has not been conducted. I conducted a systematic review of all grazing lawn journal articles made since 1984 when the first paper on the subject was published. Following the preferred reporting items for systematic reviews and meta-analyses (PRISMA) guidelines for ecology and evolutionary biology, I recorded the following for each selected paper: (1) the research location site name, protection status, ecoregion, biome, and continent, (2) each animal species present in the study area, and (3) the grazing lawn definition provided. Using these data, I examined if predators and predation effects are lacking in the study designs of previous grazing lawn journal articles, identified geographic and taxonomic patterns in grazing lawn studies, and evaluated how grazing lawn definitions have changed over time. Predators were present in most study systems but were incorporated into relatively few study designs, suggesting their influence on grazing lawn dynamics may be underappreciated. Additionally, grazing lawn research has predominantly taken place

within protected areas, the protection of which may influence the formation and maintenance of grazing lawns. Furthermore, recently identified grazing lawn maintaining herbivores are gregarious by nature, reaffirming gregariousness as an influential factor in a species ability to create and maintain grazing lawns. Modern grazing lawn definitions include bunch grasses, sedges, and forbs as possible plant types in grazing lawns as opposed to being composed of stoloniferous (lawn) grasses exclusively. My review suggests that many traditional assumptions about grazing lawns, largely based on African systems, may need revision as research expands to new ecosystems and herbivore assemblages around the world.

2.2 Introduction

Grazing lawn ecology examines the feedbacks between herbivores and vegetation that create patches of short, nutrient-rich vegetation within heterogeneous landscapes. Concentrated herbivory can cause the formation of patches with short statured vegetation and increased nutrients, distinguishable from surrounding habitats. These patches attract additional herbivores, resulting in a cyclical interaction maintaining the characteristics of these patches, known as grazing lawns. Grazing lawn ecology has been researched by ecologists for more than four decades, during which the number of ecosystems associated with grazing lawns have increased, numerous processes maintaining lawns have been examined, and alterations to the defining characteristics of lawns have been made.

Many studies have examined the ecological mechanisms that create grazing lawns. Such mechanisms include nutrient redistributions by animal dung (Johnson

and Matchett 2001) and urine (Kaspari 2020) deposits, plant production (Veldhuis et al. 2016) and nutrient (Coetsee et al. 2011) responses to herbivory, soil compaction (Veldhuis et al. 2014) effects, the influence of herbivore size (van der Plas et al. 2016) and number of individuals in a group (McNaughton 1984), ecosystem engineering by megaherbivores (Waldram et al. 2008), and predation risk (Martin et al. 2015, Sedinger et al. 2016, Fležar et al. 2019). Likewise, several reviews of grazing lawn ecology have been published. These include summaries of African savannah grazing lawn ecology (Hempson et al. 2015), comparisons of the ecological effects between browsing and grazing herbivores (du Toit and Olf 2014), proposals of mechanisms responsible for grazing lawn creation and maintenance (Howison et al. 2017, Kaspari 2020) and journal articles reviewing the impact a single species can have on grazing lawns (Voysey et al. 2023). However, a systematic review of grazing lawn ecology has not yet been conducted, creating an opportunity to synthesize the growing body of research on this topic.

I conducted a systematic review of all grazing lawn journal articles made since McNaughton's pioneering publication of the subject in 1984. I reviewed all journal articles on the subject and recorded information pertaining to where research has been conducted, what animal species are common, and how grazing lawns have been defined. I summarized the locations where grazing lawn research has taken place, the total number and frequency of animal species by taxonomic group and recorded the grazing lawn definition provided in each publication. Furthermore, I used my systematic review to examine whether predators or predation effects have been underappreciated in grazing lawn ecology, challenge the historical requirement

for lawn maintaining herbivores to be gregarious by nature, and suggest a modern definition for grazing lawns that accommodates all characteristics, habitats, and animals capable of maintaining grazing lawns worldwide. As such, I used the following research questions to guide my systematic review:

1. Have predators and predation effects been underappreciated in historical grazing lawn ecology research?
2. Where has grazing lawn ecology research been conducted?
3. Which animal species and taxonomic groups dominate the grazing lawn literature?
4. How are grazing lawns defined in ecological literature?

2.3 Methods and Materials

2.3.1 Search Strategy

I structured my systematic review following the PRISMA extension for ecology and evolutionary biology (O’Dea et al. 2021) and registered my review on Research Gate on April 28th of 2025 (Woodrich 2025). In May of 2025, the terms “grazing lawn” and “browsing lawn” were searched for in the titles, abstracts, and keyword plus fields of journal articles using the Web of Science search engine’s query builder function. I included browsing lawns in my search since browsing effects are often considered in grazing lawn research (Fornara and Toit 2007). However, during the screening process I omitted journal articles that exclusively consisted of browsing lawn research since browsing lawns and grazing lawns are not completely synonymous with one another. My search query was formatted as the

following: “((((TI=(browsing lawn)) OR TI=(grazing lawn)) OR KP=(browsing lawn)) OR KP=(grazing lawn)) OR AB=(browsing lawn)) OR AB=(grazing lawn)”.

2.3.2 Article Screening and Eligibility Criteria

I used a two-step screening process to filter qualifying journal articles from the initial search. First, the abstracts of all journal articles were read and excluded based on the presence of any of the following characteristics:

- The publication is not a peer-reviewed journal article
- The publication is an opinion article, and does not provide data analysis
- The publication is not in English
- The publication is a review article
- The topic of the publication is clearly unrelated to lawn ecology based on the title and abstract alone

Second, qualifying journal articles were read in their entirety and omitted based on the presence of any of the following characteristics:

- The publication is unrelated to grazing lawn ecology
- The publication pertains only to browsing lawn ecology, and does not include grazing lawn ecology
- The publication is related to grazing lawn ecology but involves livestock during the time research took place
- The publication uses “lawn” and not “grazing lawn” to describe its study site

- The publication briefly mentions grazing lawns but does not primarily involve lawn ecology research

2.3.3 Data Collection

I collected the following data from qualifying journal articles: (1) the trophic levels involved in research, (2) the research location, (3) the animals identified in the study area, and (4) the definition used for grazing lawns. I recorded the trophic levels or trophic interactions explicitly discussed within the methods section of each publication to quantify the percentage of grazing lawn journal articles that considered predation effects or predators in their study design. I predefined possible trophic level categories to standardize data collection, consisting of predators, herbivores, and producers. If the methods section mentioned predators, herbivores, or producers, the trophic level was recorded. Journal articles that included more than one trophic level (e.g., herbivores and producers) were recorded as involving trophic interactions.

I recorded the presence of grazing lawn herbivore preying predators as “confirmed”, “likely”, “unlikely”, “absent” or “unknown” for all journal articles using two methods of validation. First, if a journal article confirmed the presence of predators within the geospatial boundary of an area, then all subsequent journal articles that took place in the same area were recorded as “likely” to have predators present. Second, if predator presence in an area was not mentioned in any qualifying journal articles, I searched for additional literature that described predator populations and assemblages, or lack thereof, within each area.

I recorded the following geospatial data for each study site identified in the methods of all journal articles: (1) the name of the study site, (2) the ecoregion the site resides in, (3) the biome the site resides in, and (4) if the site resides in a protected area. I recorded ecoregion and biome data by cross referencing the geospatial coordinates, or general location described for each study site, to their respective location on the Ecoregions 2017 world map (Dinerstein et al. 2017). I recorded all game reserves, national parks, wildlife refuges, and nature reserves, as protected areas. If a laboratory experiment or a simulation model was used for research, I searched for information regarding where the variables used for experimentation were collected from or information regarding the location the model pertains to for recording geospatial data.

I recorded the presence of all animal species identified within the study areas of each journal article. If any animals were mentioned in a journal article, but not identified to a species level, I recorded it as “[genus name] species”. Taxa referenced at levels higher than genus were not recorded. If a journal article differed to a supplemental section or a previous journal article to disclose animal information, I did not review the additional material. I standardized common and scientific names and recorded the taxonomic classifications of all animal species using the National Center for Biotechnology Information (NCBI) database as reference (Sayers et al. 2024).

I searched for grazing lawn definitions within the entirety of qualifying journal articles. If a journal article had one or more sentences that “characterizes”, “identifies”, “refers to”, or “defines” grazing lawns or ecological phenomena related

to grazing lawn ecology, such as the “formation”, “creation”, “resulting”, or “maintenance” of grazing lawns, I recorded the sentences as a grazing lawn definition. If the grazing lawn definition referenced previous journal articles, I recorded all journal articles mentioned for subsequent analysis.

2.3.4 Data Analysis

I compared the total number of journal articles that included predators or predation effects in their study design to the total number of journal articles where grazing lawn predators were “confirmed” or “likely” to be present. I tallied the total number of study sites in qualifying journal articles by site name, ecoregion, biome, continent, and by protection status to summarize location data.

I tallied the total number of journal articles each animal species was identified, represented by the value (n), and tallied the number of instances each animal was identified by their taxonomic family, order, and class. I divided the taxonomic summaries by the total number of instances all animals were identified to determine which taxonomic groups were most common in grazing lawn ecology research. I calculated animal richness by tallying the total number of unique animals identified by species. Any animals identified by genus, and not to a species taxonomic level, were not included in my animal richness calculation.

I reviewed each definition provided and assigned thematic categories to words or phrases used within them. The two predetermined thematic categories I used were: (1) “ecological processes/interactions” and (2) “ecological characteristics”. I summarized the total number of definitions that depicted each category type.

Additionally, I identified the ten most frequently used words across all grazing lawn definitions. The words, “grazing”, “lawn(s)”, “characterized”, and “defined” were excluded from this process. Finally, I tallied the most frequently referenced journal articles cited within the definitions I identified.

I used R for all data analysis (R Core Team 2022), ggplot (Wickham et al. 2025), cowplot (Wilke 2025) packages for graph creation, the sankey (Csárdi and Weiner 2017) package for taxonomic illustrations, the tidytext (Queiroz et al. 2025) package for identifying common words in grazing lawn definitions, and the taxize (Chamberlain et al. 2025) package for assigning NCBI values to animals.

2.4 Results

My initial search for grazing lawn journal articles yielded 252 results, with 77 total journal articles qualifying for the systematic review following my screening process (Figure 2.1). Grazing lawn journal articles were rare for the first 16 years after McNaughton’s (1984) initial paper was published but increased in annual journal article frequency after the year 2000 (Figure 2.2).

I found grazing lawn journal articles rarely integrated predators or predation effects (17% of journal articles) in their study design. (Figure 2.3 A). However, predators of grazing lawn herbivores were “confirmed” (16% of journal articles) or “likely” (65% of journal articles) to be present in the areas where grazing lawn research was conducted (Figure 2.3 B).

My review indicates grazing lawn research has been conducted on four continents and is most predominant in Africa (60% of journal articles) and North America (27% of journal articles) (Figure 2.4). Each journal article had study sites

within the confines of one continent, while several journal articles had study sites across multiple ecoregions and biomes. Grazing lawn study sites were most common in tropical and subtropical grasslands, savannas, and shrublands (62% of study sites), and tundra biomes (11% of study sites) (Figure 2.4). Within these two biomes, grazing lawn study sites were most common in the Limpopo lowveld (28% of study sites), the Beringia lowland tundra (10% of study sites), and the Zambezian mopane woodland (9% of study sites) ecoregions (Appendix Table 2.1). Grazing lawn study sites were most frequently located within the boundaries of Hluhluwe–iMfolozi Park (23% of study sites) and Kruger National Park (8% of study sites) in Africa, and in the Yukon-Kuskokwim Delta Wildlife Refuge (10% of study sites) in North America. Finally, study sites of grazing lawn journal articles were predominantly within protected areas (94% of study sites), defined as national parks, wildlife refuges, game reserves, nature reserves, and areas clearly articulated by the authors as protected (Appendix Table 2.2).

I identified 121 animal species in grazing lawn journal articles, consisting of 99 species of mammals, 14 species of birds, and 8 species of insects (Appendix Tables 2.3 – 2.19), and provide the first visualization of all animals by taxonomic class, order, and family in grazing lawn ecology (Figure 2.5 A). By taxonomic class, I found mammals (Mammalia) were most common in grazing lawn research (88%), followed by birds (Aves) (8%) and insects (Insecta) (4%) to a much lesser extent. By taxonomic order, even-toed (Artiodactyla) and odd-toed ungulates (Perissodactyla) were most common, and the most common family examined were species of the cloven-hoofed ruminant mammal (Bovidae) family. Finally, I provide a

list of the ten most frequently researched animals in grazing lawn journal articles (Figure 2.5 B) and a list of all predators, consisting of species in the Carnivora order, identified in grazing lawn journal articles (Figure 2.5 C).

Most qualifying journal articles provided a grazing lawn definition (75% of journal articles), of which, a previous publication was commonly referenced in their definition (72% of journal articles with grazing lawn definitions) (Figure 2.6).

Definitions frequently described both the “ecological processes/interactions” and the “ecological characteristics” of grazing lawns, and used the terms “grass/grasses”, “short”, “species”, “plant/plants/vegetation”, “graze/grazed/grazer”, “maintain/maintained”, “herbivores”, “stoloniferous”, “patches”, and “nutrient” most frequently in their definitions. The most frequently referenced journal articles in grazing lawn definitions were McNaughton (1984) (23 references), Hempson et al., (2015) (10 references), Person et al., (2003) (6 references), and Croomsigt & Olff (2008) (5 references). Each grazing lawn definition identified, and the thematic categories I identified in them, are provided in Appendix Table 2.20, and a reference list of all qualifying journal articles for this systematic review is provided in Appendix Table 2.21.

2.5 Discussion

2.5.1 Are predators and predation effects underappreciated in grazing lawn ecology research?

I evaluated whether predators are underappreciated in grazing lawn research. Predators were confirmed in 16% and likely to be present in 65% of the study areas of journal articles while only 17% of journal articles integrated predators into the study

design. As such, predators are frequently not considered, or at the very least, unincorporated in the study designs and research questions of grazing lawn journal articles. I identified 23 predator species (members of the Carnivora order) present in the study areas of grazing lawn journal articles. In the grazing lawn journal articles that considered predators or predation effects in their study design, nine journal articles identified predation risk as influencing (Martin et al. 2015, Sedinger et al. 2016, Uher-Koch et al. 2019, Fležar et al. 2019, Voysey et al. 2021, Avila et al. 2025) or possibly influencing (Stewart et al. 2006, Thapa et al. 2022) grazing lawn herbivore behavior, two journal articles identified grazing lawns as the preferred hunting habitat by apex predators (Kittle et al. 2021, Yiu et al. 2022), and three journal articles acknowledged predators as present, but did not comment on their impact on grazing lawn ecology (Rooney 2009, Thapa et al. 2021, 2023). In 1984, McNaughton pondered whether the evolution of gregarious behavior of grazing lawn maintaining herbivores in the African savanna occurred as a mitigation strategy to predation. Furthermore, other grazing lawn reviews have emphasized predation as an integral component responsible for the creation and maintenance of grazing lawns (Hempson et al. 2015, Monk and Schmitz 2022). Why, then, are predators and predation effects overlooked so frequently in grazing lawn journal articles?

Despite widespread recognition that predators can shape herbivore behavior through behaviorally mediated trophic cascades (BMTC), most grazing lawn studies do not explicitly incorporate predators into their study design. When predators are present, behavioral responses of grazing lawn herbivores can be species dependent (Fležar et al. 2019), and may be offset by other, stronger, environmental or ecological

factors influencing herbivores, such as water availability (Martin et al. 2015), fire disturbance (Donaldson et al. 2018), or plant productivity (Veldhuis et al. 2016). Likewise, grazing lawn patches can benefit herbivores by increasing predator visibility (Hempson et al. 2015), but can also benefit predators by providing areas where prey is routinely present (Yiu et al. 2022). These confounding effects make it difficult to differentiate how predators influence grazing lawn formation and maintenance in general and restrict predator-prey effects to be site specific. Moreover, the historical emphasis of producers and herbivores interactions (70% of journal articles) in lawn journal articles also accounts for predator absence.

In general, predators may influence grazing lawn ecology by altering herbivore behavior and encouraging grazing in areas with lower perceived predation risk. For instance, predator presence can instill BMTC by increasing predation risk, causing herbivore behavior to change, resulting in habitat and landscape changes (Schmitz et al. 1997, Palmer et al. 2021). One such change may be the formation and maintenance of grazing lawns. BMTC research has been conducted in areas where grazing lawns are predominant, specifically Hluhluwe–iMfolozi Park (HIP) in Africa. In HIP, grazing lawn mesoherbivores concentrated in experimentally manipulated open spaces when predation risk was increased (le Roux et al. 2018), indicating grazing lawns may offer additional benefits to herbivores besides the usual increase in nutrient concentrations for grazing. Some predators identified in my review are already known to instill BMTCs, including leopards (*Panthera pardus*) (Ford et al. 2014, Atkins et al. 2019) and pumas (*Puma concolor*) (Donadio and Buskirk 2016, Lundgren et al. 2022). Likewise, some herbivores identified in my review are also

known to alter their feeding habits in response to increased predation risk, such as white-tailed deer (*Odocoileus virginianus*) (Cherry et al. 2016, Palmer et al. 2021), red deer (*Cervus elaphus*) (Kuijper et al. 2013), and mule deer (*Odocoileus hemionus*) (Ripple and Beschta 2008). If predators influence herbivores to graze in areas where visibility is high, subsequent grazing in these areas may promote the formation or persistence of grazing lawns. However, because herbivore predation has been historically overlooked in grazing lawn journal articles, additional research examining the direction and strength predation has on grazing lawn ecology must be conducted. In doing so, the presence of grazing lawns may be documented as an additional cascading reaction to predator presence, caused by a BMTC.

I offer three insights for future grazing lawn research based upon my review of predation in grazing lawns: (1) Previously unreported grazing lawns may persist in landscapes where herbivore predation risk is high. (2) Herbivores known to alter their feeding habits in response to increased predation risk may create grazing lawns. (3) Predation risk may influence the formation and maintenance of grazing lawns, and should be examined, or at least estimated, in future grazing lawn research.

2.5.2 Where has grazing lawn ecology research been conducted?

I found grazing lawn research has been conducted in most of Earth's biomes (11 of 15 biomes as defined by Dinerstein et al. 2017). Within these biomes, the ecoregions with the greatest number of study sites were the Limpopo lowveld and the Zambezian mopane woodland ecoregions of Africa, and the Beringia lowland tundra in North America, along with 23 other ecoregions world-wide. Grazing lawns have

been described in several new ecoregions within the previous half-decade, including the Sri Lanka dry-zone dry evergreen forests (Kittle et al. 2021), Chilean Matorral woodlands (Root-Bernstein et al. 2021), the Canadian low arctic tundra (Kellett and Alisauskas 2022), the Terai-Duar savanna and grasslands of Nepal (Thapa et al. 2021, 2023), and the humid chaco grasslands of Argentina (Avila et al. 2025, Mata et al. 2025). There are many differences between the ecoregions where grazing lawns persist, including, but not limited to, variations in species assemblages, climate, and natural disturbances. Most critically, however, almost all grazing lawn study sites were within protected areas (94% of study sites).

In their review of African grazing lawns, Hempson et al. (2015) identified frequent grazing by herbivores as the primary driver for grazing lawn formation and maintenance. I suggest that protected areas may promote more frequent and concentrated grazing patterns by herbivores in comparison to unprotected areas. For example, protected areas in Africa have grown steadily in number and size since the 1970s (Newmark 2008), and can increase the density of large herbivores significantly compared to areas with partial or no protection (Stoner et al. 2007). Likewise, geese populations rely on the protected areas of the Yukon-Kuskokwim Delta Wildlife Refuge for resources during the breeding season, including >70% of the global breeding population of the grey bellied brent goose (*Branta bernicla nigricans*) (Sedinger et al. 1993, 2019), among several other geese species. Alternatively, grazing lawns may persist as frequently in unprotected areas. For instance, protected areas can exemplify the baseline ecological conditions of the ecoregions they persist in (Arcese and Sinclair 1997), making them well-suited for research examining the

feedbacks between herbivores and their resources. As such, protected areas may be more appealing for grazing lawn research than unprotected areas, explaining their prolific abundance amongst grazing lawn journal articles. Research comparing grazing lawn characteristics in protected and unprotected areas should be conducted to identify whether a preference for protected areas is human caused, or if the protection of landscapes promotes the formation and maintenance of grazing lawns by native herbivores.

2.5.3 Which animal species and taxonomic groups dominate the grazing lawn literature?

I compiled the first comprehensive list of all animals in grazing lawn journal articles and summarized the most dominant species and taxonomic groups in grazing lawn literature. Worldwide, researchers have studied 121 animal species in grazing lawn journal articles. I found the Bovidae family had the highest species richness of all animal families identified (28 species). Additionally, several of the top ten most frequently identified animals in grazing lawn literature are members of the Bovidae family. Bovid dominance in grazing lawn literature may be due to the high proportion of grazing lawn research conducted in the African savannah where cloven-hoofed ruminant mammal richness is high. Alternatively, many African members of the Bovidae family are gregarious by nature (Estes 1991), a beneficial characteristic for maintaining grazing lawn hypothesized by McNaughton (1984). In the following paragraphs, I introduce the most recently identified lawn maintaining herbivores, provide examples of uncommon animals associated with grazing lawns, and discuss the impact these species have on grazing lawn ecology.

Gregariousness is hypothesized to be an integral component in the maintenance of grazing lawns due to the increased grazing pressure by large groups of herbivores as opposed to solitary grazers (McNaughton 1984). I found gregarious herbivores were prevalent amongst the most frequently researched animal species in grazing lawn journal articles, including animals not part of the Bovidae family, such as geese (*Branta* and *Anser* species) (Akesson and Raveling 1982), zebras (*Equus* species) (Chen et al. 2025), warthogs (*Phacochoerus* species) (Jørgensen et al. 2024), and chital (*Axis axis*) (Ghosal and Venkataraman 2013). I examined whether recently identified herbivores in grazing lawn literature were also gregarious. I compiled a list of all herbivores identified in only one or two journal articles, then revisited the journal articles they were mentioned in to identify if they were depicted by the authors as lawn maintaining. I identified 14 grazing lawn maintaining herbivores, all of which were gregarious by nature. These species consist of Canada geese (*Branta canadensis*) (Kotanen and Abraham 2006), Orinoco geese (*Neochen jubata*) (Davenport et al. 2012), Ross's geese (*Anser rossii*) (Kellett and Alisauskas 2022), emperor geese (*Anser canagicus*), cackling geese (*Branta hutchinsii minima*), white-fronted geese (*Anser albifrons frontalis*) (Lake et al. 2008, Uher-Koch et al. 2019), western swamp deer (*Rucervus duvaucelii duvaucelii*) (Karki et al. 2000), white-tailed deer (*Odocoileus virginianus*) (Rooney 2009), kob (*Kobus kob*) (Verweij et al. 2006), degu (*Octodon degus*) (Root-Bernstein et al. 2021), capybara (*Hydrochoerus hydrochaeris*) (Avila et al. 2025, Mata et al. 2025), blesbok (*Damaliscus pygargus phillipsi*), springbok (*Antidorcas marsupialis*) and black wildebeest (*Connochaetes gnou*) (Novellie and Gaylard 2013). Historically, there have been two general size

classes of grazing lawn maintaining herbivores, large-bodied mammals in Africa (McNaughton 1984) and small-bodied birds in North America (Person et al. 2003). In both size classes, the number of individuals in a group of grazing lawn maintaining herbivores can be in the thousands of individuals, such as the grey bellied brent goose (Person et al. 1998), or the brindled gnu (*Connochaetes taurinus*) (Torney et al. 2018). I found recent exceptions to this size class generality, such as the medium-bodied capybara in South America, or micro-sized termite species in Africa (Root-Bernstein et al. 2021). Furthermore, some recently identified herbivores are capable of maintaining lawns in far smaller group sizes, such as the kob, ranging from 8 - 20 individuals per group (Verweij et al. 2006). The recent identification of these species indicates herbivores capable of maintaining grazing lawns are still predominantly gregarious by nature, supporting McNaughton's (1984) hypothesis over forty years later. However, the group size and physical size of grazing lawn herbivores may not be as influential in their capabilities to maintain lawns as previously perceived.

There is an additional size class of herbivores known to maintain grazing lawns in singularity. Megaherbivores, crucial in their ability to create dynamic changes in vegetation structure and food-web dynamics as an individual, are at risk of complete collapse world-wide (Ripple et al. 2015). Since megaherbivores are at greater risk of extinction, I revisited all journal articles with megaherbivores and reviewed the impacts each species had on grazing lawn ecology. I identified nine megaherbivore species, defined as species with an average mass of >1,000 kg, as present in the study areas of 32 journal articles (42% of journal articles).

In most instances, megaherbivore species were identified within the study area of a publication, and not integral to the research being conducted. However, I identified some journal articles that specifically examined the impact megaherbivores have on grazing lawn ecology. For instance, van der Plas et al. (2016) studied the impact the white rhinoceros (*Ceratotherium simum*) has on grazing lawn plant composition in contrast to other mesoherbivores, identifying the white rhinoceros to be influential in the long-term functional composition of landscapes with grazing lawns. In other journal articles, the white rhinoceros was capable of instilling landscape-level changes by increasing the proportion and size of grazing lawns (Cromsigt and te Beest 2014), and serving as an ecological engineer for grazing lawn habitat by altering fire regimes (Waldram et al. 2008). Likewise, Fležar et al. (2019) found mesoherbivores species reacted differently to simulated browsing by another megaherbivore species, the African savannah elephant (*Loxodonta africana*), indicating the megaherbivore can influence the perceived risk of predation in grazing lawns. Another megaherbivore, the hippopotamus (*Hippopotamus amphibius*), can alter the structure of plant communities, soil nutrient composition, and influence other herbivores' use of grazing lawns (McCauley et al. 2018). In Nepal, the greater Indian rhinoceros (*Rhinoceros unicornis*) and the Asian elephant (*Elephas maximus*) are likely to open up tall grassland areas for mesoherbivores to establish as grazing lawns (Karki et al. 2000). Finally, American bison (*Bison bison*) increase grasshopper abundance in grazing lawns (Ozment et al. 2021) and can provide habitat for aquatic insects and amphibians in the wallows they generate within them (Horne et al. 2024). The remaining megaherbivores I identified, consisting of giraffes (*Giraffa*

camelopardalis & *G. giraffa*) and the black rhinoceros (*Diceros bicornis*) were identified in the study area of grazing lawn journal articles, but have not had their ecological impacts on grazing lawns assessed.

Generally, I found megaherbivores influence where grazing lawns persist by disturbing or removing non-grazing lawn associated plants, resulting in the subsequent use of disturbed patches by smaller herbivores. While the white rhinoceros has already been labeled as an ecological engineer for grazing lawns (Waldram et al. 2008), the other megaherbivore species I identified may also serve as ecological engineers in mosaic landscapes by influencing where grazing lawn patches form. If megaherbivore populations decrease in the study areas they were previously identified in, or if megaherbivores are reintroduced in study areas where they have been absent in previous journal articles, opportunities to examine the magnitude of their effects on grazing lawn ecology will present themselves.

2.5.4 How are grazing lawns defined?

I found journal articles that provided a grazing lawn definition commonly described both the ecological characteristics, and the ecological processes/interactions of grazing lawns. To paraphrase, grazing lawns were commonly depicted as “patches” of “short statured” “stoloniferous” “grasses” or “plant” “species” with increased “nutrients” “maintained” by “grazing” “herbivores.” I recognized a deviation in the types of plants mentioned in grazing lawn definitions over time, with earlier definitions specifying the presence of stoloniferous grasses (Bond et al. 2001, Archibald et al. 2005, Waldram et al. 2008, and several others) and later definitions

including bunch grasses (Veldhuis et al. 2018), sedges (R. W. Ruess et al. 2019, Foley et al. 2022, Saunders et al. 2025), and forbs (Horne et al. 2024). Grazing lawn plant composition also differed by continent, with African grazing lawn definitions mentioning grasses commonly (29 of 46 African journal articles with grazing lawn definitions) while non-African grazing lawn definitions rarely mentioned grasses (3 of 31 non-African journal articles with grazing lawn definitions). Instead, definitions for non-African grazing lawns commonly used plant heights, increased nutrient concentrations, or grazing tolerance to describe grazing lawns (Johnson and Matchett 2001, Brian D. Uher-Koch et al. 2019, Kellett and Alisauskas 2022, Thapa et al. 2022, Mata et al. 2025). It is difficult to differentiate whether the inclusion of non-grass vegetation resulted from new grazing lawns being identified in ecoregions where grasses are, generally, less prevalent compared to grassland, savanna, and shrubland biomes, or if the inclusion of additional herbaceous vegetation types has occurred over time as grazing lawn journal articles increased in frequency. Nonetheless, recent definitions of grazing lawns represent a stark contrast to earlier ones in plant composition descriptions. As such, modern grazing lawn definitions provide greater breadth in the number of ecoregions that could possibly host grazing lawns compared to historical ones.

Grazing lawn size descriptions were absent in 25 of the 58 journal articles that provided grazing lawn definitions. Three journal articles described the size of their grazing lawns explicitly, including areas “a few square meters to a couple of hectares” (Veldhuis et al. 2018), “>400 m²” (Johnson and Matchett 2001), and “<1 hectare (<10,000 m²)” (Thapa et al. 2021) in size. Of the remaining 30 journal

articles that provided a definition for grazing lawns, 11 journal articles used “patches” to describe their area, 6 used “swards” (Person et al. 2003, Stewart et al. 2006, Bonnet et al. 2010, Martin et al. 2015, Voysey et al. 2021, Hempson et al. 2022), 6 used “area” (Shackleton 1992, Coetsee et al. 2011, Donaldson et al. 2018, Awuah et al. 2020, Yiu et al. 2022, Thapa et al. 2023), 3 used “community” or “communities” (Archibald 2008, Gosling et al. 2012, Singh et al. 2024), 2 used “grasslands” or “grassland type” (Fleazar et al. 2019, Hyvarinen et al. 2023), 1 used “expanse” (Verweij et al. 2006), and 1 used “habitat type” (Archibald et al. 2005) as descriptive language for grazing lawn sizes. The lack of a consistent spatial definition for grazing lawns results from speculation regarding the expanse in which herbivores can maintain their resources and/or is influenced by the ability to recognize grazing lawns in certain ecoregions. For instance, identifying a grazing lawn in a forest is easier than in a grassland or tundra biome due to the stark contrast in plant structure and composition between lawn and forested habitat. Likewise, while grassland ecoregions are generally associated with communities of grazing herbivores, grazing lawns within the ecoregion can persist where herbivory pressure is concentrated, and the nutritional value of plants is increased, but may be harder to detect visually. The maintenance of grazing lawns in grasslands may also be influenced by many other ecological factors such as predation risk, megaherbivore presence, herd migration, fire regimes, which create gradual transitions between lawn patches and surrounding grassland habitats. Grazing lawns are likely to have been described in other ecological journal articles but not identified as such. For instance, Roberts et al., (2011) identified lawns maintained by native herbivores in the Tasmanian Central

Highland forests but did not label them as grazing lawns explicitly. As such, grazing lawns may persist in ecoregions beyond the ones identified in my systematic review, the boundaries of which may be easier to identify in some ecoregions while harder in others.

I define a grazing lawn as “a patch with increased nutritional value and short statured vegetation maintained by grazing herbivores,” omitting any description of plant species in my definition and reinforcing their spatial size as patches. However, I advocate grazing lawn plant assemblages and area estimations should continue to be described in subsequent journal articles. My definition grants the opportunity for new grazing lawns to be discovered and researched in habitats they have not been previously associated with. Furthermore, the omission of grasses as a characteristic of grazing lawns may coincide with the structural description of grazing lawns presented in McNaughton’s initial definition in 1984. Although grasses and grasslands are mentioned frequently throughout McNaughton (1984), grasses were used primarily to describe African savanna habitats and the coevolution between vegetation and herbivores within the region and may not have intended to be used as a defining feature for all grazing lawns. Instead, I suggest the interaction between grazing animals and their resources, increased nutrient availability, and short statured plants (regardless of plant type) define grazing lawns globally.

2.6 Conclusion

Grazing lawn ecology represents the study of a fascinating collection of diverse herbivores creating and maintaining nutrient hotspots distinguishable in

characteristic and function to surrounding landscapes. The number of journal articles related to the subject has steadily increased in the last two decades, revealing numerous additional ecoregions with grazing lawns and several new herbivore species capable of maintaining them. However, the influence predators and predation have on grazing lawn dynamics is overlooked in previous grazing lawn journal articles and should be examined further in future studies. Furthermore, I found grazing lawn research has expanded both in the number of animals capable of maintaining lawns and in the number of ecoregions where lawns persist. As such, I provide a modern definition for grazing lawns that encompasses the habitat characteristics of these recently identified ecoregions and encourages the search for additional habitats and animals that maintain them. I hope my review of the topic inspires research into the intricacies involved in grazing lawn ecology, sparks investigation for new grazing lawn maintaining animals, and generates discussion regarding the ecological characteristics and interactions necessary for grazing lawn creation and maintenance.

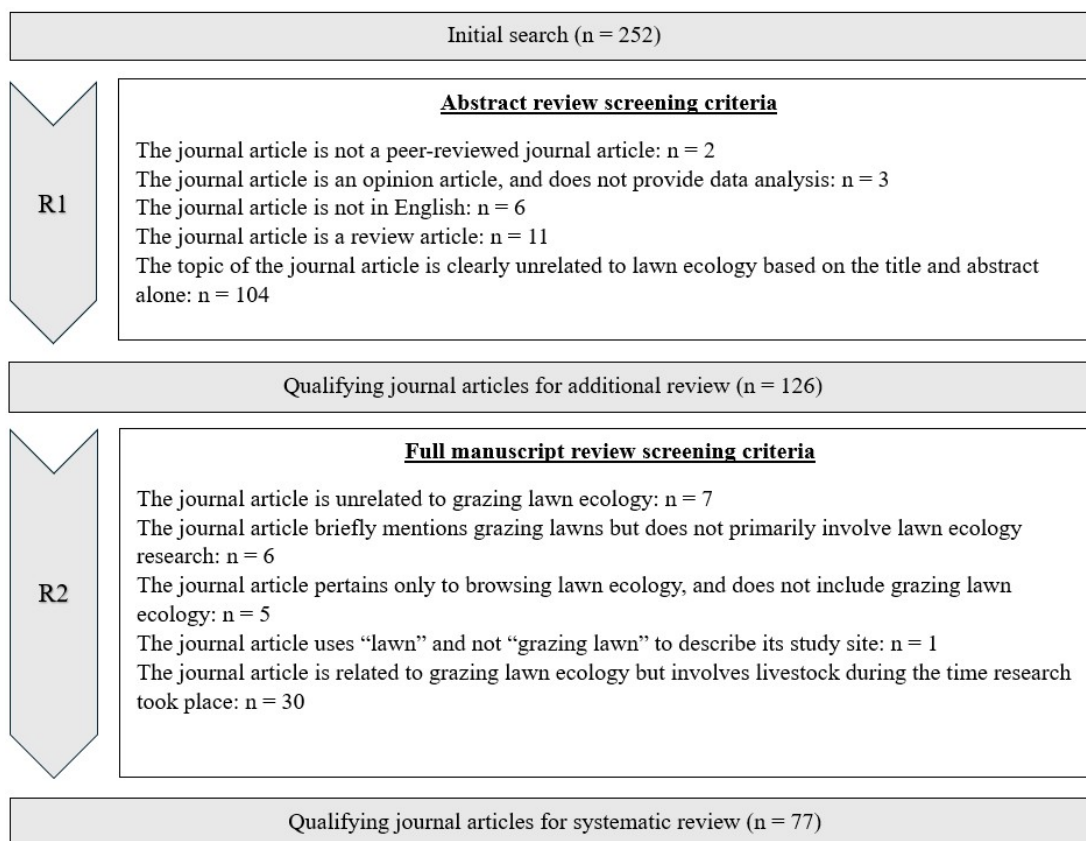


Figure 2.1: Flowchart of systematic review screening process. The flowchart depicts the screening process for grazing lawn ecology journal articles in my systematic review following my initial web of science search. The reason for omission for a paper and the number of journal articles omitted from systematic review for each reason are provided in text boxes to the right in each step of review process (R1 = first review of abstracts only, R2 = full manuscript review).

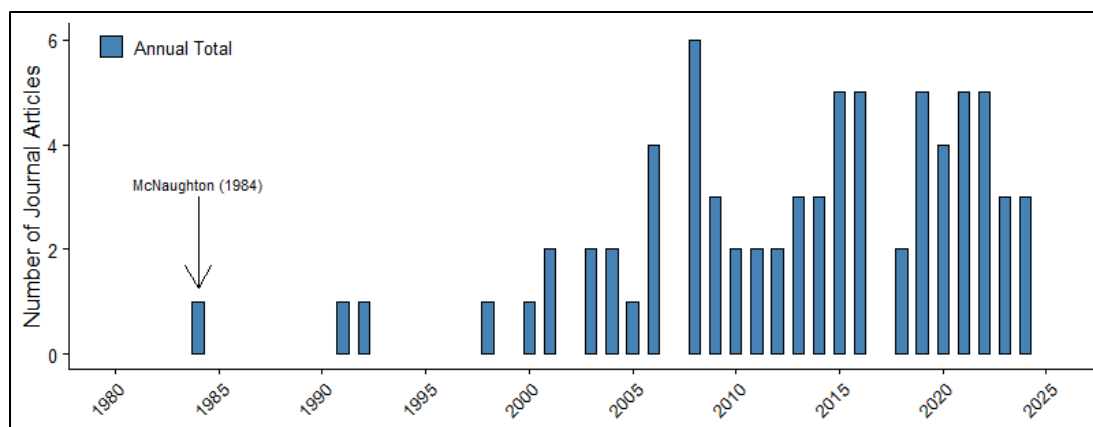


Figure 2.2: Bar graph of grazing lawn journal articles over time. The annual total of grazing lawn ecology journal articles per year is shown since McNaughton's first publication of the subject in 1984.

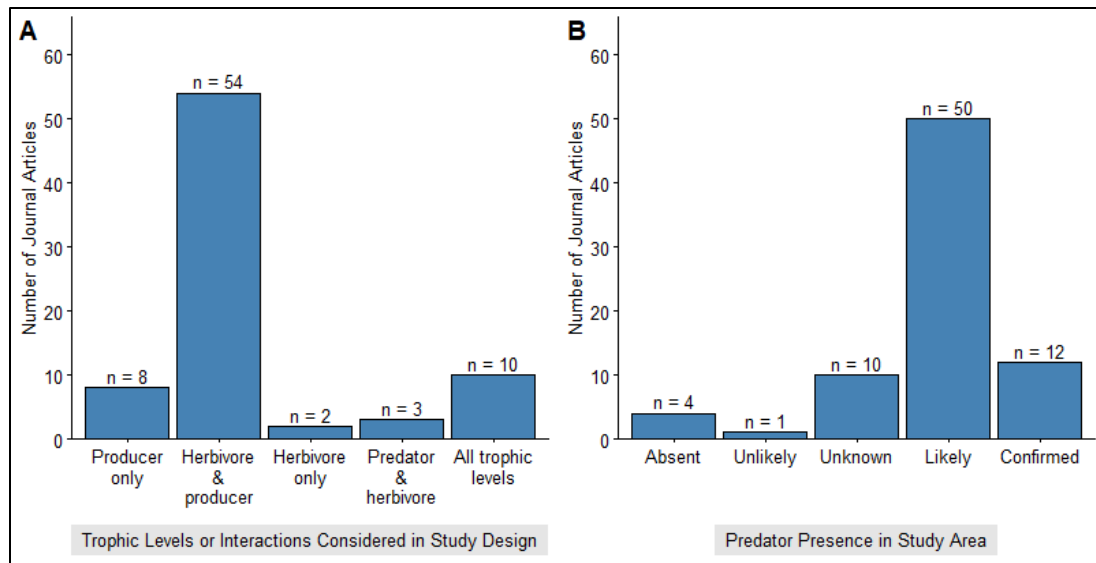


Figure 2.3: Bar graphs depicting trophic levels and predator presence likelihood in grazing lawn journal articles. (A) Distribution of grazing lawn journal articles by the trophic levels or trophic interactions considered in the study design of all journal articles. (B) Distribution of predator presence likelihood in the study sites in all grazing lawn journal articles.

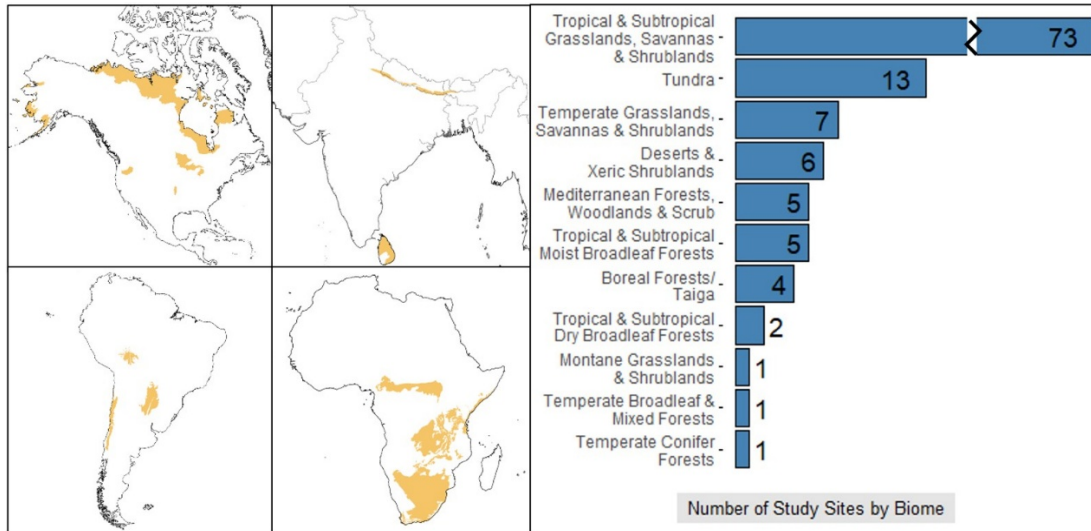


Figure 2.4: Maps and bar graphs depicting grazing lawn research locations. On the left side, maps of each continent and the boundaries of each ecoregion (in yellow) where grazing lawn study sites were identified in my systematic review. In clockwise order from the top left map, North America had 21, Asia had 6, Africa had 46, and South America had 4 total grazing lawn journal articles. Ecoregion boundaries are supplied by Dinerstein et al., 2017. On the right side, a bar graph representing the total number of study sites by biome where grazing lawn research has been conducted.

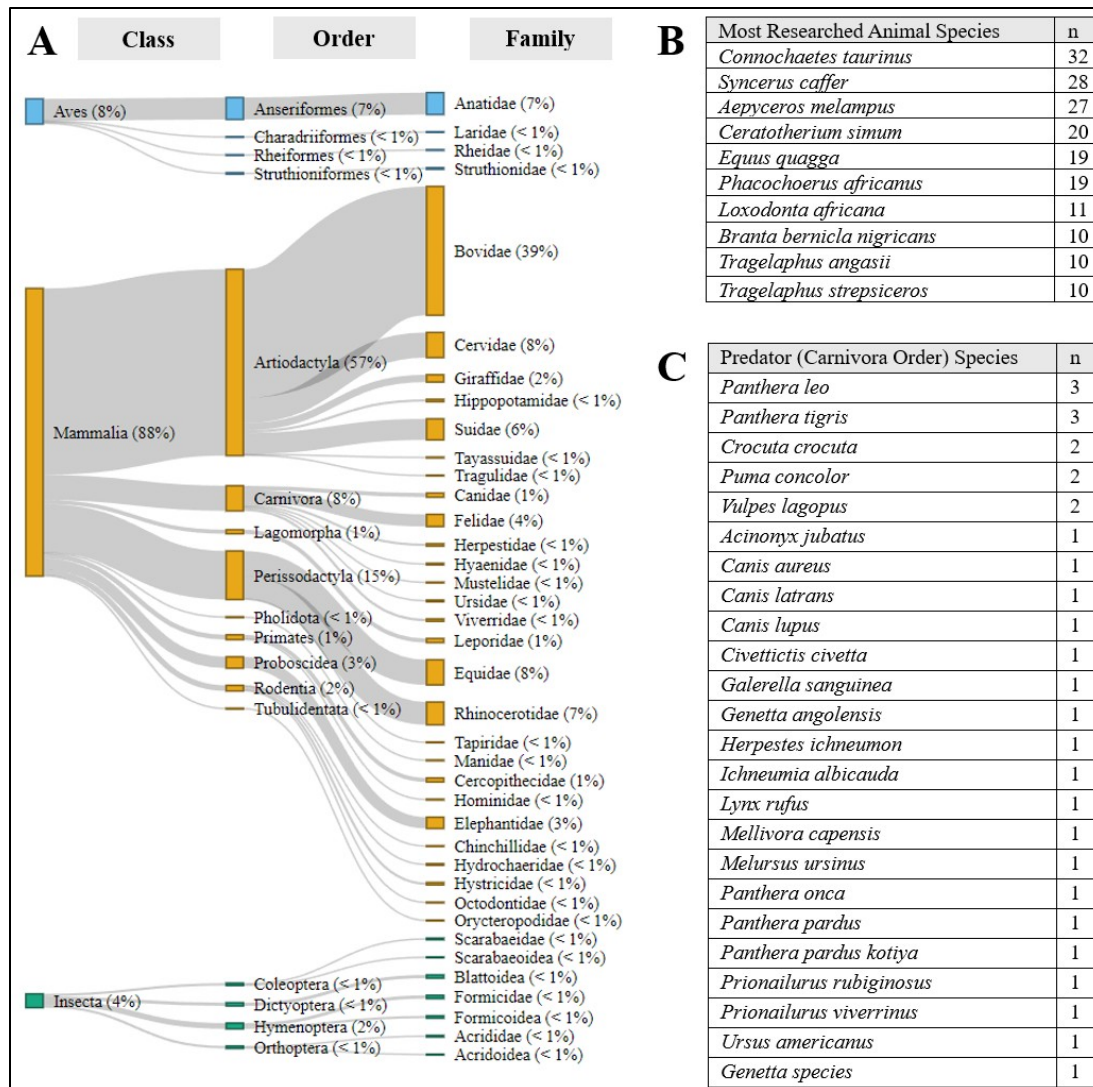


Figure 2.5: Sankey diagram and summary tables depicting all animal species and taxonomic groups identified in grazing lawn journal articles. (A) A Sankey diagram depicting the names and distributions of taxonomic groups identified in grazing lawn journal articles. Percentage values represent the total proportion of animals identified across all journal articles that belong to each respective taxonomic group. (B) A table of the ten most frequently identified animal species in grazing lawn journal articles. (C) A table of all 23 predator species (members of the Carnivora order) identified in grazing lawn journal articles. In tables (B) and (C), the total number of journal articles each species was identified is represented by (n).

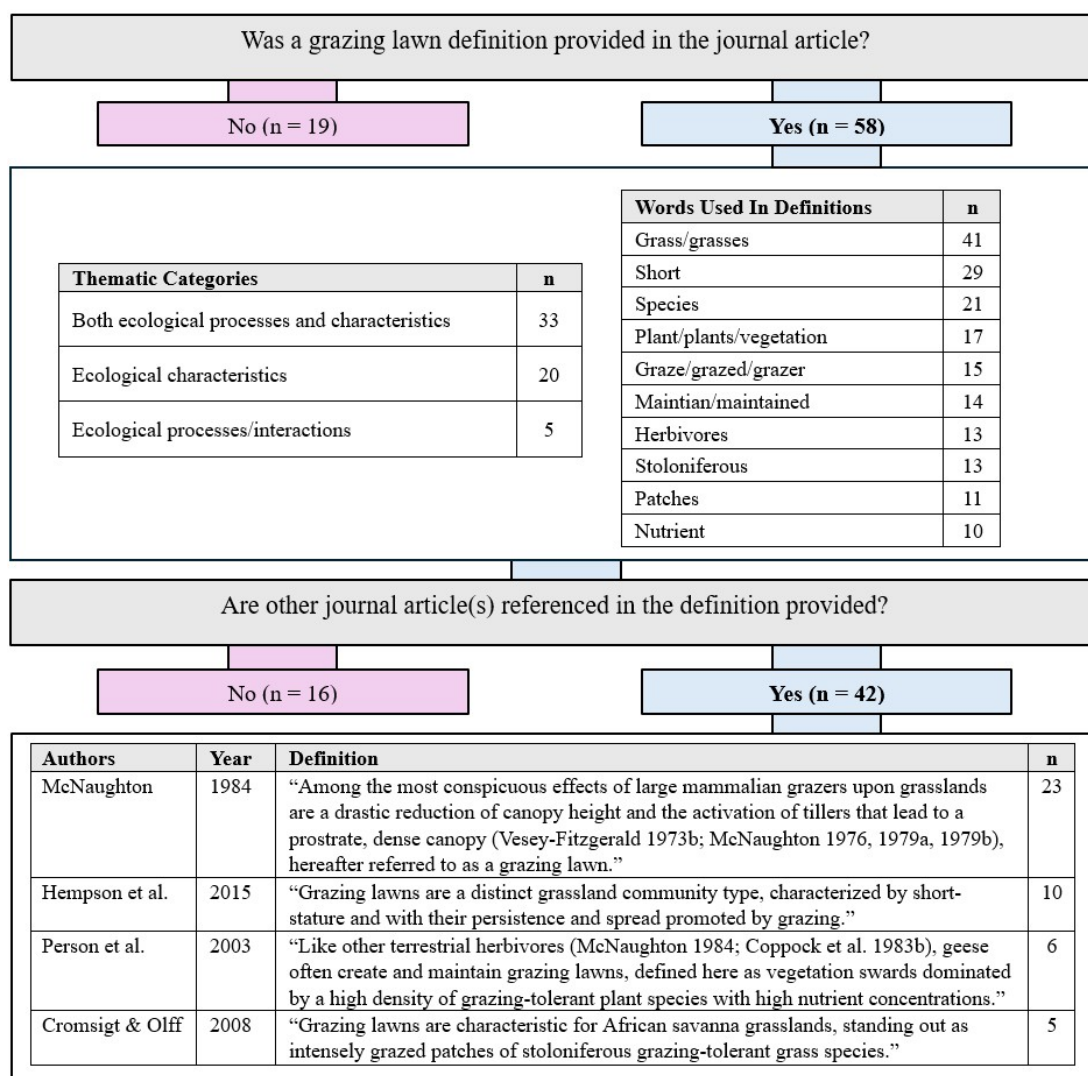


Figure 2.6: Flowchart diagram depicting commonalities in grazing lawn definitions. At the top of the flowchart, the number of journal articles that did and did not provide a grazing lawn definition, followed below by the distribution of thematic categories identified in lawn definitions and the most frequently-used words used in definitions. At the bottom of the flowchart, the number of journal articles that provided a citation of a previous source in their grazing lawn definition, followed by the four most referenced journal articles in grazing lawn definitions.

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3. CHAPTER 3: Ungulate-Maintained Grazing Lawns in a Temperate Coastal Rainforest

3.1 Abstract

Grazing lawns have not previously been documented in temperate coastal rainforests, but their formation may result from concentrated herbivory by native ungulates. I searched for grazing lawn habitats in forests along the central Oregon Coast and identified their distinguishing structural and compositional features. I conducted exclosure experiments to assess the effect that continual maintenance by native ungulates has on grazing lawns, and the ecological effects associated with grazing lawn presence in the ecosystem. I found that grazing lawns within my study area were frequented and maintained by Roosevelt elk (*Cervus canadensis roosevelti*) and black-tailed deer (*Odocoileus hemionus columbianus*). These lawns demonstrated three consistent characteristics: (1) shrub heights were predominantly at or below the general upper level of ungulate browsing, (2) fruit production was generally absent from fruit-producing shrubs, and (3) plants with browsing resistant traits were not frequently browsed. Furthermore, frequent herbivory by native ungulates has contributed to a recruitment gap in native shrubs, while less palatable plants persist. Grazing lawns were also associated with increased cover of alien plants in the herbaceous layer, suggesting that repeated herbivory gives these species a competitive advantage. When ungulate herbivory was eliminated by exclosures, vegetation composition and structure of grazing lawns began to return toward those typical of the surrounding forest. Although alien plants were dominant in grazing lawns, they were absent inside long-term ungulate exclosures, indicating the cessation

of herbivory can shift plant structure and composition towards conditions of the surrounding forest. My results indicate that frequent native ungulate herbivory can maintain grazing lawns in a rainforest ecosystem, where they had not previously been identified. Subsequently, I suggest that land managers wanting to increase native plant abundance and structural complexity of grazing lawns should consider ungulate population reductions, ungulate exclusion, or predator restoration as potential viable strategies for reducing alien plants.

3.2 Introduction

For decades, ecologists have studied the facilitation of positive consumer–resource loops between native herbivores and vegetation, examining patterns that span spatial, temporal, and trophic scales (Bell 1971, McNaughton 1976, 1984). Interest in the subject stems from a desire to gain a comprehensive understanding of nutrient exchanges between plants and animals, and how these interactions influence ecological conditions, wildlife populations, animal behavior, and most commonly vegetation composition (Person et al. 2003, Radloff et al. 2014, Hempson et al. 2015, Purdon et al. 2019). The ecological literature includes a myriad of studies and experiments from around the world examining the formation, maintenance, and ecological effects of grazing lawns. The growing interest in grazing lawn ecology likely stems in part from the similarity of ecological functions observed across diverse ecosystems, where herbivore activity often leads to comparable changes in vegetation and habitat structure (du Toit and Olff 2014).

Increased interest in the formation, maintenance, and ecological characteristics of grazing lawns has occurred since McNaughton's initial description of the habitat type in 1984 (McNaughton 1984). Subsequent contributions to the subject led to changes in the criteria required for a habitat to be considered a grazing lawn (Cromsigt and Kuijper 2011). In modern journal articles, it is common for a grazing lawn to be characterized by its habitat characteristics. For instance, Mayengo et al. (2020) defined grazing lawns by their presence of short grass species and high grazing intensity in the miombo woodland ecosystem. Stewart et al. (2006) defined grazing lawns as vegetation swards dominated by high densities of grazing-tolerant plants with high nutrient concentrations maintained by herbivory in a temperate conifer forest. Browsing ungulates, trophic dynamics, and macrodetritivores have all been associated with the maintenance of grazing lawns (Fornara and Toit 2007, Howison et al. 2016, Fležar et al. 2019), thereby increasing the geographical range of this research and the types of plants and animals involved in lawn ecology. These developments led me to study and describe grazing lawns in a temperate coastal rainforest, an ecosystem where they were previously undocumented.

If left unchecked by predators, herbivores are capable of triggering trophic cascades by creating recruitment gaps in woody browse species, affecting net aboveground primary productivity, plant diversity, and phenology (Beschta and Ripple 2008, Woodward et al. 2021, Beckett et al. 2022). One cascading effect may be the emergence of grazing lawns in forests with high herbivore densities, such as the hardwood forests of the United States (Rooney 2009). The extirpation of the gray wolf (*Canis lupus*) in the early 1900s likely caused herbivore predation risk in the

Pacific Northwest to decline. While gray wolves remain absent in the Oregon Coast range (ODFW 2023), the cougar population has since recovered to a stable state (ODFW 2017), though the population of cougar in the region likely remains below the number it was historically. The historical decline in predator populations, combined with various hunting bans and reintroduction programs for native ungulates (Harper 1971, 1987) led to an increase in ungulate populations on the Oregon Coast range throughout the 20th century. In return, Roosevelt elk (*Cervus canadensis roosevelti*) and black-tailed deer (*Odocoileus hemionus columbianus*) herbivory effects increased throughout the region as ungulate populations grew. Like other grazing lawn maintaining herbivores such as blue wildebeest (*Connochaetes taurinus*) (Martin et al. 2015) and geese (*Anser sp.* & *Branta sp.*) (Person et al. 1998, Kellett and Alisauskas 2022), Roosevelt elk make repeated visits to areas based on nutrient availability (Thomas et al. 1982). Therefore, the herbivory effects of these native ungulates may be elevated in areas where nutrient availability is greatest, such as grazing lawns.

While grazing lawns are well-documented in tropical and arid ecosystems, their existence and ecological dynamics in highly productive rainforest environments remain largely unexplored. I hypothesize grazing lawns in the Pacific Northwest coast range persist and are being maintained by the herbivory effects instilled by black-tailed deer, and Roosevelt elk. To guide my investigation, I addressed the following research questions:

1. Do native ungulates maintain grazing lawns through repeated herbivory in a temperate coastal rainforest?

2. What are the distinguishing structural and compositional features of ungulate-maintained grazing lawns?
3. Are grazing lawns associated with the invasion of alien plants?
4. Does excluding ungulates lead to a reversal in vegetation composition and structure in grazing lawns over time?

3.3 Methods and Materials

3.3.1 Study Sites

I designated two study areas on the central Oregon Coast to address these research questions: (1) the Tenmile Creek valley bottom (44.221162, -124.094811) and (2) the Cummins Ridge Wilderness area (44.252014, -124.049077). In the Tenmile Creek valley, I had two study sites, one managed by the Audubon society and the other managed by Siuslaw National Forest. Both study areas occur within the Volcanics sub-ecoregion of the marine west coast forest (US EPA 2015) and support mixed conifer and deciduous vegetation in a highly productive temperate rainforest ecosystem. Predominant overstory species consist of Sitka spruce (*Picea sitchensis*), western hemlock (*Tsuga heterophylla*), Douglas fir (*Pseudotsuga menziesii*), red alder (*Alnus rubra*), and western redcedar (*Thuja plicata*) (McCain 2002).

Disturbance events that could create grazing lawn characteristics in my study area, such as wildfire or commercial timber harvests, have not occurred recently. Historical records of fires depict the areas being burned as recently as 1868 (Gannett 1902), however there is no record of commercial timber harvests within the Cummins Ridge Wilderness area or in the lower Tenmile Creek valley bottom (Siuslaw

National Forest 1995). Sections of the Audubon property were used for livestock ranching during the early 1900s, subjecting the site to periodic anthropocentric disturbances such as selective timber harvests and controlled burns to clear patches of land.

Historical records of native ungulate and predator populations in my study area is limited. Roosevelt elk (hereafter elk) populations in the coast range increased dramatically following a 1947-71 reintroduction program by Oregon Department of Fish and Wildlife (ODFW) (Harper 1971) and have been stable since 1985 (Harper 1987, Cafferata 2013) at 60,000 – 70,000 individuals statewide. Currently, the population density of elk in the Alsea unit 18 hunting area, an area that incorporates both study areas, is estimated to be between 4800 – 5800 elk for the unit (0.95 - 1.1 elk per km²) (Oregon Department of Fish and Wildlife 2023). The most recent population density estimates for Black tailed deer (hereafter deer) in Oregon was 320,000 for the state (1.26 black-tailed deer per km²) in 2004 (Kuzyk and Heffelfinger 2015), but there are no estimates in Alsea unit 18. Cougar, a natural predator of both ungulates, has an estimated population density of 1,200 across the coast and northern cascade range (0.03 cougar per km²) (Broman et al. 2026). Wolves were extirpated from the region in the early 1900's and have not returned to the central Oregon Coast.

3.3.2 Grazing Lawn Study Design

For this study, I defined grazing lawns as patches dominated by grass or forbs that are short in stature and maintained by herbivory (McNaughton 1984, Hempson et

al. 2015). Using topographical maps, I designated search areas in each region by identifying sections of land with near flat terrain that could be accessed repeatedly. Designated search areas on near flat terrain was selected to minimize confounding ecological variables that could create lawn like characteristics, such as aspect or erosion susceptibility. I traversed the search areas during the summer of 2024 in search of grazing lawns. Any areas 500 m² or greater in size on near-flat terrain with minimal shrub and western sword fern (*Polystichum munitum*), and predominant grass and forb cover were identified as grazing lawns. Lawn perimeters were distinguishable by abrupt changes in plant composition from grasses and forbs to sword ferns and shrubs. Grazing lawn area estimates were generated using stem maps. I identified three grazing lawns at least 500 m² in area on near-flat terrain with low shrub and western sword fern cover and high forb and grass cover. One lawn was found in the Cummins Ridge Wilderness and two in the Tenmile Creek valley bottom, named the Audubon lawn and USFS lawn based on parcel ownership.

I deployed motion detection game cameras in the Audubon and USFS grazing lawns from July 15th, 2024, to January 18th, 2025, to record visiting ungulates and their behavior. A third game camera was not deployed in the Cummins Ridge lawn due to wilderness laws forbidding game camera use. Each camera was set to photograph any movement in 10 second intervals. To reduce the likelihood of inflating animal counts, I grouped photos taken within thirty-minute intervals and recorded the maximum number of animals by species seen in each photo group. Ungulate behavior data was noted by categorizing the vigilance and eating habits of all individuals in all photos. I recorded an individual as being “vigilant” if the animal

was standing and had their head raised or as “not vigilant” if their head was lowered in a photo. When possible, I recorded foraging behavior of standing ungulates such as grazing, browsing, or not eating in each photo group. To calculate ungulate visitation frequency, I divided the number of days ungulates were photographed by the total number of days cameras were deployed at each grazing lawn.

I recorded ungulate scat encounters in each grazing lawn every two months for a year, starting in July 2024. Across each lawn, I counted the number of ungulate pellet groups within one-meter-wide belt transects located 5 m apart from one another. A coin flip was used to determine the orientation of transects traversed within a lawn during each survey. If I encountered multiple pellet groups within half a meter of each other, one pellet group was recorded.

Vegetation and herbivory data were collected and summarized to identify common ecological characteristics throughout each grazing lawn. Vegetation data consisted of recording all live trees, shrubs, saplings, and seedlings by species within a lawn. I recorded any trees below 2 m in height as saplings and any saplings below 0.05 m in height as seedlings. All shrub species in grazing lawns, excluding Pacific blackberry (*Rubus ursinus*), had their heights measured. The presence or absence of fruit, browsing evidence, and nurselog facilitation for all woody plants were recorded. Any plant growing at or on the base of a standing live tree or snag was recorded as being facilitated by a nurse log.

To estimate herbaceous cover and height, I used 0.25 m² subplots systematically placed within grazing lawns. I determined the number of subplots per lawn by dividing the estimated area of the lawn by 50 m² and rounding to the nearest

whole number of subplots. Subplots were evenly spaced in four cardinal directions, starting at the center of each lawn. If a subplot was obstructed by logs, trees, or shrubs, I relocated it to the nearest unobstructed area.

At each subplot, I identified all herbaceous plants by species and estimated cover (%). Each plant species was later classified as a forb, vine, fern, bunch grass, or lawn grass. I deemed bunch grasses as caespitose grasses and lawn grasses as rhizomatous or stoloniferous grasses following Oregon Flora (Meyers et al. 2015). Pacific blackberry was recorded as a vine. Any ground cover not composed of live vegetation was recorded as non-vegetative cover (%). I estimated the average herbaceous plant height (m) by placing four .1 m by .05 m by .15 m lightweight Styrofoam pieces on top of plants in subplots and measuring the mean distance of all pieces from the ground.

3.3.3 Exclosure Study Design

To assess whether native ungulate grazing could account for the maintenance of the grazing lawns, I used two types of exclosures: (1) long-term exclosures and (2) short-term exclosures. Long-term exclosures were used to evaluate the effects herbivory removal has on lawn vegetation composition and structure after 27 years in grazing lawns. Short-term exclosures (hereafter herbivory cages) were used to evaluate the effects of herbivory removal on lawn vegetation heights and phenology after one year in grazing lawns.

Native saplings were planted in each long-term exclosure at the time of their installation to aid ecological succession in areas previously used for livestock

ranching. I selected three exclosures adjacent to current grazing lawns and established paired lawn plots 2 m distance from each exclosure. Herbaceous plant heights were measured using the Styrofoam method described above. To determine the number of subplots used to measure herbaceous plant heights, I divided the area of exclosure by 15 m², with a minimum of four subplots. I made herbaceous plant cover estimates within the entire area of each exclosure and adjacent paired plots and recorded the heights of naturally established saplings and shrubs. Pacific blackberry was classified as a shrub for this study since the collection of height, fruit presence, and browsing evidence data for all individuals was less time consuming in the exclosure experiment compared to the grazing lawn characteristic study.

For the short-term herbivory cage study, I installed four cages in a grazing lawn located at the Audubon property to observe the effects ungulate exclusion had on herbaceous plant height and phenology over the course of a year. Each cage was 0.9 m in height and diameter and was paired with a nearby plot of the same size accessible by native animals. I determined the installation location of each cage through random selection among the subplot locations used when characterizing the lawn that had $\leq 5\%$ non-vegetative cover. Paired plots 3 m from each cage were designated by randomly generating a cardinal or intercardinal direction from the cage's location. If the location of the plot differed greatly in herbaceous richness and cover from the cage it was paired with, I randomly generated another location for the paired plot. Every two months, I recorded the average herbaceous plant height using the Styrofoam measurement method described above and counted the number of blooming forbs by species.

3.3.4 Statistical Analysis

I used a binomial logistic regression to evaluate whether ungulate vigilance differed between deer and elk in grazing lawns where game cameras were deployed. No outliers were detected, and residual normality was not assessed because residuals from binomial generalized linear models are not expected to be normally distributed. A chi-square test of independence was used to examine if ungulate vigilance differed between lawns where predators were and were not observed on game cameras. I used a one-sample binomial test to examine whether ungulate browsing limited the heights of shrubs by comparing the number of shrubs below and above the 2 m upper level of browsing capable by elk (Beschta et al. 2023), and to determine whether shrubs were more likely to show evidence of browsing than not in the three identified grazing lawns. The three long-term exclosure-control pairs served as the independent experimental units. The four short-term herbivory cages served as independent experiment units as well. Because only three pairs of exclosures, and only four pairs of herbivory cages were available, statistical power to detect treatment effects was inherently low for both herbivory exclusion types. I therefore emphasize the magnitude and consistency of paired responses rather than significance tests. However, I used a negative binomial generalized linear model with herbivory cage installation and sampling month as fixed effects to evaluate if herbivory exclusion was associated with increased blooming forb abundance during spring and summer months. Residual normality was not assessed because residuals from generalized linear models are not expected to be normally distributed, and common yarrow

(*Achillea millefolium*) blooms were observed as an outlier in forb bloom amount for herbivory cages in September 2024. To evaluate its influence, the model was rerun with the observation removed. The manipulation effect remained significant, indicating that the primary result was robust to the outlier.

I used R for all data analysis (R Core Team 2022). Ggplot (Wickham et al. 2025), cowplot (Wilke 2025), and ggrepel (Slowikowski et al. 2026) packages were used for graph creation. For statistical analysis, I used the nlme (Pinheiro and Bates 2026) and MASS (Ripley et al. 2025) packages for linear models, the emmeans (Lenth et al. 2026) package for pairwise comparisons, and the integrated stats package for all other tests in R.

3.4 Results

I identified three grazing lawns that exhibited evidence of intense grazing and browsing by native herbivores (Figure 3.1 A-C). For the two sites where it was possible to place game cameras, elk and deer visited grazing lawns on Audubon (25% of days) and USFS (10% of days), and deer vigilance (73%) was significantly higher than elk vigilance (24%) ($p < 0.001$) (Appendix Table 3.1). In the Audubon lawn, cougar and black bear (*Ursus americanus*) were observed on game cameras. Ungulate vigilance frequency differed significantly between lawns where predators were observed (39%) and where predators were not observed (32%) ($p = 0.002$). Ungulates were photographed most frequently grazing on herbaceous vegetation ($n = 905$, 60%), or not eating ($n = 579$, 38%), and were infrequently observed browsing woody plants ($n = 29$, 2%) in grazing lawns photos (Figure 3.2 A-D).

In the herbaceous layer of all three grazing lawns, plants of alien origin (42%) occupied similar ground cover as the native vegetation (41%) (Figure 3.3A). However, alien plant cover was more prevalent in the USFS (59%) and Audubon (63%) lawns in the Tenmile Creek valley bottom than in the Cummins Ridge lawn (1%). When considering differences by vegetation type, I found that forbs were the most common vegetation in the herbaceous layer of all grazing lawns (49%), followed by grasses (28%), ferns (3%), and vines (2%). Within the grasses, bunch grasses occupied a greater cover (16%) than lawn grasses (12%).

Above the herbaceous layer, shrubs were the most predominant vegetation type; 93% of which did not have fruit present (Figure 3.3 B). The average height of all shrubs (0.71 m) was below the 2 m upper level of browsing by elk with only 4% shrubs exceeding 2 m ($p < 0.001$) (Figure 3.4 A & 3.4 B), and 60% of all shrubs had evidence of browsing ($p < 0.001$) (Appendix Table 3.2). Sitka spruce was the most predominant sapling species in grazing lawns (46 total spruce saplings), but I found no evidence of browsing on any spruce saplings.

Comparison of the vegetation in the long-term exclosures to that of the adjacent grazing lawn plots showed that alien plants were absent and non-vegetative cover was prevalent inside the exclosures (Figure 3.5 A). Shrub density inside the exclosures was on average three times greater and had an average height that was 0.91 m taller than those in grazing lawns (Figure 3.5 B & 3.5 C). Exclosures were composed of dense understory woody vegetation while adjacent grazing lawns lacked shrub and sapling recruitment (Figure 3.6). While my small sample size prohibited me from analyzing the statistical significance of these results, I noted a biologically

meaningful contrast in vegetation. Specifically, exclosures exhibited higher shrub density and height than grazing lawns, indicating that herbivore exclusion alters plant structure and composition.

In addition to having a greater average height across each bi-monthly survey after cage installation, the average height of herbaceous vegetation in my short-term herbivory cages one year after installation was 0.28 m taller than the herbaceous vegetation in my paired lawn plots (Figure 3.7 A & 3.7 B). The total number of blooming forbs during the spring and summer months was greater in herbivory cages (96 plants) compared to grazing lawn plots (25 plants), and the number of blooming forbs for each month sampled significantly increased when herbivory was excluded ($p = 0.005$) (Appendix Table 3.3).

3.5 Discussion

3.5.1 *Can native ungulates maintain grazing lawns in the temperate coastal rainforest?*

My vegetation, herbivory, and game camera results indicate elk, and deer can maintain grazing lawns in a highly productive forest environment, resulting in lawns characterized by short-statured forbs, lawn grasses, and bunch grasses with evidence of grazing. My short-term herbivory cages resulted in increased herbaceous plant heights and number of blooming forbs, indicating ungulates are not only maintaining short-saturated herbaceous plants in grazing lawns, but also limiting sexual reproduction of forbs. Plants of alien origin were predominant in two of the three grazing lawns I identified. Following herbivory exclusion for 27 years, the long-term exclosures had an absence of alien plants indicating ungulates are facilitating alien

plants in grazing lawns. My findings support my hypothesis that ungulate-caused grazing lawns can persist in the coastal temperate rainforest ecosystem, a region previously not associated with such modified plant communities. Furthermore, the ecological effects associated with grazing lawn presence in my study area facilitate discussion regarding the effects native ungulates have on the habitats they frequently utilize for resources.

Previous journal articles have utilized various metrics to define or characterize grazing lawns, including: predominant stoloniferous or rhizomatous (lawn) grasses in the herbaceous layer (Archibald et al. 2005), increased nutrient availability for herbivores (Person et al. 2003), and herbivore/herbivory evidence (Milchunas and Lauenroth 1989, Kotanen and Abraham 2006, Leonard et al. 2010, McCauley et al. 2018). While I did not measure nutrient availability in my lawns, the presence of herbaceous plant dominated patches in an ecosystem otherwise associated with rapid woody plant establishment and growth renders the testing for increased nutrient availability unnecessary, since it is evident these patches are being maintained in their current state by herbivory. Furthermore, recent grazing lawn journal articles depicting grazing lawns mostly composed of forbs (Horne et al. 2024) and cespitose (bunch) grasses (Hempson et al. 2022) as the dominant herbaceous species of grazing lawns provide additional justification for my study sites to qualify as such. Above all else, these grazing lawns demonstrate a positive consumer-resource loop like that observed in earlier journal articles (McNaughton 1984), as evident by the reversion of plant composition and structure in my exclosures following herbivory exclusion and photographic evidence of native ungulates frequenting my sites.

3.5.2 *Distinguishing features of temperate coastal rainforest grazing lawns*

Three ecological characteristics were common in the grazing lawns I identified: (1) shrubs were predominantly at or below the typical maximum browsing height of elk, (2) most fruit producing shrubs did not produce fruit, and (3) plants with browsing resistant traits were not frequently browsed by ungulates.

The predominance of shrubs at or below the general upper browse level of elk signifies ungulates are a limiting factor in the recruitment of deciduous plants in grazing lawns. Woody plants below the maximum browsing height of herbivores in nutrient rich environments are more likely to perish when browsing is heavy due to their investment in rapid growth (Makhabu et al. 2006). I found shrub species capable of clonal reproduction, like *Vaccinium* and *Rubus*, were isolated and rarely exhibited new shoot growth from the soil surrounding a browsed plant. This shrub structure is atypical for an ecosystem associated with rapid plant recruitment, in which shrubs and saplings species dominant the forest floor as canopies close overhead. Repeated browsing of plants capable of clonal reproduction such as rowan (*Sorbus aucuparia*) have resulted in the formation of dense shrub layers, deemed ‘browsing lawns’ in other forest ecosystems (Speed et al. 2013). I did not observe a similar plant response in my grazing lawns, perhaps due to differences in plant species-specific tolerances to browsing (Jager and Pastor 2010), or due to differing herbivory rates in my study area compared to that observed by Speed et al. (2013). In my lawns, plants below the general upper browsing level of large herbivores may not have energy to invest in non-sexual reproduction after being browsed repeatedly, resulting in short, browsed

shrubs isolated from one another rather than a relatively tall and dense shrub layer of uniform height.

Fruit producing shrubs in my grazing lawns were observed to rarely produce fruit. Investment of nutrients towards rapid growth is often at the expense of sexual reproduction capability for plants, resulting in reduced fruit production when browsed (Crawley 1997). When sampling, I observed two “groups” of shrubs in grazing lawns; infrequent tall fruit producing shrubs, and more commonly, short non-fruit producing shrubs. The presence of tall shrubs indicates herbivory pressure may have been lower in the past, or that a select few plants were able to escape the maximum browsing heights of elk and produce fruit (Figure 3.8 A). In browsing lawns, Fornara and Toit (2007) found the likelihood of woody plants reproducing sexually decreases as the frequency of browsing increases, and hypothesized the few plants that are above the browsing heights of herbivores become the suppliers of propagules within lawns, ultimately resulting in a long-term negative effect on plant reproductive success (Fornara and Toit 2007). I tested this hypothesis by sampling shrub characteristics in two plots one kilometer east of the Audubon Lawn location that I deemed an ungulate refugia, or an area with minimal evidence of ungulate herbivory (Appendix Table 3.4). I found shrubs above the general browsing height of elk were three times more likely to bear fruit in the elk refugia than in grazing lawns, signifying herbivory as a limiting factor of sexual reproduction. If shrub establishment in grazing lawns is dependent upon the fruit of plants that have escaped the browsing height of native ungulates, then grazing lawn shrub density may be decreasing throughout time, supporting Fornara and Toit’s (2007) hypothesis. However, unlike the browsing

lawns of the African savanna, my grazing lawns are surrounded by densely vegetated forests with plants capable of supplying fruit for shrub recruitment. If I assume shrub seeds are supplied from surrounding forest habitats, then ungulate browsing in my grazing lawns remains a limiting factor in shrub recruitment. Furthermore, since browsing also limits the capability of shrubs to reproduce non-sexually, this can lead to the ungulate-caused recruitment gap in shrubs I observed in my grazing lawns.

Plants with browsing resistance traits are more capable of growing above the reach of ungulates than are plants without such traits (Tallamy and Raupp 1991), granting browse resistant plants a greater chance to become established and persist in grazing lawns. I did not find browsing evidence on any of the 46 spruce saplings growing in my grazing lawns consistent with the findings of Cowan (1945) and Arno (1977) where they indicated Sitka spruce has low nutritional value and is rarely browsed by ungulates. It is possible that spruce saplings were browsed at an earlier stage of growth, since spruce are capable of withstanding frequent browsing at a young age before eventually escaping above the browse heights of ungulates through rapid growth (Vila et al. 2002). If spruce saplings were browsed before my study took place, the saplings may have developed greater physical defenses such as increased needle length (Duncan et al. 1998), decreasing their palatability over time. The heightened defenses of spruce may enable other woody and herbaceous plants to persist in grazing lawns by creating miniature refugia from herbivory as spruce saplings mature.

In the USFS lawn, I observed an assemblage of spruce saplings and trees growing on a nurse log with several other woody and herbaceous plants (Figure 3.8

B). I assessed whether spruce created a physical barrier protecting surrounding plants from herbivory by grouping all shrubs and saplings, excluding spruce, together and compared browsing percentages of the plants protected to plants not protected in the USFS lawn. I found shrubs and saplings protected by spruce defenses, or palatable plants near plants with defensive traits, were five times more likely to be unbrowsed, increasing their likelihood of surviving in grazing lawns (Appendix Table 3.5). If herbivory remains prevalent in these grazing lawns, over time, the community assemblage of shrubs and trees may consist of plants with browse resistant traits, and of palatable plants in areas protected from browsing by unpalatable ones.

3.5.3 *Are grazing lawns associated with the invasion of alien plants?*

I found ungulate herbivory in grazing lawns to promote the invasion of alien herbaceous plants. Furthermore, when palatable shrubs such as huckleberry or red elderberry (*Sambucus racemosa*) were present in grazing lawns I often observed browsing evidence on plants with browsing defenses such as English holly (*Ilex aquifolium*), Cascade Oregon grape (*Berberis nervosa*), salmonberry (*Rubus spectabilis*), Himalayan blackberry (*Rubus armeniacus*), and cut leaved blackberry (*Rubus laciniatus*). My findings support the hypothesis made by Cromsigt and Kuijper (2011), in which they suggest that lawns can persist in resource-rich environments where herbivores show strong patch selection, while having a weak selection for plants within a patch. If elk and deer have a weak selection of plants within a lawn, plants that have rapid growth rates or are adapted to frequent herbivory are likely to dominant grazing lawns, similar to how invasive plants can gain a

competitive advantage over native plants by livestock overgrazing (Kimball and Schiffman 2003). In many cases, grasses and forbs benefit the most from continual herbivory pressure due to their ability to reproduce vegetatively when sexual reproduction is unlikely (Briske 1996). My short-term herbivory cage results demonstrated that sexual reproduction of forbs is inhibited by frequent herbivory by deer and elk. This situation is likely to benefit alien herbaceous plants by giving them a competitive and reproductive advantage over native ones, creating the vegetative characteristics I identified, thus facilitating alien plant dominance in the herbaceous layer of grazing lawns in the temperate coastal rainforest ecosystem.

While alien plant dominance is not inherently associated with all grazing lawns, herbivores known for maintaining grazing lawns can create alien plant dominated patches through repeated grazing in habitats otherwise not associated with annual grasses and grazing herbivores (Best and Arcese 2009). While deer and elk are native to my study area, temperate coastal rainforest ecosystems with a predator rich trophic level would have been unlikely to be associated with intensive grazing or a predominant cover of native annual grasses in the herbaceous layer (Martin et al. 2024). An increase in grazing is likely to favor alien plants that are adapted to or benefit from herbivory, creating a unique grazing lawn habitat in which the herbaceous layer is dominated by alien plants.

3.5.4 Does excluding ungulates lead to a reversal in vegetation composition and structure in grazing lawns over time?

Following the exclusion of ungulates, I found a general reversal in vegetation composition and structure in my long-term ungulate exclosures, resulting in increased

shrub recruitment and fruit production. It is possible the developing canopy may have reduced alien plant cover in the herbaceous layer by decreasing light availability in the understory (McDaniel and Ostertag 2010), or by increasing litter deposits on the floor (Huebner 2010). Additionally, these soil conditions may have facilitated native plant establishment. For example, Sabo et al. (2017) demonstrated deer exclusion made soil conditions more favorable for understory development of native plants by reducing soil compaction and light availability. Nevertheless, alien plants in my grazing lawns likely maintained a dominance in the herbaceous layer by responding favorably to continuous herbivory, lack of overstory cover, or perhaps by benefiting from the indirect effects caused by frequent ungulate grazing such as seed dispersal or trampling. I recognize that the planting of native saplings in each enclosure obfuscates my enclosure results, making it more difficult to confirm ungulate herbivory as the main driver for the observed differences between ungulate enclosures and adjacent grazing lawns. The development of an overstory canopy in an enclosure likely promotes the establishment of plants in the understory (Gautam et al. 2016), perhaps creating more suitable conditions for native plants than for alien ones. Nonetheless, the exclusion of ungulates facilitated the rapid growth of planted saplings by eliminating browsing pressure, which in turn affected understory conditions. Thus, the exclusion of herbivory appears to represent the main factor affecting observed differences between enclosures and grazing lawns, and that it is the exclusion of ungulates that led, over time, to a recovery of understory species and structure.

3.5.5 *Apex predator effects*

Finally, a comment on the potential effect apex predators, or lack thereof, may have on the maintenance of coastal forest grazing lawns. I found the presence of predators, such as cougar, did not deter ungulates from visiting the lawn, but did increase ungulate vigilance in lawns. Increased visibility provided by the short statured vegetation may reduce predation risk for herbivores in grazing lawns (Burkepile et al. 2013). Additionally, lowered predation risk may allow ungulates to maintain areas of increased nutrient concentrations (Stewart et al. 2006), thus reducing the distance they need to travel to satisfy nutritional requirements. Elk are known to make repeated visits to areas for resources as opposed to spending long periods of time grazing or browsing in a single area (Anderson et al. 2008). My game camera photos revealed elk residing in grazing lawns for several minutes to eat, rest, and feed calves (Figure 3.8 C & 3.8 D). As the time an ungulate spends in a grazing lawn increases, I suspect this also increases herbivory pressure on plants. This situation, in turn, increases the likelihood that less preferred plants will be eaten, further benefiting plant species with rapid responses in growth after grazing or browsing occurs, or plants capable of non-sexual reproduction like alien lawn grasses. As evident in my exclosures, the exclusion of herbivory leads to a recovery in vegetation composition and structure. I suggest that if predation from large carnivores were to increase in the coastal temperate rainforest ecosystem, or if ungulate populations were to decrease because of increased human hunting, grazing lawns will begin to exhibit similar changes as demonstrated in the exclosures.

3.6 Conclusion

The identification of grazing lawns in the temperate rainforest ecosystem of the Oregon coast range is novel, and my research provides crucial evidence of grazing lawns maintained by elk and deer persisting in an ecosystem where they are otherwise deemed unlikely to occur, expanding the geographical range of grazing lawn ecology. I found that grazing lawns can form and be maintained in these highly productive forests through concentrated herbivory by native ungulates, with consequences to plant community composition and the invasion of alien species. Characteristics of grazing lawns in this study revealed the facilitation of alien plants in the herbaceous layer, a reduced likelihood of fruit production by shrubs, and a competitive advantage of plants with herbivory resistant traits over those lacking such traits. Ecologically, continuous maintenance of these lawns by native ungulates are likely to exacerbate these observations, whereas ungulate exclusion causes their reversal. Thus, I encourage subsequent research into the temporal changes associated with herbivory exclusion in grazing lawns of the region, including the use of exclosures to evaluate trends in plant community succession. Since elevated levels of herbivory by native ungulates can result in unfavorable habitat conditions, such as alien plant dominance in the herbaceous layer, I suggest that land managers wanting to restore grazing lawn habitats should consider ungulate population reductions and/or ungulate exclusion as viable strategies for reducing alien plants.

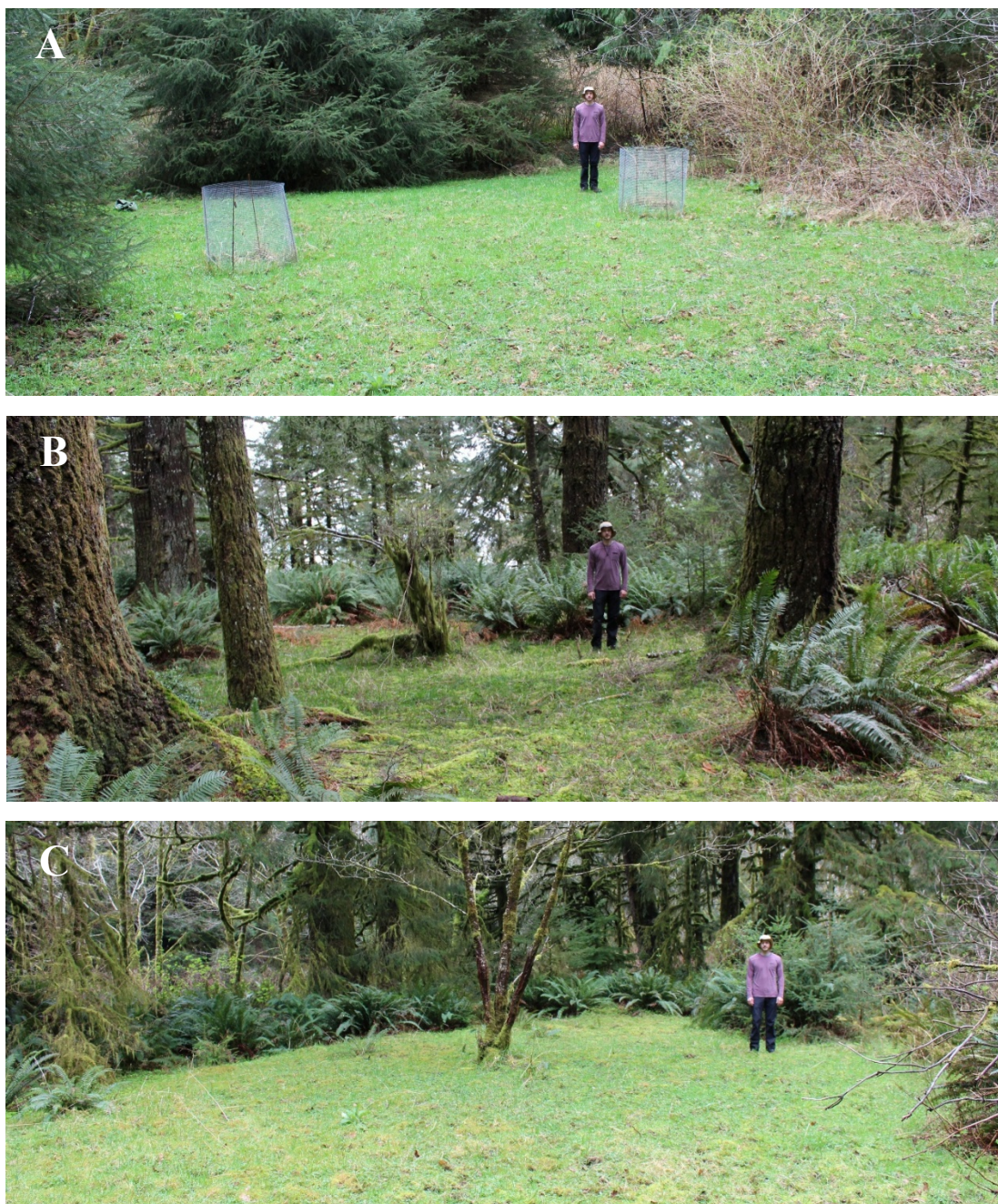


Figure 3.1: Overview photos of grazing lawns identified in the Tenmile Creek watershed and Cummins Ridge Wilderness: (A) Audubon lawn, (B) Cummins lawn, and (C) the USFS lawn taken on March of 2025. The researcher in (A), (B), and (C) is 1.85 m tall (Photo credits: Kendall Elmer).

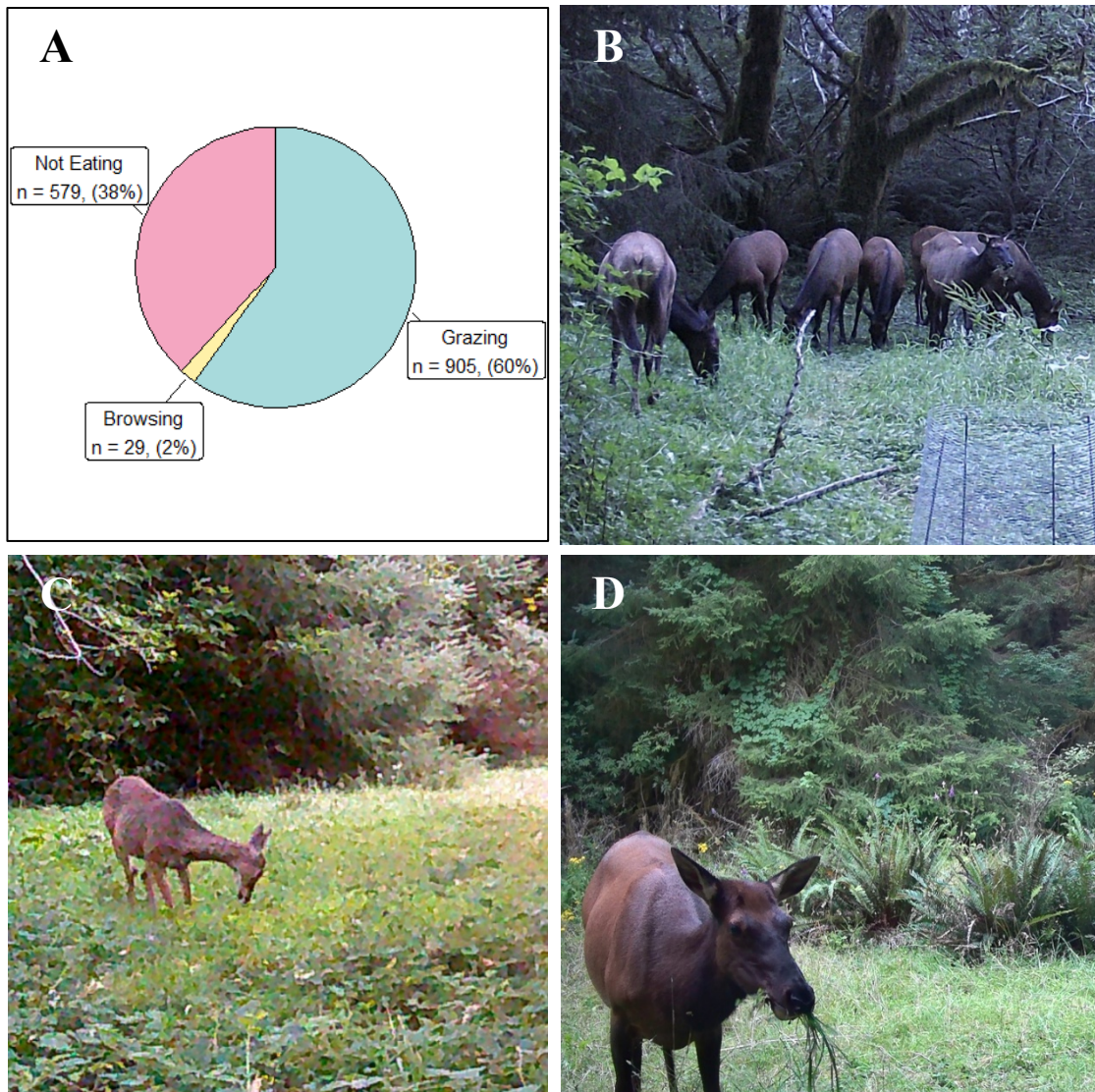


Figure 3.2: Pie graph summary and game camera photos of ungulate behavior. (A) A pie graph summary of the foraging behaviors observed in game camera photos in the USFS and Audubon lawns where game cameras were deployed. Game camera photos show: (B) six elk grazing, July 2024, Audubon site, (C) one deer grazing September 2024, Audubon site, and (D) an elk with grass in its mouth, August 2024, USFS site.

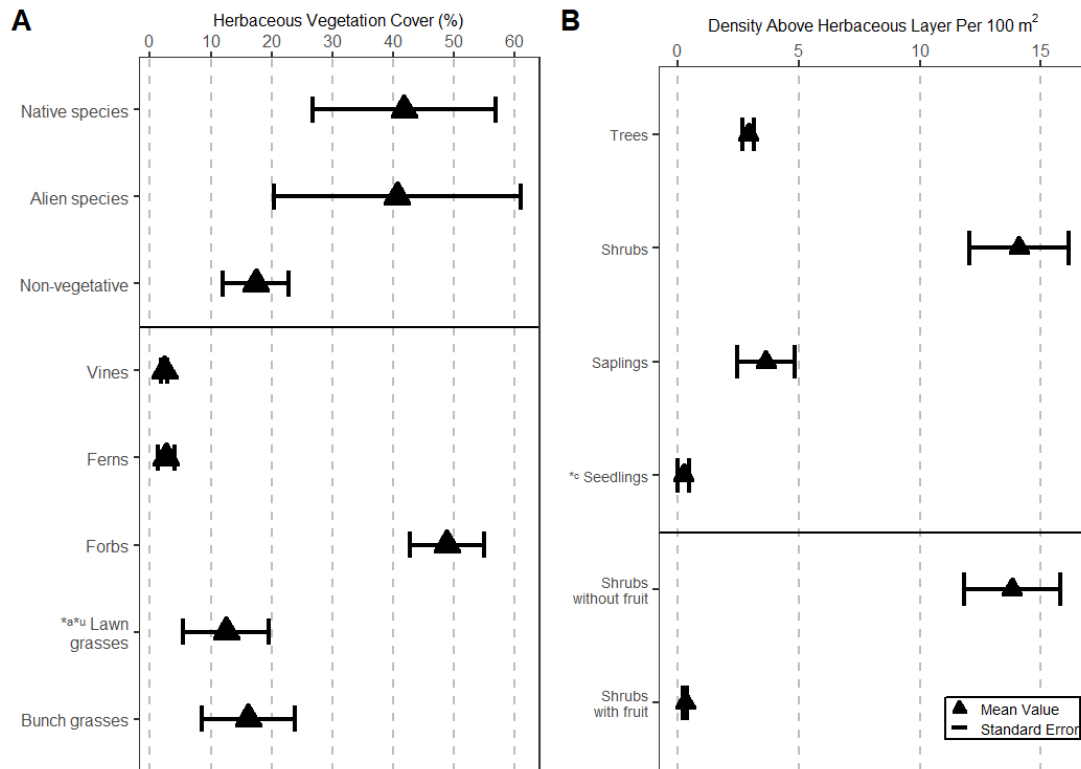


Figure 3.3: Point (triangle) and standard error (capped line) plots depicting common vegetation characteristics of identified grazing lawns. (A) Herbaceous cover by native versus alien origin and vegetation type, and (B) woody vegetation density by plant type and shrub fruiting status. Vegetation not found in all three lawns have superscript letters indicating the lawns in which the traits were found (*a = Audubon, *c = Cummins, *u = USFS). Mean and standard error are shown.

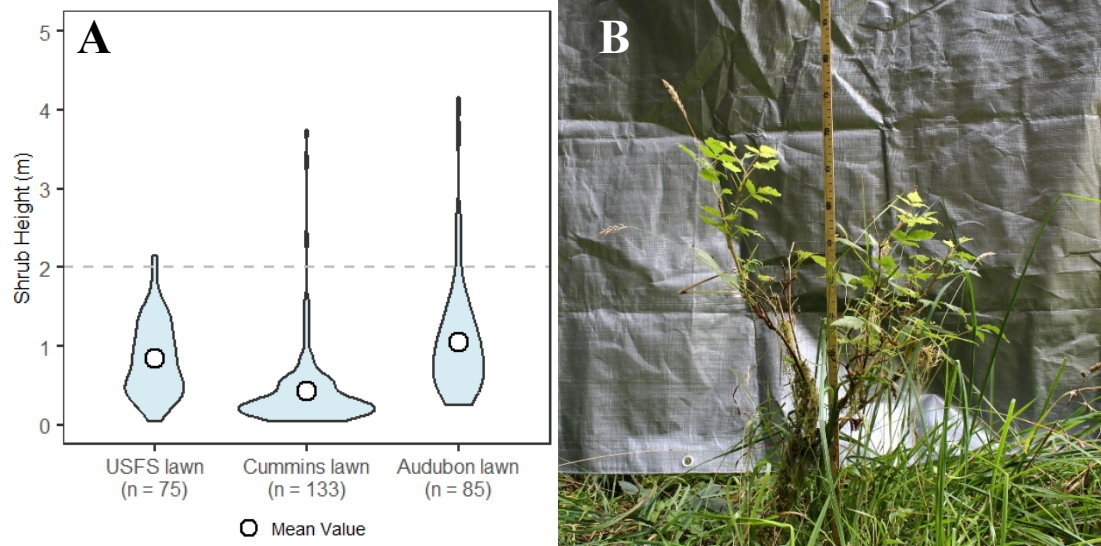


Figure 3.4: Violin plot and photo of shrub characteristics in grazing lawns. (A) Violin plots of shrub heights in identified grazing lawns. The general upper browse level of elk is depicted by a horizontal grey line at 2 m (Beschta et al. 2023). Shrub heights in grazing lawns rarely exceeded 2 m (4% of shrubs, $p < 0.001$), and 60% of all shrubs had evidence of browsing ($p < 0.001$). The average height of all shrubs in grazing lawns was 0.71 m ($n = 293$). (B) A 0.69 m tall salmonberry (*Rubus spectabilis*) shrub in the USFS lawn with evidence of browsing, September 2025 (Photo credit: Samuel Woodrich).

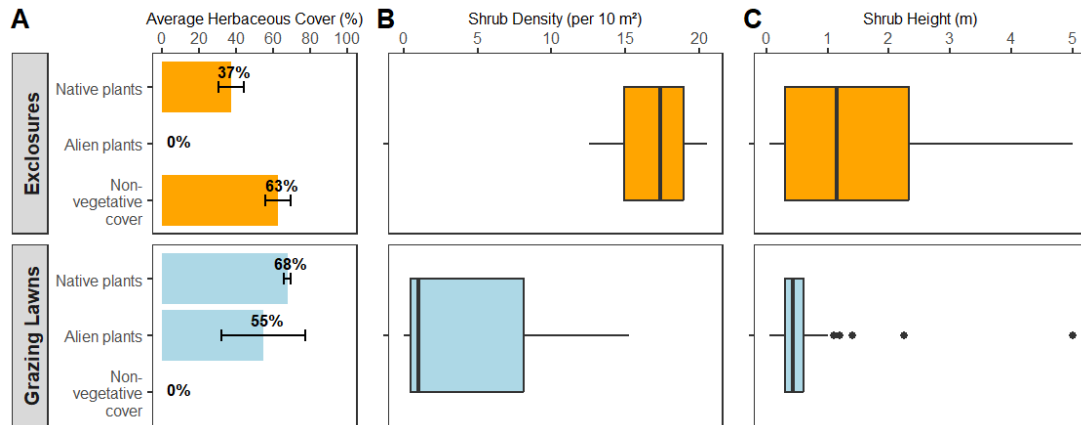


Figure 3.5: Box plots of the vegetation characteristics of all long-term herbivory exclosures and paired grazing lawns. (A) Herbaceous plant cover by species origin in grazing lawns (blue, lower panel) compared to long-term exclosures (orange, upper panel, the mean and standard error are shown). (B) Shrub density per 10 m² in grazing lawns compared to exclosures. (C) Shrub height (m) in grazing lawns compared to exclosures. The boxplots in (B) and (C), show the median (bold line) and the interquartile range (the lower and upper ends of each box), with whiskers extending to 1.5 times the interquartile range and points depicting the outliers.



Figure 3.6: Photo demonstrating vegetative response to herbivory exclusion in a long-term exclosure. The researcher illustrates the dense understory of woody vegetation following herbivory exclusion for 27 years by touching the fence of a long-term exclosure behind him. In the foreground, a grazing lawn lacking woody plant recruitment, composed of herbaceous vegetation (Photo credit: Robin Comforto).

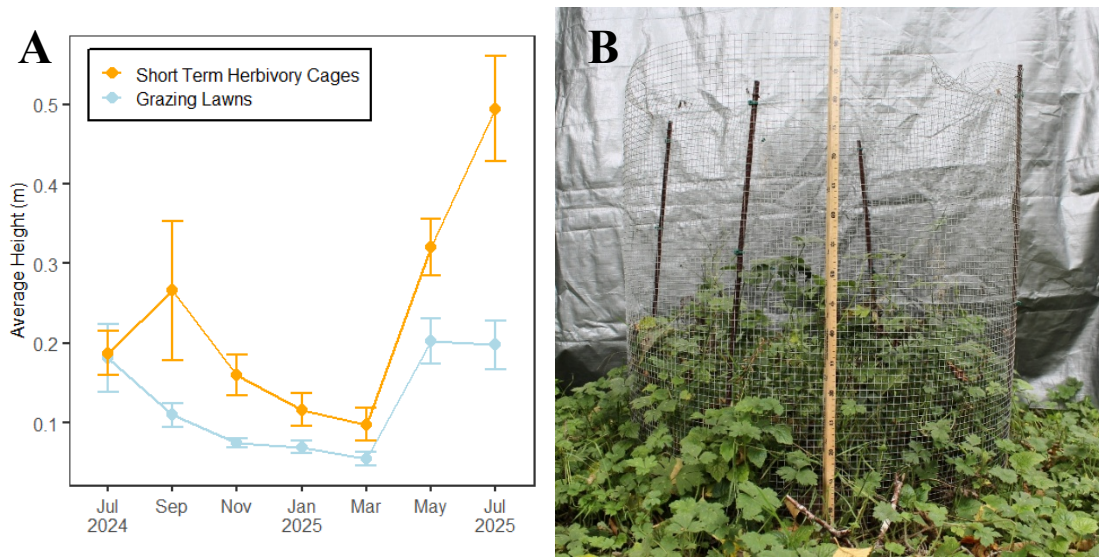


Figure 3.7: Line graph and photo depicting short-term herbivory cage results. (A) Herbaceous plant height in short-term herbivory cages and paired grazing lawn plots after cage installation in July 2024 (average and standard error estimates shown). (B) Example height difference of herbaceous plants in a cage compared to the surrounding herbaceous plants in a grazing lawn, September of 2025. In the center of the image is a 1 m tall Biltmore stick. The average height of herbaceous plants within the herbivory cage was 0.39 m and their average height on the paired plot (not pictured) was 0.11 m (Photo credit: Samuel Woodrich).

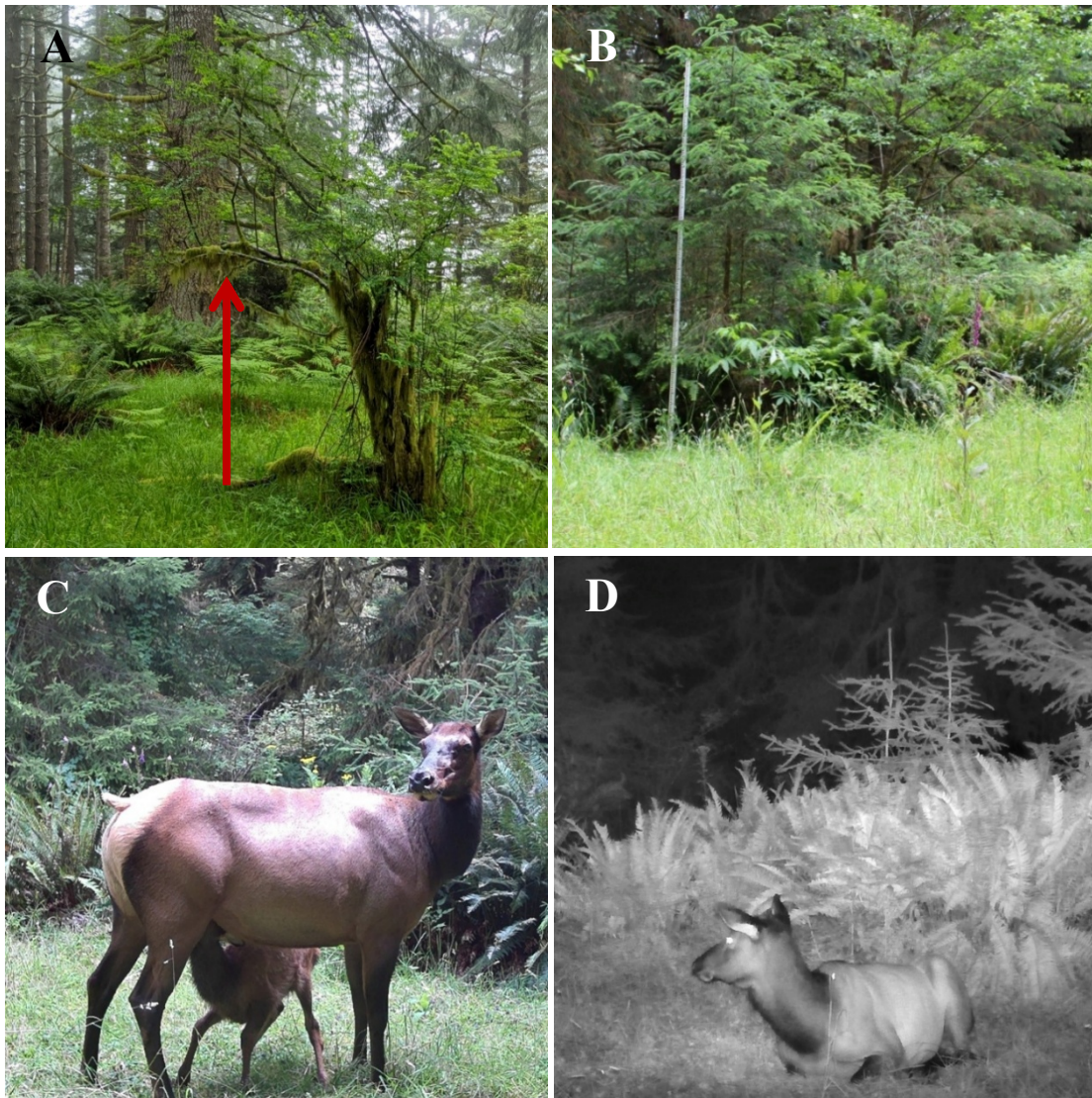


Figure 3.8: Photos of the ecological characteristics of grazing lawns and behavior of elk within them. (A) A 3.75 m tall, fruit-bearing, red huckleberry (*Vaccinium parvifolium*) located in the center of the Cummins Ridge lawn, May 2024. The maximum browse height of elk is represented by the lack of foliage in the lower half of the shrub, indicated by the red arrow pointing to the main branch of the plant and the lack of foliage underneath the main branch. (B) Sitka spruce (*Picea sitchensis*) saplings and trees creating a herbivory refugia on a nurse log located in the USFS lawn for surrounding plants, June 2024. The measuring stick shown is 4 m in height (Photo credits: (A) and (B) Samuel Woodrich). (C) and (D) Photos demonstrate elk reside in lawns for extended periods of time. Elk were frequently photographed participating in maternal care such as feeding young and photographed resting in the USFS lawn, both photos from August 2024.

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4. CHAPTER 4: Conclusion

Grazing lawn ecology has been researched for over four decades and continues to gain interest amongst ecologists as the number of herbivores known to maintain grazing lawns and the number of ecosystems where lawns persist increases. Numerous researchers have examined the functions, influences, and characteristics of grazing lawns, contributing to our collective understanding of the impacts herbivores have on the systems they inhabit. My dissertation contributes to the growth and development of the subject in its entirety by providing a systematic review on the subject and by identifying an additional habitat where grazing lawns persist. Modern ecological research necessitates the bridge between theory and practice (Turner et al. 2002). The need for applying ecological theory to practice is prominent in restoration ecology (Silliman et al. 2024), especially when considering if modern systems should function similar to baseline conditions (Harris et al. 2006), or as novel systems (Hobbs et al. 2006). In the remaining portion of this chapter, I provide commentary on the influences and effects herbivory has on the degraded habitats of the temperate coastal rainforest ecosystem and make suggestions on how to mitigate these effects. I start by drawing comparisons between the functions and characteristics of grazing lawn patches and the degraded sites actively undergoing ecological restoration in the Tenmile Creek watershed. Then, I revisit the results of my field experiments and use them to conceptualize the processes preventing passive restoration from occurring in the degraded habitats of the system. Finally, I use the information gathered from my systematic review and field study to prescribe more effective restoration practices that

encourage the passive transition from degraded conditions to improve structure and function at various spatial scales.

4.1 Degraded Habitat Characteristics of The Tenmile Creek Watershed

Several parcels of land purchased during the Forest Homestead Act of 1906 are now owned and managed by various environmental groups in the Tenmile Creek watershed (Sleeper 2003, Engelmeyer 2003, Suring and Lance 2021). These properties consist of various habitat types ranging from late successional forest patches to homogenous fields of alien plants. Examples of alien plant species common in degraded areas include, but are not limited to, common bent grass (*Agrostis capillaris*), rough meadow grass (*Poa trivialis*), Yorkshire fog (*Holcus lanatus*), sweet vernal grass (*Anthoxanthum odoratum*), English ryegrass (*Lolium perenne*), orchard grass (*Dactylis glomerata*), ornamental mint (*Mentha spp.*), creeping buttercup (*Ranunculus repens*), foxglove (*Digitalis purpurea*), common bird's-foot trefoil (*Lotus corniculatus*), and sheep sorrel (*Rumex acetosella*). Some alien plants in degraded sites are invasive, including reed canary grass (*Phalaris arundinacea*), false-brome (*Brachypodium sylvaticum*), tansy ragwort (*Senecio jacobaea*), and Himalayan black blackberry (*Rubus armeniacus*) (Oregon Noxious Weed Control Program 2024). In areas with homogenous alien plant assemblages, passive native shrub and sapling establishment is infrequent to impossible due to several ecological interactions and feedbacks present within them. While the ecological processes that maintain the degraded characteristics of these sites seem bottom-up orientated due to the abundance of alien plants, there are additional top-

down forces responsible for the degraded habitats and lack of native habitat recovery in the restoration sites of the Tenmile Creek watershed.

Native ungulate herbivory is a top-down force persistent in both the degraded restoration sites and grazing lawns of this system. In chapter three, I demonstrated the power and influence Roosevelt elk (*Cervus canadensis roosevelti*) (hereafter elk) and black tailed deer (*Odocoileus hemionus columbianus*) (hereafter deer) have in maintaining grazing lawn patches in the Tenmile Creek and Cummins Ridge watersheds. These ungulates provide favorable conditions for alien plants by suppressing native shrub and tree establishment, by granting competitive advantages to fast growing herbaceous vegetation, and by benefiting plants that can reproduce asexually when grazed. I suggested native ungulates experience lower predation risk in lawns due to the increased visibility provided by the absence of a dense understory layer, allowing ungulates to select for palatable plants when foraging. Since many species of alien plants in degraded sites are unpalatable or avoided by elk and deer (Duby 1975, Corrigan et al. 1978, Marten 1985), or respond favorably to herbivory through compensatory mechanisms (Järemo et al. 1996), native plants have increased herbivory pressure and growth suppression compared to alien plants. Like grazing lawns, I hypothesize elk and deer are utilizing the increased visibility for prey within the degraded restoration sites, allowing them to spend longer periods of time foraging for preferred food contributing to alien plant dominance. Degraded habitats in restoration sites were originally maintained by livestock ranching and have not reverted in characteristic or structure like the forested areas surrounding them despite the implementation of bottom-up oriented restoration initiatives. The lack of change

in plant species assemblage, characteristic, and structure since livestock removal indicates native herbivores have replaced livestock in function and effect for the system, sustaining the degraded characteristics of the restoration sites in the Tenmile Creek watershed.

4.2 Limitations of Passive Restoration in The Degraded Habitats of The Tenmile Creek Watershed

Herbivory exclusion in grazing lawns of the temperate coastal rainforest ecosystem resulted in the complete reversion of plant species assemblage and structure from a lawn to forest habitat. After 27 years of herbivory exclusion, alien plants were replaced by native ones and structural complexity increased as forest layers developed. Numerous exclosure experiments that measure responses in habitat characteristics, structure, and function of grazing lawns following herbivory exclusion have been conducted worldwide, each yielding various results. In the temperate broadleaf forests of Wisconsin, the 16-year exclusion of deer reverted the homogeneous structure of sedge and grass grazing lawns to mixed assemblages of woody plants, ferns, and sugar maple saplings (*Acer saccharum*) (Rooney 2009). However, grazing lawns in grassland systems can exhibit alternative changes to herbivory exclusion. In the Humid Chaco grasslands of South America, 4-year capybara (*Hydrochoerus hydrochaeris*) exclusion generated new species assemblages of herbaceous vegetation, increasing ecosystem heterogeneity (Avila et al. 2025), but did not promote shrub or sapling regeneration. Likewise, in the Terai-Duar grasslands of Nepal, the 150 day exclusion of native ungulates reduced plant diversity and reverted grazing lawns to bunchgrass assemblages indistinguishable from neighboring

grasslands, and did not promote the creation of a shrubland habitat (Karki et al. 2000). These studies indicate herbivory exclusion can instill a myriad of changes in grazing lawns across various ecosystems and temporal spans.

Passive restoration initiatives, such as herbivory exclosures, may not promote the establishment of woody plants in the degraded habitats of restoration sites alone. Instead, large areas inhabited by alien plants may be unaffected by herbivory exclusion if forested areas are not nearby to supply the seeds of native plants, as observed in previous grazing lawn exclosure studies in grassland habitats. Degraded habitat in forest ecosystems can develop complex vegetation structures through increased woody plant recruitment and alien plant exclusion when herbivory pressure is eliminated (Shen et al. 2016). However, the effects of herbivory exclusion on the ecological conditions and functions of degraded habitats in the Tenmile Creek watershed may be spatially dependent and inversely related to the area of degraded habitat. While the drivers responsible for forest succession are system dependent, the planting of native saplings and proximity of degraded habitat to forested habitat contributed to the ecological effects observed in herbivory exclosures of grazing lawns in the Tenmile Creek watershed. Grazing lawn exclosures ranged from 15 m² to 82 m² in area and were constructed in lawns surrounded by forested habitat while the degraded habitats of restoration sites in the Tenmile Creek watershed are composed of alien vegetation for hundreds of meters in area. The presence of forested habitat surrounding grazing lawn patches created favorable conditions for the passive establishment of native plants. Noonan et al. (2021) demonstrated canopy gap size as a determinant factor for passive restoration success in the heavily browsed boreal

forests of Newfoundland, revealing active restoration prescriptions such as native sapling planting is required in addition to ungulate exclosure installation in areas with large canopy gaps. Lack of overstory canopy cover, increased distance from forest habitats, and alien plant competition would prevent the passive establishment of native shrubs and saplings in the herbivory exclosures of degraded habitats if unaccompanied by the planting of native saplings. Therefore, passive restoration strategies, such as the installation of herbivory exclosures, should be accompanied by active restoration strategies, such as the planting of native saplings to increase the effectiveness of ecological restoration in the degraded habitats of the Tenmile Creek watershed.

4.3 Restoration Prescriptions For The Degraded Habitats of The Tenmile Creek Watershed

Ecological restoration requires multifaceted approaches for improving the characteristics and functions of degraded habitats. Effective restoration requires ecological prescriptions that improve several conditions and functions of a degraded habitat simultaneously while mitigating the primary drivers that caused habitat degradation in the first place (Gann et al. 2019). Passive restoration efforts can cost less than active restoration prescriptions when the ecological functions of a healthy system assist in restoring degraded habitats within a system (Zahawi et al. 2014). However, in ecosystems where ecological forces contribute to the environmental degradation of habitats, active restoration is necessary to improve the degraded conditions of restoration sites. In the restoration sites of the Tenmile Creek watershed, a mix of active and passive restoration techniques are required to improve habitat

quality, mitigate the drivers maintaining the degraded ecological condition and functions of current restoration sites, and prevent subsequent degradation from occurring.

To mitigate the absence of a functional apex predator population in the system, I propose the use of herbivory exclosures and simulated predator presence mechanisms, the mechanical management of alien herbaceous plants, and subsequent planting of native sapling and shrub species may improve the habitat quality and function of degraded sites managed by restoration practitioners within the system (Figure 1). These prescriptions address the primary forces maintaining the degraded qualities of restoration sites within the system and expedite the recovery of native vegetation that instill passive recovery processes. In the interest of improving habitat structure and function in the degraded sites of the Tenmile Creek watershed, I outline the active restoration prescriptions that enable passive ecological processes capable of improving habitat quality and function in the remainder of this chapter.

Herbivory exclusion, or the reduction of herbivory pressure by native ungulates can be achieved several ways within restoration sites of the Tenmile Creek watershed. Installing herbivory exclosures will mitigate the herbivory effects instilled by native ungulates, although variables such as the locations, sizes, and length of time they are installed for must be considered before their implementation. A common tradeoff restoration practitioners weigh when considering exclosure installations involves the cost of exclosure installation compared to the size the exclosure must be to facilitate the passive establishment of native plants. The grazing lawn exclosures in the Tenmile Creek watershed demonstrated small (15 m²) exclosures provide suitable

conditions for the passive establishment of native understory vegetation, indicating the exclusion of native ungulates across an entire restoration site is unnecessary. In fact, exclosure experiments conducted in completely deforested sites heavily herbivorized by deer in the temperate broadleaf and mixed forest ecoregion of the northeastern United States demonstrated small area (9 m² to 36 m²) exclosures were effective in increasing the survival rate of native trees and shrubs (Sweeney and Dow 2019). I propose multiple evenly spaced exclosures should be installed within the restoration sites of the Tenmile Creek watershed as opposed to one large exclosure surrounding the entirety of a restoration site. Exclosures should be prioritized in areas furthest from forested habitats to maximize canopy complexity and to establish ecological feedbacks that passively improve the conditions of degraded sites. Exclosures need to be large enough to accommodate understory development of native sapling, shrubs, ferns, and forbs while remaining within the budget provided by the environmental groups that manage the restoration sites.

Within each exclosure, an assemblage of fast-growing native saplings capable of increasing canopy cover and leaf litter deposition should be planted. Based on the growing conditions of degraded sites in the watershed, western redcedar (*Thuja plicata*), bigleaf maple (*Acer macrophyllum*), Douglas fir (*Pseudotsuga menziesii*), Sitka spruce (*Picea sitchensis*), and cascara (*Rhamnus purshiana*) are optimal tree species for planting within exclosures. Native shrubs and ferns should also be planted in exclosures to increase layer complexity and provide seeds for nearby areas outside of exclosures. Red elderberry (*Sambucus racemosa*), evergreen huckleberry (*Vaccinium ovatum*), salmonberry (*Rubus spectabilis*), thimbleberry (*Rubus*

parviflorus), twinberry honeysuckle (*Lonicera involucrata*), salal (*Gaultheria shallon*), and common sword fern (*Polystichum munitum*) are suitable understory plants for exclosures. Over time, these species will provide ecological conditions favorable for the passive establishment of additional native plants by increasing canopy cover, decreasing surface temperature, facilitating the spread of native plants by seeds, and reducing soil compaction. These conditions will encourage the development of understory plant communities within the exclosures (Sabo et al. 2017), while establishing seedbanks of woody plants to be spread in the areas surrounding each exclosure as growing conditions across the restoration site improve (Tamura 2016).

Alien plants can prohibit native tree regeneration through competition for resources, instill chemical, physical, and structural changes to patches that impact regeneration, and dominate an area through indirect interactions with other alien plant species (Langmaier and Lapin 2020). Inversely, low light conditions generated by overstory canopy development reduces alien grass biomass while facilitating woody plant establishment (McDaniel and Ostertag 2010). Early active restoration prescriptions for managing alien plants can increase native plant richness in exclosures (Bourg et al. 2017) and should be considered when exclosures are installed to restoration sites in the Tenmile Creek watershed. The diversity of alien plants within degraded sites of the system prevents a unanimous restoration prescription for the reduction of alien plants in exclosures. Instead, restoration managers should derive alien plant management strategies pertaining to each individual exclosure based on the alien species assemblages present within them. For instance, the mowing

of alien plants should be considered in exclosures where fast growing, tall grasses are present to reduce interspecific competition for resources with native species.

However, if the growth of alien plants in exclosures are accelerated by herbivory effects (Paige and Whitham 1987), then mowing should be avoided to prevent further facilitation of alien plant dominance. Likewise, in areas where mechanical alien plant removal is inexpensive, swift, and effective, restoration practitioners should opt to remove them when exclosures are installed. Regardless of the exclosure-specific decisions made for alien plant management, herbivory exclusion coupled with the planting of native species will result in improvements in habitat complexity, plant species assemblages, and ecological functions of the restoration site over time as plants become established. Active alien plant management facilitates rapid improvements to the plant species assemblages and structures of exclosures, expediting the passive effects that improve the habitat quality and function of the restoration site in its entirety.

To facilitate native plant dominance and layer complexity throughout the restoration sites of the Tenmile Creek watershed, restoration prescriptions that manage ungulates outside of exclosures should be implemented in tandem with the restoration initiatives I have mentioned. The formation of tight, irregular corridors through the installation of multiple exclosures in restoration sites can influence the behaviors of ungulates within the system by decreasing visibility for predators. Restoration practitioners can use these corridors to increase the perceived risk of predation on native ungulates, decreasing herbivory pressure by generating landscapes of fear in the absence of consumptive effects instilled by predators.

Common restoration initiatives that increase perceived risk include the mimicry of predator presence through audio broadcasts (Watve et al. 2019) and odor deposits (Sullivan et al. 1985). Audio broadcasts of predator sounds can be very effective in instilling behavioral changes in their prey (Adams and Kitchen 2020) decreasing the herbivory effects of native ungulates at a restoration site. Additionally, mesopredators may not be affected by these broadcasts (Root-Gutteridge et al. 2024), allowing the direct and indirect predation effects of species like coyote (*Canis latrans*) and bobcat (*Lynx rufus*) to continue without interference, maintaining food web dynamics in the system. Predator odor, whether through urine, feces, anal gland secretions, or skin, can also deter herbivores (Apfelbach et al. 2005). In theory, the simulated presence of an apex predator, like wolves or cougar, would instill nonconsumptive herbivory limiting effects that promote the establishment of native plants outside of exclosures (van Beeck Calkoen et al. 2021, 2026). Whereas exclosures are impassable by most fauna, simulating predator risk does not restrict access to habitat or limit animal movements within the system. Coupled with the installation of herbivory exclosures, the planting of native vegetation, and alien plant management, these restoration initiatives are likely to improve the ecological function and habitat quality of degraded sites in the Tenmile Creek watershed.

Continuous tracking, analysis, and review of how each prescription contributes to the ecological restoration of degraded habitats must be implemented, allowing for adaptive management practices that improve the function and quality of the system in its entirety. As climate change, natural disturbance, and alien species continue to influence the ecological integrity and functions of both the restoration

sites and natural habitats of the Tenmile Creek watershed, it is imperative active management practices are incorporated into the plans for restoration sites in the system. Weaknesses identified in the active and passive restoration prescriptions must be addressed as they are identified so that fluid and effective improvements of the conditions and functions of a restoration site can be made as they occur. Simple and effective methods for measuring the success of each prescription consist of collecting plant diversity, mortality rates, fruit presence or absence, and stem density data within exclosures. Along the perimeter of exclosures, pioneering native plant species should be identified and considered for subsequent planting to expedite subsequent effects for habitat improvement. Game cameras measuring the frequency, time spent, and behaviors of native ungulates should be installed throughout the site to gauge the effectiveness of predator mimicry and simulated predation risk on herbivory effects. Collecting ecological data after the implementation of restoration initiatives allows restoration managers to review the effectiveness of each individual prescription and actively adjust their methods for restoration, increasing the likelihood of achieving restoration goals.

Finally, system-wide restoration can be achieved through the management of apex predators and/or herbivore populations. To decrease herbivory pressure and facilitate the passive establishment of native plants and the development of complex vegetation layers within the forests and degraded sites of the Tenmile Creek watershed, top-down oriented management decisions, such as the restoration of wolves or culling of native ungulates could be considered by the governmental agencies that manage wildlife in the region. Apex predators create dynamic

landscapes of fear (Laundré et al. 2001, Ganz et al. 2024), and directly limit herbivore populations (Ripple and Beschta 2012), thereby decreasing herbivory effects throughout a system (Ripple and Beschta 2008, Beschta and Ripple 2010, 2016). The knock-on effects associated with increased predation and predation risk could facilitate the ecological feedbacks and mechanisms in the active and passive restoration initiatives I outlined, potentially improving the structure and habitat of the system (Figure 2).

4.4 Concluding Remarks

The study of grazing lawn ecology provides remarkable insight into the ability, processes, and effects associated with herbivores and how herbivory can affect systems in their entirety. By systematically reviewing journal articles on the subject and through the identification and experimental manipulation of grazing lawns in the temperate rainforest ecosystem, I gained an understanding of the power herbivores have in shaping the functions and characteristics of ecosystems worldwide, and at a local level. Furthermore, my research provides significant insight into the processes and functions herbivory has in maintaining the degraded habitats of the Tenmile Creek watershed and methods for instilling effective restoration prescriptions that mitigate these effects, improve habitat quality, and restore ecosystem functions of the temperate coastal rainforest ecosystem.

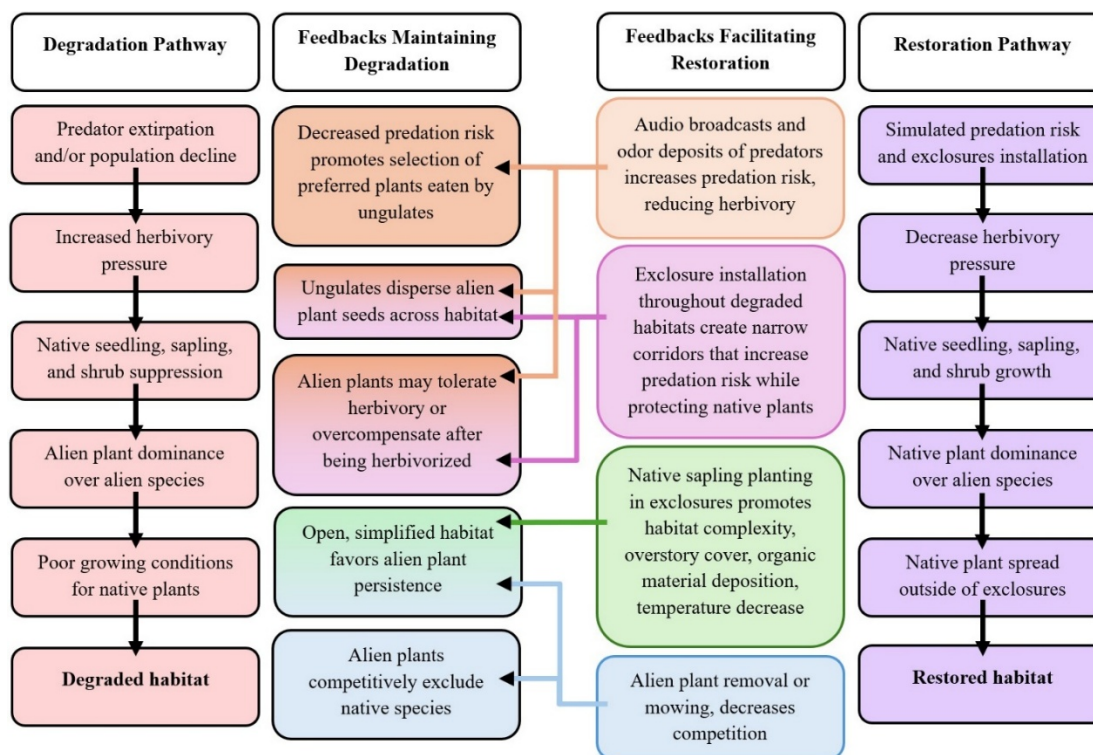


Figure 4.1: Conceptual framework diagram of the ecological pathways and feedbacks associated with degraded and restored habitats in the Tenmile Creek watershed. Starting in the left column, a flow chart depicting the pathway for degraded habitat creation followed the feedbacks that maintain degraded habitats in the watershed. On the right side, the feedbacks that facilitate ecological restoration by mitigating the processes that maintain degraded habitats followed by a flow chart depicting the pathway for restored habitat creation in the watershed. Colored arrows connecting restoration feedbacks with degradation feedbacks illustrate how each restoration initiative mitigates the processes that maintain degraded habitats.

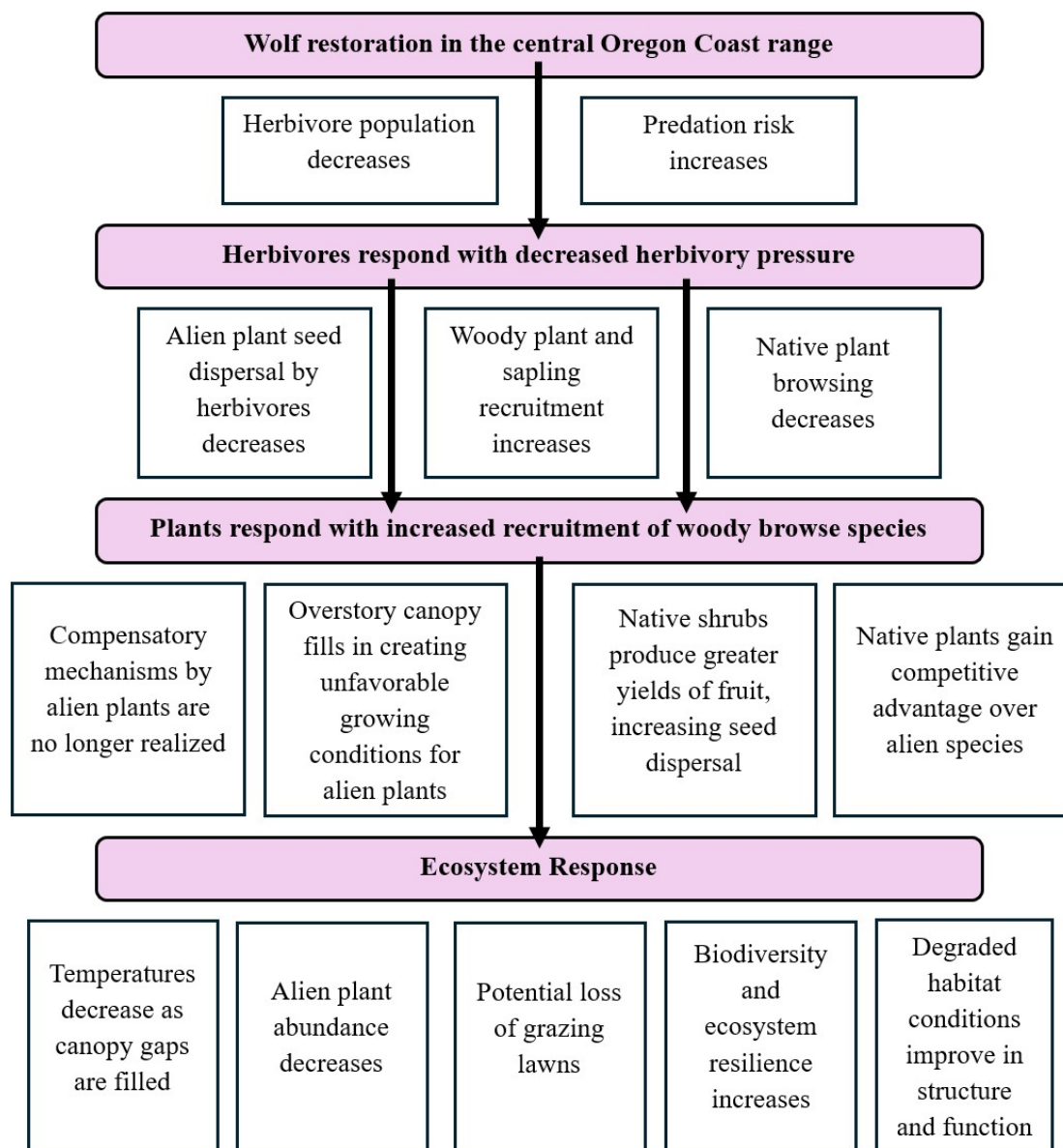


Figure 4.2: Conceptual framework diagram of the cascading ecological effects associated with wolf restoration in the central Oregon Coast range.

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APPENDICIES

Appendix A: Chapter 2 Supplementary Material

Tropical & Subtropical Grasslands, Savannas & Shrublands		Number of study sites in biome = 73
	Limpopo lowveld	Study sites in ecoregion = 33
	Zambezian mopane woodlands	Study sites in ecoregion = 11
	Southern Acacia-Commiphora bushlands and thickets	Study sites in ecoregion = 8
	Drakensberg grasslands	Study sites in ecoregion = 7
	Terai-Duar savanna and grasslands	Study sites in ecoregion = 5
	Central bushveld	Study sites in ecoregion = 3
	Central Zambezian wet miombo woodlands	Study sites in ecoregion = 2
	Humid Chaco	Study sites in ecoregion = 2
	Beni savanna	Study sites in ecoregion = 1
	Northern Congolian Forest-Savanna	Study sites in ecoregion = 1
Tundra		Number of study sites in biome = 13
	Beringia lowland tundra	Study sites in ecoregion = 12
	Canadian Low Arctic tundra	Study sites in ecoregion = 1
Tropical & Subtropical Moist Broadleaf Forests		Number of study sites in biome = 7
	Maputaland coastal forests and woodlands	Study sites in ecoregion = 4
	Kwazulu Natal-Cape coastal forests	Study sites in ecoregion = 2
	Northern Swahili coastal forests	Study sites in ecoregion = 1
Deserts & Xeric Shrublands		Number of study sites in biome = 6
	Nama Karoo shrublands	Study sites in ecoregion = 5
	Kalahari xeric savanna	Study sites in ecoregion = 1
Mediterranean Forests, Woodlands & Scrub		Number of study sites in biome = 5
	Albany thickets	Study sites in ecoregion = 3
	Chilean Matorral	Study sites in ecoregion = 1
	Fynbos shrubland	Study sites in ecoregion = 1
Montane Grasslands & Shrublands		Number of study sites in biome = 5
	Highveld grasslands	Study sites in ecoregion = 5
Temperate Grasslands, Savannas & Shrublands		Number of study sites in biome = 4
	Flint Hills tallgrass prairie	Study sites in ecoregion = 4
Boreal Forests/Taiga		Number of study sites in biome = 2
	Southern Hudson Bay taiga	Study sites in ecoregion = 2
Temperate Broadleaf & Mixed Forests		Number of study sites in biome = 1
	Western Great Lakes forests	Study sites in ecoregion = 1
Temperate Conifer Forests		Number of study sites in biome = 1
	Blue Mountains forests	Study sites in ecoregion = 1
Tropical & Subtropical Dry Broadleaf Forests		Number of study sites in biome = 1
	Sri Lanka dry-zone dry evergreen forests	Study sites in ecoregion = 1

Appendix Table 0.1: Study site locations of grazing lawn journal articles by biome and ecoregion. Grazing lawn study site locations by biomes and ecoregions organized from top to bottom in decreasing number of study sites identified in my review. In bold, the name and the number of study sites within each biome identified. In the subsections below each biome name, the name and number of study sites for each ecoregion within of a biome.

Appendix Table 0.2: Study site locations of grazing lawn journal articles by status of protection and site name. Grazing lawn study site locations by status of protection and study site name. In bold, the name of the protection status category and the number of study sites within each category. In the subsections below each protection status category, the name of each study site and number of studies that took place there. Study sites with an (*) were named by Hempson et al., (2022) and identified as protected areas by the authors.

Appendix Table 2.2 (Continued)

Game Reserve		Total = 35
	Hluhluwe–iMfolozi Park	27
	Manyaleti Game Reserve	2
	Asante Sana Game Reserve	1
	Ithala Game Reserve	1
	Masai Mara Game Reserve	1
	Mkambati Game Reserve	1
	Samara Karoo Reserve	1
	Welgevonden Private Game Reserve	1
Protected Area		Total = 32
	Berg-en-Dal *	1
	Black Rock *	1
	Camdeboo *	1
	Cape Vidal *	1
	Commando Drift *	1
	Gariep Dam *	1
	Giants Castle *	1
	Golden Gate *	1
	Great Fish *	1
	Ithala *	1
	Kirawira *	1
	Kogatende *	1
	Koppies Dam *	1
	Lower Sabie *	1
	Luchaba *	1
	Mara *	1
	Marakele *	1
	Maria Moroka *	1
	Mkambati *	1
	Mokala *	1
	Mountain Zebra *	1
	Mpofu *	1
	Ndumo *	1
	Pilanesberg *	1
	Satara *	1
	Seronera *	1
	Soit *	1
	Spioenkop *	1
	St Lucia *	1
	Tembe *	1
	Tsolwana *	1
	Willem Pretorius *	1

Appendix Table 2.2 (Continued)

National Park		Total = 24
	Kruger National Park	10
	Royal Bardia National Park	5
	Iberá National Park	2
	Serengeti National Park	2
	Benue National Park	1
	Manu National Park	1
	Mountain Zebra National Park	1
	Saadani National Park	1
	Wilpattu National Park	1
Wildlife Refuge		Total = 13
	Yukon-Kuskokwim Delta Wildlife Refuge	12
	Queen Maud Gulf (Ahiak) Migratory Bird Sanctuary	1
Nature Reserve		Total = 7
	Konza Prairie Biological Station	4
	Timbavati Private Nature Reserve	2
	De Hoop Nature Reserve	1
Unknown		Total = 4
	Akimiski Island, Nunavut Territory, Canada	1
	La Perouse Bay, Manitoba, Canada	1
	Rinconada de Maipú Experimental Station	1
	Starkey Experimental Forest and Range	1
Unprotected Area		Total = 3
	Issa valley western Tanzania	2
	Vilas County, Wisconsin USA	1

Anseriformes				
<i>Species</i> Common Name	Taxonomy ID	Number of Journal articles (n)	Genus	Family
<i>Branta bernicla nigricans</i> grey bellied brent goose	184714	10	Branta	Anatidae
<i>Anser caerulescens</i> caerulescens snow goose	70340	3	Anser	Anatidae
<i>Branta hutchinsii</i> cackling goose	371860	3	Branta	Anatidae
<i>Anser albifrons frontalis</i> greater white-fronted goose	184919	2	Anser	Anatidae
<i>Anser canagicus</i> emperor goose	107021	2	Anser	Anatidae
<i>Branta hutchinsii minima</i> small cackling goose	184597	2	Branta	Anatidae
<i>Anser rossii</i> Ross' goose	56281	1	Anser	Anatidae
<i>Branta bernicla</i> brent goose	184712	1	Branta	Anatidae
<i>Branta canadensis</i> Canada goose	8853	1	Branta	Anatidae
<i>Branta canadensis maxima</i> giant Canada goose	184599	1	Branta	Anatidae
<i>Neochen jubata</i> Orinoco goose	8882	1	Neochen	Anatidae

Appendix Table 0.3: Anseriformes species identified in the systematic review of grazing lawn journal articles. Species and common names, taxonomic ID numbers, and taxonomic information are assigned based on the taxonomic classifications provided by the National Center for Biotechnology Information (NCBI) database as reference (Sayers et al. 2024). Species are organized from top to bottom by decreasing number of journal articles the animal was identified in.

Appendix Table 0.4: Artiodactyla species identified in the systematic review of grazing lawn journal articles. Species and common names, taxonomic ID numbers, and taxonomic information are assigned based on the taxonomic classifications provided by the National Center for Biotechnology Information (NCBI) database as reference (Sayers et al. 2024). Species are organized from top to bottom by decreasing number of journal articles the animal was identified in.

Appendix Table 2.4 (Continued)

Artiodactyla				
<i>Species</i> Common Name	Taxonomy ID	Number of Journal articles (n)	Genus	Family
<i>Connochaetes taurinus</i> brindled gnu	9927	32	Connochaetes	Bovidae
<i>Syncerus caffer</i> African buffalo	9970	28	Syncerus	Bovidae
<i>Aepyceros melampus</i> impala	9897	27	Aepyceros	Bovidae
<i>Phacochoerus africanus</i> common warthog	41426	19	Phacochoerus	Suidae
<i>Tragelaphus angasii</i> nyala	66437	10	Tragelaphus	Bovidae
<i>Tragelaphus strepsiceros</i> greater kudu	9946	10	Tragelaphus	Bovidae
<i>Axis axis</i> chital	30531	7	Axis	Cervidae
<i>Giraffa camelopardalis</i> giraffe	9894	7	Giraffa	Giraffidae
<i>Kobus ellipsiprymnus</i> waterbuck	9962	7	Kobus	Bovidae
<i>Axis porcinus</i> hog deer	57737	5	Axis	Cervidae
<i>Sylvicapra grimmia</i> bush duiker	119562	5	Sylvicapra	Bovidae
<i>Alcelaphus buselaphus</i> hartebeest	59517	4	Alcelaphus	Bovidae
<i>Bison bison</i> American bison	9901	4	Bison	Bovidae
<i>Rucervus duvaucelii</i> barasingha	43328	4	Rucervus	Cervidae
<i>Hippopotamus amphibius</i> hippopotamus	9833	3	Hippopotamus	Hippopotamidae
<i>Oreotragus oreotragus</i> klipspringer	66444	3	Oreotragus	Bovidae
<i>Potamochoerus larvatus</i> bushpig	273792	3	Potamochoerus	Suidae
<i>Redunca redunca</i> bohar reedbuck	59556	3	Redunca	Bovidae
<i>Rusa unicolor</i> sambar	662561	3	Rusa	Cervidae
<i>Tragelaphus oryx</i> common eland	9945	3	Tragelaphus	Bovidae
<i>Antidorcas marsupialis</i> springbok	59523	2	Antidorcas	Bovidae
<i>Blastocerus dichotomus</i> marsh deer	248133	2	Blastocerus	Cervidae
<i>Connochaetes gnou</i> black wildebeest	59528	2	Connochaetes	Bovidae
<i>Damaliscus pygargus</i> <i>phillipsi</i> blesbok	37172	2	Damaliscus	Bovidae
<i>Giraffa giraffa</i> southern giraffe	439325	2	Giraffa	Giraffidae
<i>Hippotragus equinus</i> roan antelope	37186	2	Hippotragus	Bovidae

Appendix Table 2.4 (Continued)

Artiodactyla				
<i>Muntiacus muntjak</i> muntjak	9888	2	Muntiacus	Cervidae
<i>Ozotoceros bezoarticus</i> pampas deer	63824	2	Ozotoceros	Cervidae
<i>Philantomba monticola</i> blue duiker	907742	2	Philantomba	Bovidae
<i>Raphicercus campestris</i> steenbok	59544	2	Raphicercus	Bovidae
<i>Sus scrofa</i> pig	9823	2	Sus	Suidae
<i>Tragelaphus scriptus</i> bushbuck	66440	2	Tragelaphus	Bovidae
<i>Boselaphus tragocamelus</i> nilgai	9917	1	Boselaphus	Bovidae
<i>Bubalus bubalis</i> water buffalo	89462	1	Bubalus	Bovidae
<i>Cervus elaphus</i> red deer	9860	1	Cervus	Cervidae
<i>Damaliscus korrigum</i> korrigum sassaby	9930	1	Damaliscus	Bovidae
<i>Damaliscus pygargus</i> pygargus bontebok	143918	1	Damaliscus	Bovidae
<i>Eudorcas thomsonii</i> Thomson's gazelle	69308	1	Eudorcas	Bovidae
<i>Kobus kob</i> kob	59530	1	Kobus	Bovidae
<i>Mazama gouazoubira</i> gray brocket	52407	1	Subulo	Cervidae
<i>Moschiola meminna</i> Indian spotted chevrotain	439461	1	Moschiola	Tragulidae
<i>Muntiacus vaginalis</i> northern red muntjac	9887	1	Muntiacus	Cervidae
<i>Nanger granti</i> Grant's gazelle	27591	1	Nanger	Bovidae
<i>Odocoileus hemionus</i> mule deer	9872	1	Odocoileus	Cervidae
<i>Odocoileus virginianus</i> white-tailed deer	9874	1	Odocoileus	Cervidae
<i>Oryx gazella</i> gemsbok	9958	1	Oryx	Bovidae
<i>Pecari tajacu</i> collared peccary	9829	1	Dicotyles	Tayassuidae
<i>Phacochoerus aethiopicus</i> desert warthog	85517	1	Phacochoerus	Suidae
<i>Potamochoerus porcus</i> red river hog	273791	1	Potamochoerus	Suidae
<i>Redunca arundinum</i> southern reedbuck	59554	1	Redunca	Bovidae
<i>Rucervus duvaucelii</i> duvaucelii western swamp deer	2170375	1	Rucervus	Cervidae

Appendix Table 0.5: Carnivora species identified in the systematic review of grazing lawn journal articles. Species and common names, taxonomic ID numbers, and taxonomic information are assigned based on the taxonomic classifications provided by the National Center for Biotechnology Information (NCBI) database as reference (Sayers et al. 2024). Species are organized from top to bottom by decreasing number of journal articles the animal was identified in.

Appendix Table 2.5 (Continued)

Carnivora				
<i>Species</i> Common Name	Taxonomy ID	Number of Journal articles (n)	Genus	Family
<i>Panthera leo</i> lion	9689	3	Panthera	Felidae
<i>Panthera tigris</i> tiger	9694	3	Panthera	Felidae
<i>Crocuta crocuta</i> spotted hyena	9678	2	Crocuta	Hyaenidae
<i>Puma concolor</i> puma	9696	2	Puma	Felidae
<i>Vulpes lagopus</i> Arctic fox	494514	2	Vulpes	Canidae
<i>Acinonyx jubatus</i> cheetah	32536	1	Acinonyx	Felidae
<i>Canis aureus</i> golden jackal	68724	1	Canis	Canidae
<i>Canis latrans</i> coyote	9614	1	Canis	Canidae
<i>Canis lupus</i> wolf	9612	1	Canis	Canidae
<i>Civettictis civetta</i> African civet	94186	1	Civettictis	Viverridae
<i>Galerella sanguinea</i> slender mongoose	210644	1	Galerella	Herpestidae
<i>Genetta angolensis</i> Angolan genet	235200	1	Genetta	Viverridae
<i>Herpestes ichneumon</i> Egyptian mongoose	9700	1	Herpestes	Herpestidae
<i>Ichneumia albicauda</i> white-tailed mongoose	205599	1	Ichneumia	Herpestidae
<i>Lynx rufus</i> bobcat	61384	1	Lynx	Felidae
<i>Mellivora capensis</i> ratel	9664	1	Mellivora	Mustelidae
<i>Melursus ursinus</i> sloth bear	9636	1	Melursus	Ursidae
<i>Panthera onca</i> jaguar	9690	1	Panthera	Felidae
<i>Panthera pardus</i> leopard	9691	1	Panthera	Felidae
<i>Panthera pardus kotiya</i> Sri Lankan leopard	2816903	1	Panthera	Felidae
<i>Prionailurus rubiginosus</i> rusty-spotted cat	61387	1	Prionailurus	Felidae
<i>Prionailurus viverrinus</i> fishing cat	61388	1	Prionailurus	Felidae
<i>Ursus americanus</i> American black bear	9643	1	Ursus	Ursidae
<i>Genetta species</i> genet species	94189	1	Genetta	Viverridae

Charadriiformes				
<i>Species</i> Common Name	Taxonomy ID	Number of Journal articles (n)	Genus	Family
<i>Larus hyperboreus</i> glaucous gull	119637	1	Larus	Laridae

Appendix Table 0.6: Charadriiformes species in the systematic review of grazing lawn journal articles. Species and common names, taxonomic ID numbers, and taxonomic information are assigned based on the taxonomic classifications provided by the National Center for Biotechnology Information (NCBI) database as reference (Sayers et al. 2024).

Coleoptera				
<i>Species</i> Common Name	Taxonomy ID	Number of Journal articles (n)	Genus	Family
<i>Gymnopleurus humanus</i> dung beetle species	1165003	1	Gymnopleurus	Scarabaeidae
<i>Kheper nigroaeneus</i> diurnal copper dung beetle	205299	1	Kheper	Scarabaeidae
<i>Allogymnopleurus species</i> beetle species	206789	1	Allogymnopleurus	Scarabaeoidea

Appendix Table 0.7: Coleoptera species identified in the systematic review of grazing lawn journal articles. Species and common names, taxonomic ID numbers, and taxonomic information are assigned based on the taxonomic classifications provided by the National Center for Biotechnology Information (NCBI) database as reference (Sayers et al. 2024).

Dictyoptera				
<i>Species</i> Common Name	Taxonomy ID	Number of Journal articles (n)	Genus	Family
<i>Macrotermes species</i> termite species	46572	2	Macrotermes	Blattoidea
<i>Odontotermes species</i> termite species	60587	1	Odontotermes	Blattoidea
<i>Trinervitermes species</i> termite species	70854	1	Trinervitermes	Blattoidea

Appendix Table 0.8: Dictyoptera species identified in the systematic review of grazing lawn journal articles. Species and common names, taxonomic ID numbers, and taxonomic information are assigned based on the taxonomic classifications provided by the National Center for Biotechnology Information (NCBI) database as reference (Sayers et al. 2024). Species are organized from top to bottom by decreasing number of journal articles the animal was identified in.

Hymenoptera				
<i>Species</i> Common Name	Taxonomy ID	Number of Journal articles (n)	Genus	Family
<i>Pheidole species</i> ant species	190769	2	Pheidole	Formicoidea
<i>Anoplolepis custodiens</i> common pugnacious ant	602572	1	Anoplolepis	Formicidae
<i>Myrmecaria natalensis</i> Natal droptail ant	612924	1	Myrmecaria	Formicidae
<i>Ocymyrmex fortior</i> hot-rod ant species	613407	1	Ocymyrmex	Formicidae
<i>Tetramorium sericeiventris</i> common fierce ant	628885	1	Tetramorium	Formicidae
<i>Odontomachus species</i> ant species	43086	1	Odontomachus	Formicoidea

Appendix Table 0.9: Hymenoptera species identified in the systematic review of grazing lawn journal articles. Species and common names, taxonomic ID numbers, and taxonomic information are assigned based on the taxonomic classifications provided by the National Center for Biotechnology Information (NCBI) database as reference (Sayers et al. 2024). Species are organized from top to bottom by decreasing number of journal articles the animal was identified in.

Lagomorpha				
<i>Species</i> Common Name	Taxonomy ID	Number of Journal articles (n)	Genus	Family
<i>Lepus saxatilis</i> scrub hare	63224	3	Lepus	Leporidae
<i>Lepus nigricollis</i> Indian hare	2791018	1	Lepus	Leporidae
<i>Lepus species</i> hare species	9980	1	Lepus	Leporidae

Appendix Table 0.10: Lagomorpha species identified in the systematic review of grazing lawn journal articles. Species and common names, taxonomic ID numbers, and taxonomic information are assigned based on the taxonomic classifications provided by the National Center for Biotechnology Information (NCBI) database as reference (Sayers et al. 2024). Species are organized from top to bottom by decreasing number of journal articles the animal was identified in.

Orthoptera				
<i>Species</i> Common Name	Taxonomy ID	Number of Journal articles (n)	Genus	Family
<i>Orphulella speciosa</i> slant-faced pasture grasshopper	1034424	1	Orphulella	Acrididae
<i>Syrbula admirabilis</i> admirable grasshopper	1034413	1	Syrbula	Acrididae
<i>Mermiria species</i> grasshopper species	1034429	1	Mermiria	Acridoidea

Appendix Table 0.11: Orthoptera species identified in the systematic review of grazing lawn journal articles. Species and common names, taxonomic ID numbers, and taxonomic information are assigned based on the taxonomic classifications provided by the National Center for Biotechnology Information (NCBI) database as reference (Sayers et al. 2024).

erissodactyla				
<i>Species</i> Common Name	Taxonomy ID	Number of Journal articles (n)	Genus	Family
<i>Ceratotherium simum</i> white rhinoceros	9807	20	Ceratotherium	Rhinocerotidae
<i>Equus quagga</i> plains zebra	89248	19	Equus	Equidae
<i>Equus quagga burchellii</i> Burchell's zebra	89252	9	Equus	Equidae
<i>Diceros bicornis</i> black rhinoceros	9805	5	Diceros	Rhinocerotidae
<i>Rhinoceros unicornis</i> greater Indian rhinoceros	9809	3	Rhinoceros	Rhinocerotidae
<i>Equus zebra</i> mountain zebra	9791	1	Equus	Equidae
<i>Equus zebra zebra</i> Cape mountain zebra	301482	1	Equus	Equidae
<i>Tapirus terrestris</i> Brazilian tapir	9801	1	Tapirus	Tapiridae
<i>Equus species</i> odd-toed ungulate species	9789	1	Equus	Equidae

Appendix Table 0.12: Perissodactyla species identified in the systematic review of grazing lawn journal articles. Species and common names, taxonomic ID numbers, and taxonomic information are assigned based on the taxonomic classifications provided by the National Center for Biotechnology Information (NCBI) database as reference (Sayers et al. 2024). Species are organized from top to bottom by decreasing number of journal articles the animal was identified in.

Pholidota				
<i>Species</i> Common Name	Taxonomy ID	Number of Journal articles (n)	Genus	Family
<i>Manis crassicaudata</i> Indian pangolin	1496684	1	Manis	Manidae

Appendix Table 0.13: Pholidota species identified in the systematic review of grazing lawn journal articles. Species and common names, taxonomic ID numbers, and taxonomic information are assigned based on the taxonomic classifications provided by the National Center for Biotechnology Information (NCBI) database as reference (Sayers et al. 2024).

Primates				
<i>Species</i> Common Name	Taxonomy ID	Number of Journal articles (n)	Genus	Family
<i>Chlorocebus pygerythrus</i> vervet	60710	1	Chlorocebus	Cercopithecidae
<i>Macaca sinica</i> toque macaque	9552	1	Macaca	Cercopithecidae
<i>Pan troglodytes</i> <i>schweinfurthii</i> eastern chimpanzee	37010	1	Pan	Hominidae
<i>Papio cynocephalus</i> yellow baboon	9556	1	Papio	Cercopithecidae
<i>Papio ursinus</i> chacma baboon	36229	1	Papio	Cercopithecidae
<i>Semnopithecus priam</i> tufted gray langur	1208733	1	Semnopithecus	Cercopithecidae

Appendix Table 0.14: Primate species identified in the systematic review of grazing lawn journal articles. Species and common names, taxonomic ID numbers, and taxonomic information are assigned based on the taxonomic classifications provided by the National Center for Biotechnology Information (NCBI) database as reference (Sayers et al. 2024).

Proboscidea				
<i>Species</i> Common Name	Taxonomy ID	Number of Journal articles (n)	Genus	Family
<i>Loxodonta africana</i> African savanna elephant	9785	11	Loxodonta	Elephantidae
<i>Elephas maximus</i> Asian elephant	9783	3	Elephas	Elephantidae

Appendix Table 0.15: Proboscidea species identified in the systematic review of grazing lawn journal articles. Species and common names, taxonomic ID numbers, and taxonomic information are assigned based on the taxonomic classifications provided by the National Center for Biotechnology Information (NCBI) database as reference (Sayers et al. 2024). Species are organized from top to bottom by decreasing number of journal articles the animal was identified in.

Rheiformes				
<i>Species</i> Common Name	Taxonomy ID	Number of Journal articles (n)	Genus	Family
<i>Rhea americana</i> greater rhea	8797	1	Rhea	Rheidae

Appendix Table 0.16: Rheiformes species identified in the systematic review of grazing lawn journal articles. Species and common names, taxonomic ID numbers, and taxonomic information are assigned based on the taxonomic classifications provided by the National Center for Biotechnology Information (NCBI) database as reference (Sayers et al. 2024).

Rodentia				
<i>Species</i> Common Name	Taxonomy ID	Number of Journal articles (n)	Genus	Family
<i>Hydrochoerus hydrochaeris</i> capybara	10149	2	Hydrochoerus	Hydrochaeridae
<i>Hystrix africaeaustralis</i> Cape porcupine	10138	2	Hystrix	Hystricidae
<i>Hystrix indica</i> Indian crested porcupine	192883	1	Hystrix	Hystricidae
<i>Lagostomus maximus</i> plains viscacha	10154	1	Lagostomus	Chinchillidae
<i>Octodon degus</i> degu	10160	1	Octodon	Octodontidae

Appendix Table 0.17: Rodentia species identified in the systematic review of grazing lawn journal articles. Species and common names, taxonomic ID numbers, and taxonomic information are assigned based on the taxonomic classifications provided by the National Center for Biotechnology Information (NCBI) database as reference (Sayers et al. 2024). Species are organized from top to bottom by decreasing number of journal articles the animal was identified in.

Struthioniformes				
<i>Species</i> Common Name	Taxonomy ID	Number of Journal articles (n)	Genus	Family
<i>Struthio camelus</i> African ostrich	8801	2	Struthio	Struthionidae

Appendix Table 0.18: Struthioniformes species identified in the systematic review of grazing lawn journal articles. Species and common names, taxonomic ID numbers, and taxonomic information are assigned based on the taxonomic classifications provided by the National Center for Biotechnology Information (NCBI) database as reference (Sayers et al. 2024).

Tubulidentata				
<i>Species</i> Common Name	Taxonomy ID	Number of Journal articles (n)	Genus	Family
<i>Orycteropus afer</i> aardvark	9818	1	Orycteropus	Orycteropodidae

Appendix Table 0.19: Tubulidentata species identified in the systematic review of grazing lawn journal articles. Species and common names, taxonomic ID numbers, and taxonomic information are assigned based on the taxonomic classifications provided by the National Center for Biotechnology Information (NCBI) database as reference (Sayers et al. 2024).

Appendix Table 0.20: Grazing lawn definitions identified in qualifying journal articles of the systematic review. Definitions are organized by increasing year of publication. Rows in blue and bold font are journal articles most frequently referenced in other grazing lawn definitions I identified. Words or phrases highlighted in yellow signify language used to describe the ecological processes or interactions involved with grazing lawns. Words or phrases of definitions highlighted in green signify language used to describe the ecological characteristics involved with grazing lawns.

Appendix Table 2.20 (Continued)

Article Author(s) YEAR	Grazing Lawn Definition Provided in Journal Article. "Ecological Process/Interaction" descriptions are highlighted in yellow. "Ecological Characteristic" descriptions are highlighted in green.	Are "Ecological Process/Interactions", "Ecological Characteristics", Or Both Categories Present in Definition?	Previous Journal Article(s) Referenced in Definition?
McNaughton, SJ 1984	"Among the most conspicuous effects of large mammalian grazers upon grasslands are a drastic reduction of canopy height and the activation of tillers that lead to a prostrate, dense canopy (Vesey-Fitzgerald 1973b; McNaughton 1976, 1979a, 1979b), hereafter referred to as a grazing lawn."	Both present in definition	Y
Shackleton, CM 1992	"In this context a grazing lawn was any localized area in which marked grazing was evident through reduction in plant height and cover."	Ecological characteristic present in definition	N
Person, BT; Babcock, CA; Ruess, RW 1998	"Along the coastal flats of the Yukon-Kuskokwim (Y-K) delta in south-western Alaska, pacific black brant (<i>Branta bernicla nigricans</i> Lawrence) maintain grazing lawns of <i>C. subspatheacea</i> ."	Ecological process/interaction present in definition	N
Karki, JB; Jhala, YV; Khanna, PP 2000	"Areas in east Africa where ungulates maintain short grasses during periods of high utilization have been referred to as grazing lawns (McNaughton 1984)."	Ecological process/interaction present in definition	Y
Bond, WJ; Smythe, KA; Balfour, DA 2001	"We noted the dominant grass species and classified the grass cover as 'grazing lawn' (sensu McNaughton, 1984) or 'bunch grassland'. Grazing lawns are usually short (< 10 cm), with a low grass biomass and often with patches of bare ground. They are identified qualitatively by the presence of stoloniferous grasses (e.g. <i>Digitaria longifora</i> , <i>Panicum coloratum</i>). They indicate areas of prolonged heavy grazing and burn at very low intensities or not at all."	Both present in definition	Y
Johnson, LC; Matchett, JR 2001	"Grazing lawns are >400 m ² , within which are smaller 4-m ² grazed patches (Knapp et al. 1999). Lawns are characterized by foliage removal to ~5 cm above the soil."	Ecological characteristic present in definition	Y
Person, BT; Herzog, MP; Ruess, RW; Sedinger, JS; Anthony, RM; Babcock, CA 2003	"Like other terrestrial herbivores (McNaughton 1984; Coppock et al. 1983b), geese often create and maintain grazing lawns, defined here as vegetation swards dominated by a high density of grazing-tolerant plant species with high nutrient concentrations."	Both present in definition	Y
Archibald, S; Bond, WJ; Stock, WD; Fairbanks, DHK 2005	"We define a grazing lawn as a specific habitat type: an area dominated by stoloniferous 'lawn grass' species, as opposed to caespitose 'bunch grass' species."	Ecological characteristic present in definition	N

Appendix Table 2.20 (Continued)

Article Author(s) YEAR	Grazing Lawn Definition Provided in Journal Article. “Ecological Process/Interaction” descriptions are highlighted in yellow. “Ecological Characteristic” descriptions are highlighted in green.	Are “Ecological Process/Interactions”, “Ecological Characteristics”, Or Both Categories Present in Definition?	Previous Journal Article(s) Referenced in Definition?
Lake, BC; Lindberg, MS; Schmutz, JA; Anthony, RM; Broerman, FJ 2006	“In this paper, we use the term “grazing lawn” to refer both to <i>C. subspathacea</i> and to <i>C. ramenskii</i> that has reverted to a form indistinguishable from <i>C. subspathacea</i> .”	Ecological characteristic present in definition	N
Kotanan, PM; Abraham, KF 2006	“Intense grazing by vertebrates can convert grassland vegetation to “grazing lawns” characterized by short stature, reduced flowering, increased tillering, and increased growth rate, nutrient content, and palatability (Jameson 1963; McNaughton 1984; Coughenour 1985).”	Both present in definition	Y
Stewart, KM; Bowyer, RT; Ruess, RW; Dick, BL; Kie, JG 2006	“We define grazing lawns as vegetation swards dominated by high densities of grazing-tolerant plants with high nutrient concentrations that are maintained by herbivory (Person et al. 2003).”	Both present in definition	Y
Verweij, RJT; Verrelst, J; Loth, PE; Heitkönig, IMA; Brunsting, AMH 2006	“A grazing lawn, defined as an expanse of short grass in an immature state, has in addition grass with a higher leaf to stem ratio and a higher bulk density than that of tall grass stands.”	Ecological characteristic present in definition	N
Archibald, S 2008	“The term was first made fashionable by McNaughton (1984) when he used it to describe a community of low statured grass species that could support and maintain heavy grazing, which animals selected due to the increased yield provided per bite.”	Both present in definition	Y
Cromsigt, JPGM; Olf, H 2008	“Grazing lawns are characteristic for African savanna grasslands, standing out as intensely grazed patches of stoloniferous grazing-tolerant grass species.”	Both present in definition	N
Moe, SR; Wegge, P 2008	“Another important effect of ungulates on vegetation is grazing and browsing. Several studies have shown that herbivory may enhance the quality of the food and induce further feeding, eventually resulting in grazing and browsing lawns (McNaughton 1984; Fornara and du Toit 2007).”	Ecological process/interaction present in definition	Y

Appendix Table 2.20 (Continued)

Article Author(s) YEAR	Grazing Lawn Definition Provided in Journal Article. “Ecological Process/Interaction” descriptions are highlighted in yellow. “Ecological Characteristic” descriptions are highlighted in green.	Are “Ecological Process/Interactions”, “Ecological Characteristics”, Or Both Categories Present in Definition?	Previous Journal Article(s) Referenced in Definition?
Waldram, MS; Bond, WJ; Stock, WD 2008	“Grazing lawns, characterized by grazing-tolerant short and/or stoloniferous grasses (for example, <i>Digitaria longiflora</i> , <i>Digitaria argyrograptia</i> , <i>Urochloa mossambicensis</i> , <i>Panicum coloratum</i> , <i>Sporobolus nitens</i>), occur as patches in the taller bunch-grass matrix.”	Ecological characteristic present in definition	N
Cromsigt, JPGM; Prins, HHT; Olf, H 2009	“We defined grazing lawn as present when lawn grass species dominated (> 75%) a 5-m plot and extended for several metres away on both sides of a transect (at least 5 m), i.e. if lawn species only covered the transect it was not recorded as grazing lawn.”	Ecological characteristic present in definition	N
Rooney, TP 2009	“It is plausible that deer are creating grazing lawns—productive, graminoid-dominated vegetation perpetuated by repeated grazing (McNaughton 1984).”	Both present in definition	Y
Bonnet, O; Fritz, H; Gignoux, J; Meuret, M 2010	“By enhancing local resource attractiveness, repeated grazing can lead to the formation of intensively grazed, short-cropped sward areas called grazing lawns (McNaughton 1984).”	Both present in definition	Y
Stock, WD; Bond, WJ; van de Vijver, CADM 2010	“Grazing lawns are heavily used by herbivores but are dominated by stoloniferous and very short-statured grasses.”	Ecological characteristic present in definition	N
Coetsee, C; Stock, WD; Craine, JM 2011	“Grazing lawns can be defined as areas where grasses are kept in a short, actively growing, palatable state by the action of grazers.”	Both present in definition	N
Gosling, CM; Cromsigt, JPGM; Mpanza, N; Olf, H 2012	“Grazing lawns consist of a community of short statured, stoloniferous, grazing tolerant grasses that are key resource areas for savannah herbivores (Grant and Scholes 2006; Verweij and others 2006; Stock and others 2009).”	Ecological characteristic present in definition	Y

Appendix Table 2.20 (Continued)

Article Author(s) YEAR	Grazing Lawn Definition Provided in Journal Article. “Ecological Process/Interaction” descriptions are highlighted in yellow. “Ecological Characteristic” descriptions are highlighted in green.	Are “Ecological Process/Interactions”, “Ecological Characteristics”, Or Both Categories Present in Definition?	Previous Journal Article(s) Referenced in Definition?
Novellie, P; Gaylard, A 2013	“In a detailed review of this phenomenon, Cromsigt and Kuijper (2011:212) defined a grazing lawn as ‘a grazed or browsed patch in which intense foraging leads to increased resource availability and a proportional increase in palatable species in the patch’. Thus, grazing lawns are characterised by a positive association between grazers and grazing tolerant grasses; intensive grazing has the effect of increasing the productivity and quality of the food resource (Cromsigt & Kuijper 2011).”	Both present in definition	Y
van der Plas, F; Zeinstra, P; Veldhuis, M; Fokkema, R; Tielens, E; Howison, R; Olff, H 2013	“... and ‘grazing lawns’, dominated by short, clonally reproducing grass species such as <i>Digitaria longiflora</i> and <i>Sporobolus nitens</i> (Archibald et al. 2005; Stock et al. 2010).”	Ecological characteristic present in definition	Y
Cromsigt, JPGM; te Beest, M 2014	“We defined the 2-m plot as a grazing lawn state if >50% of the plot was covered with stoloniferous, prostrate growing grass species, and the grass height was not higher than 4 cm (see Cromsigt & Kuijper 2011 for a more extensive definition).”	Ecological characteristic present in definition	Y
Veldhuis, MP; Howison, RA; Fokkema, RW; Tielens, E; Olff, H 2014	“They are characterized by high productivity and different plant species composition; with higher abundances of nutrient- and mineral-rich species that form keystone resources for grazing herbivores (McNaughton 1979, 1984; Augustine, McNaughton & Frank 2003).”	Both present in definition	Y
Fox, HV; Bonnet, O; Cromsigt, JPGM; Fritz, H; Shrader, AM 2015	“Grazing lawns are conspicuous patches of short, mat-forming, grazing-tolerant grass species resulting from repeated cropping by herbivores.”	Both present in definition	N
Martin, J; Benhamou, S; Yoganand, K; Owen-Smith, N 2015	“Furthermore, herbivores may maintain local grass swards in a short-cropped state of high nutritional value, referred to as “grazing lawns” McNaughton (1984); Bonnet et al. (2010).”	Both present in definition	Y

Appendix Table 2.20 (Continued)

Article Author(s) YEAR	Grazing Lawn Definition Provided in Journal Article. “Ecological Process/Interaction” descriptions are highlighted in yellow. “Ecological Characteristic” descriptions are highlighted in green.	Are “Ecological Process/Interactions”, “Ecological Characteristics”, Or Both Categories Present in Definition?	Previous Journal Article(s) Referenced in Definition?
Owen-Smith, N; Martin, J; Yoganand, K 2015	“Within regions of the gabbro uplands, short grasses predominated, representing grazing lawns maintained by herbivore pressure (Hempson et al. 2014).”	Both present in definition	Y
Zwerts, JA; Prins, HHT; Bomhoff, D; Verhagen, I; Swart, JM; de Boer, WF 2015	“Grazing lawns consist of grass species which by means of continuous cropping remain in a highly productive state, resulting in the availability of nutritious, low fibre regrowth (Verweij et al. (2006); Georgiadis & McNaughton (1990)).”	Both present in definition	Y
Howison, RA; Berg, MP; Smit, C; van Dijk, K; Olff, H 2016	“The main mechanisms that have been put forward for grazing lawn formation are co-evolution of plants that tolerate or resist grazing (McNaughton 1984; Milchunas and others 1988) in the presence of large herbivores which optimize foraging strategies (Westoby 1974; McNaughton 1979) and positive nutrient feedbacks (Ruess and McNaughton 1987; Croomsigt and Olff 2008).”	Ecological process/interaction present in definition	Y
Sedinger, JS; Nicolai, CA; VanDellen, AW; Leach, AG; Wilson, HM; Anthony, RM 2016	“Numerous species of grazers maintain grazing lawns in which plants are more nutritious than in surrounding areas (Bazely and Jefferies 1986), and reduced grazing pressure can result in loss of grazing lawns (McNaughton 1985, Person et al. 2003, Archibald 2008).”	Both present in definition	Y
Veldhuis, MP; Fakkert, HF; Berg, MP; Olff, H 2016	“Grazing lawn patches, existing of short (0–20 cm) stoloniferous grass species with high foliar nutrient concentrations (McNaughton 1984; Stock et al. 2010; Hempson et al. 2014)...”	Ecological characteristic present in definition	Y
Donaldson, JE; Archibald, S; Govender, N; Pollard, D; Luhdo, Z; Parr, CL 2018	“We defined grazing lawns as areas where high grazing pressure limits grass height below levels required to carry fire, that is, grazing lawns are formed when positive feedbacks promote their maintenance and restrict fire.”	Both present in definition	N

Appendix Table 2.20 (Continued)

Article Author(s) YEAR	Grazing Lawn Definition Provided in Journal Article. “Ecological Process/Interaction” descriptions are highlighted in yellow. “Ecological Characteristic” descriptions are highlighted in green.	Are “Ecological Process/Interactions”, “Ecological Characteristics”, Or Both Categories Present in Definition?	Previous Journal Article(s) Referenced in Definition?
Veldhuis, MP; Gommers, MI; Olf, H; Berg, MP 2018	“The grassy vegetation is characterized by a high heterogeneity with short grazing lawns ranging in size between a few square meters to a couple of hectares that alternate with tall grasslands dominated by bunch/tussock-forming species.”	Ecological characteristic present in definition	N
Fleazar, U; le Roux, E; Kerley, GIH; Kuijper, DPJ; te Beest, M; Druce, DJ; Prinsloo, D; Cromsigt, JPGM 2019	“We used natural grazing lawns (short stoloniferous, high-quality grasslands, highly preferred by grazers (Cromsigt and Olf, 2008)) as our experimental sites to ensure high visitation by ungulates and to standardize food availability (grass quality and quantity) among sites.”	Ecological characteristic present in definition	Y
Hyvärinen, O; Hoffman, MT; Reynolds, C 2019	“Grazing lawns are characterized by nutritious stoloniferous grasses that are highly adapted to grazing (McNaughton 1984; Cromsigt and Olf 2008; Hempson et al. 2015), and their formation is often associated with large grazing herds (McNaughton 1984) or megaherbivores such as White Rhinoceros (Owen-Smith 1988; Cromsigt and te Beest 2014).”	Both present in definition	Y
Purdon, J; Parr, CL; Somers, MJ 2019	“Under intense grazing, bunch grasses (e.g., Themeda triandra, Eragrostis curvula and Hyparrhenia filipendula, in southern Africa), which are less resilient to disturbances (Coughenour, 1985), are replaced by patches of short, stoloniferous grasses (e.g., Urochloa mosambicensis, Digitaria longifolia, and Sporobolus nitens)—also called grazing lawns (McNaughton, 1985; Archibald et al., 2005; Cromsigt, 2006; Cromsigt et al., 2017).”	Both present in definition	Y
Ruess, RW; McFarland, JW; Person, B; Sedinger, JS 2019	“On the Yukon–Kuskokwim River Delta in western Alaska, geese maintain short-statured grazing lawns of Carex subspathacea, a rhizomatous sedge whose growth form, high N content, and rapid growth rate are dependent on frequent grazing and the nutrients recycled by geese (Ruess et al. 1997, Person et al. 2003).”	Both present in definition	Y

Appendix Table 2.20 (Continued)

Article Author(s) YEAR	Grazing Lawn Definition Provided in Journal Article. “Ecological Process/Interaction” descriptions are highlighted in yellow. “Ecological Characteristic” descriptions are highlighted in green.	Are “Ecological Process/Interactions”, “Ecological Characteristics”, Or Both Categories Present in Definition?	Previous Journal Article(s) Referenced in Definition?
Uher-Koch, BD; Schmutz, JA; Wilson, HM; Anthony, RM; Day, TL; Fondell, TF; Person, BT; Sedinger, JS 2019	“Grazing lawns (which are patches of grazing-tolerant plants with high nutrient value) provide important resources for adult herbivores and their young (McNaughton 1984).”	Ecological characteristic present in definition	Y
Awuah, KT; Aplin, P; Marston, CG; Powell, I; Smit, IPJ 2020	“... grazing lawns were identified as short grass areas with stoloniferous growth forms and height <20 cm.”	Ecological characteristic present in definition	N
Mayengo, G; Piel, AK; Treydte, AC 2020	“Alocally high grazing pressure has also been associated with grazing lawns, which are characterized by the presence of short grass species, as well as maintained and influenced by high grazing intensity Hempson et al. (2015); Archibald (2008); Croomsigt & Olff (2008); McNaughton (1984).”	Both present in definition	Y
Ozment, KA; Welti, EAR; Shaffer, M; Kaspari, M 2021	“Some mammals create grazing lawns: areas regularly grazed by large herbivores creating patches of short-grass swards which can act as nutrient hotspots (Hempson et al., 2015; Mayengo et al., 2020; McNaughton, 1984).”	Both present in definition	Y
Thapa, SK; de Jong, JF; Subedi, N; Hof, AR; Corradini, G; Basnet, S; Prins, HHT 2021	“We categorized grazing lawns as small (<1 hectare) areas containing short grasses with low stature growth form, growing closer to the ground with prostrate leaves and tillers established and maintained through intensive grazing by wild herbivores.”	Both present in definition	N
Voysey, MD; Archibald, S; Bond, WJ; Donaldson, JE; Staver, AC; Greve, M 2021	“In African savannas, large mammal grazers create grazing lawns through frequent defoliation of the grass sward (Hempson et al., 2015; McNaughton, 1984).”	Ecological process/interaction present in definition	Y
Foley, KM; Beard, KH; Atwood, TB; Waring, BG 2022	“By returning to the same sites year after year, Pacific black brant create mosaic patches of shorter, more nutritious <i>C. subspathacea</i> , described as ‘grazing lawns’ (Person, Babcock, & Ruess (1998); Uher-Koch et al. (2019).”	Both present in definition	Y

Appendix Table 2.20 (Continued)

Article Author(s) YEAR	Grazing Lawn Definition Provided in Journal Article. “Ecological Process/Interaction” descriptions are highlighted in yellow. “Ecological Characteristic” descriptions are highlighted in green.	Are “Ecological Process/Interactions”, “Ecological Characteristics”, Or Both Categories Present in Definition?	Previous Journal Article(s) Referenced in Definition?
Hempson, GP; Parr, CL; Lehmann, CER; Archibald, S 2022	“Grazing lawns are short-grass communities with dense leafy swards that provide high-quality forage for grazers (McNaughton, 1984).”	Ecological characteristic present in definition	Y
Kellett, DK; Alisauskas, RT 2022	“Under optimal levels of grazing, herbivores often create and maintain “grazing lawns” characterized by short, decumbent, and dense growth forms and high nutritional content, with significant reduction of aboveground vegetative biomass compared with ungrazed states (Cargill & Jefferies, 1984b; McNaughton, 1984; Person et al., 2003).”	Both present in definition	Y
Thapa, SK; de Jong, JF; Hof, AR; Subedi, N; Joshi, LR; Prins, HHT 2022	“Grazing lawns are “nutrient hotspots” from where herbivores can maximize their energy intake (Thapa, de Jong, et al., 2021) and require frequent grazing to persist (Hempson et al., 2015; McNaughton, 1984).”	Both present in definition	Y
Yiu, SW; Owen-Smith, N; Cain, JW 2022	“In west-central Kruger National Park, South Africa, where our study was located, wildebeest herds confine their movements mostly to grazing lawns, that is, short grass areas also having little woody plant cover, which encompass less than 10% of the study area, for most of the wet season and into the early dry season (Yoganand and Owen-Smith 2014; Owen-Smith and Traill 2017).”	Ecological characteristic present in definition	Y
Hyvarinen, O; te Beest, M; le Roux, E; Kerley, GIH; Findlay, N; Schenkeveld, WDC; Trouw, V; Cromsigt, JPGM 2023	“Grazing lawns are functionally distinct grassland types, characterized by a specific community of grazing-tolerant, often prostrate growing stoloniferous grass species of high nutritional quality (Hempson et al. 2015) that only develop following prolonged (at least 2–3 years) continuous grazing (Cromsigt and Olff 2008).”	Both present in definition	Y
Saunders, T; Adkins, J; Beard, KH; Atwood, TB; Waring, BG 2023	“Grazing of <i>C. subspathacea</i> by Pacific black brant and cackling geese results in ‘grazing lawns’ in which the <i>C. subspathacea</i> is converted to a shortstatured form of ~ 1 cm (Person et al. 2003; Kelsey et al. 2016).”	Both present in definition	Y

Appendix Table 2.20 (Continued)

Article Author(s) YEAR	Grazing Lawn Definition Provided in Journal Article. “Ecological Process/Interaction” descriptions are highlighted in yellow. “Ecological Characteristic” descriptions are highlighted in green.	Are “Ecological Process/Interactions”, “Ecological Characteristics”, Or Both Categories Present in Definition?	Previous Journal Article(s) Referenced in Definition?
Thapa, SK; de Jong, JF; Hof, AR; Subedi, N; Prins, HHT 2023	“We further expect that the availability of high-quality forage will attract herbivores to graze in the area, resulting in increased grazing pressure that initiates the process of grazing lawn formation (Cromsigt and Olff, 2008; Hempson et al., 2015; McNaughton, 1984).”	Both present in definition	Y
Horne, EA; Blackmore, P; Bello, NM; Taylor, J; Skibbe, A 2024	“A grazing lawn was defined by the presence of short-cropped grasses and abundant forbs.”	Ecological characteristic present in definition	N
McGregor, S; Cromsigt, JPGM; te Beest, M; Chen, J; Roy, DP; Hawkins, HJ; Kerley, GIH 2024	“... grazing lawn patches (intensely-grazed and dominated by short-statured, prostrate-growing <i>Cynodon dactylon</i>)...”	Ecological characteristic present in definition	N
Singh, J; Donaldson, JE; Archibald, S; Parr, CL; Voysey, MD; Davies, AB 2024	“Grazing lawns are characterized by communities of short-statured grasses with a high leaf to stem ratio and increased bulk density, which are predominantly maintained by the frequent concentration of high grazing activity (Cromsigt & Olff, 2008; Hempson et al., 2015).”	Both present in definition	Y
Mata, JC; Svenning, JC; Chequín, RN; Davalos, M; Salas, R; Vucko, A; Buitenwerf, R 2025	“A prominent pathway by which grazers affect ecosystem structure, processes and plant communities are grazing lawns, a short lawn-like vegetation state maintained through repeated grazing (Cingolani et al. 2003; Hempson et al. 2015). Grazing lawns have positive consumer-resource feedbacks, where continuous grazing maintains high forage quality (low C:N ratios) by preventing plants from accumulating structural biomass such as stems, while at the same time receiving increased nutrient input through dung deposition.”	Both present in definition	Y

Appendix Table 2.20 (Continued)

Article Author(s) YEAR	Grazing Lawn Definition Provided in Journal Article. “Ecological Process/Interaction” descriptions are highlighted in yellow. “Ecological Characteristic” descriptions are highlighted in green.	Are “Ecological Process/Interactions”, “Ecological Characteristics”, Or Both Categories Present in Definition?	Previous Journal Article(s) Referenced in Definition?
Saunders, T; Adkins, J; Atwood, TB; Waring, BG; Beard, KH 2025	“Geese in these ecosystems primarily feed on the abundant sedge species, <i>Carex subspathacea</i> (hereafter referred to as <i>Carex</i>), creating ‘grazing lawns’ of short-statured <i>Carex</i> with higher quality forage (decreased C:N ratios and lignin content) compared to immediately adjacent ungrazed <i>Carex</i> meadows (Person et al. 2003; Beard et al. 2019; Ruess et al. 2019).”	Both present in definition	Y

Appendix Table 0.21: Reference list of all qualifying journal articles in the systematic review.

Appendix Table 2.21 (Continued)

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Appendix Table 2.21 (Continued)

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Appendix Table 2.21 (Continued)

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Appendix Table 2.21 (Continued)

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Appendix B: Chapter 3 Supplementary Material

Appendix Table 0.22: Summary table of herbaceous plant data, ungulate herbivory evidence, and camera footage of ungulate behavior, and woody plant data in identified grazing lawns. Values with (X) represent unobtainable data summaries due to wilderness laws forbidding game camera use.

Appendix Table 3.1 (Continued)

	Audubon lawn (A)	Cummins lawn (B)	USFS lawn (C)	All lawns combined
Herbaceous Vegetation Cover by Plant Type in Lawns				
Bunch grasses (%)	4%	14%	30%	16%
Lawn grasses (%)	24%	13%	0%	12%
Forbs (%)	58%	38%	52%	49%
Ferns (%)	2%	5%	1%	3%
Vines (%)	2%	2%	3%	2%
Plant Cover Type by Origin in the Herbaceous Layer of Lawns				
Alien plants (%)	63%	1%	59%	41%
Native plants (%)	27%	71%	27%	42%
Non-vegetative cover (%)	10%	28%	14%	17%
Herbaceous Plant Heights in Lawns				
Average height of quadrats (m)	.14 m	.1 m	.14 m	.13 m
Ungulate Herbivory Evidence in Lawns				
Quadrats with grazing evidence (%)	85%	33%	92%	68%
Quadrats without grazing evidence (%)	15%	67%	8%	32%
Plants with browsing evidence (%)	48%	33%	62%	45%
Plants without browsing evidence (%)	52%	67%	38%	55%
Average ungulate pellet groups per survey	n = 6	n = 4	n = 4	n = 5
Game Camera Summaries of Ungulate Behavior				
Visitation frequency (% of days ungulates visited lawns)	25%	X	10%	27%
Elk encounters on cameras (n)	n = 33	X	n = 6	n = 39
Deer encounters on cameras (n)	n = 18	X	n = 19	n = 37
Elk vigilance (% of animals with head up)	30%	X	20%	24%
Deer vigilance (% of animals with head up)	72%	X	76%	73%
Elk grazing (% of individuals in photos)	52%	X	41%	44%
Deer grazing (% of individuals in photos)	19%	X	13%	17%
Elk browsing (% of individuals in photos)	4%	X	1%	1%
Deer browsing (% of individuals in photos)	0%	X	5%	2%
Vegetation Data Above Herbaceous Layer				
Seedling density (per 100 m ²)	0	1	0	< 1
Sapling density (per 100 m ²)	2	6	4	4
Shrub density (per 100 m ²)	13	18	11	14
Tree density (per 100 m ²)	2	3	3	3
Average Tree DBH (m)	0.25 m	0.77 m	0.04 m	0.39 m
Nurse log facilitation (% of plants)	0%	37%	32%	26%
Fruit Presence (% of fruit producing plants)	2%	2%	1%	2%

Common Name	Scientific Name	Average Height (m)	n	Browsing (%)
All conifer tree saplings		0.90 m	66	18%
Douglas fir	<i>Pseudotsuga menziesii</i>	0.2 m	2	0%
Sitka spruce	<i>Picea sitchensis</i>	0.99 m	46	0%
Western hemlock	<i>Tsuga heterophylla</i>	0.76 m	17	65%
Western red cedar	<i>Thuja plicata</i>	0.4 m	1	100%
All deciduous tree saplings		0.75 m	10	0%
Bigleaf maple	<i>Acer macrophyllum</i>	1.2 m	2	0%
Cascara	<i>Rhamnus purshiana</i>	0.75 m	5	0%
Red alder	<i>Alnus rubra</i>	0.9 m	1	0%
Unknown species	<i>Unknown species</i>	0.2 m	2	0%
All shrub species		0.71 m	293	60%
Cascade Oregon grape	<i>Berberis nervosa</i>	0.32 m	11	9%
English holly *	<i>Ilex aquifolium</i> *	0.38 m	4	25%
Himalayan blackberry *	<i>Rubus armeniacus</i> *	1.01 m	18	50%
Cut leaved blackberry *	<i>Rubus laciniatus</i> *	0.93 m	10	10%
Evergreen huckleberry	<i>Vaccinium ovatum</i>	0.48 m	26	12%
Osoberry	<i>Oemleria cerasiformis</i>	1.4 m	1	100%
Red elderberry	<i>Sambucus racemosa</i>	1.15 m	40	25%
Red huckleberry	<i>Vaccinium parvifolium</i>	0.49 m	66	82%
Salal	<i>Gaultheria shallon</i>	0.17 m	18	0%
Salmonberry	<i>Rubus spectabilis</i>	0.81 m	99	96%

Appendix Table 0.23: Summary table of the average height, total number of individuals (n), and percentage of individuals with browsing evidence present across all identified grazing lawns by plant species. Plant species of alien origin have a (*) next to their names.

Blooming forb summaries by month		
Month & Year	Short-Term Herbivory Cages	Grazing Lawns
July 2024 * Cage Installation *	n = 34 S = 5	n = 18 S = 6
September 2024	n = 45 S = 2	n = 0 S = 0
November 2024	n = 0 S = 0	n = 0 S = 0
January 2025	n = 0 S = 0	n = 0 S = 0
March 2025	n = 0 S = 0	n = 0 S = 0
May 2025	n = 34 S = 3	n = 11 S = 4
July 2025	n = 17 S = 4	n = 14 S = 6
Blooming forb summaries after cage installation in July 2024		
Total	n = 96 S = 7	n = 25 S = 7

Appendix Table 0.24: Summary table of forb blooms and richness in short-term herbivory cages. The number of blooming forbs (n) and blooming forb richness (S) in all short-term herbivory cages and paired grazing lawn plots are summarized by sample month, followed by total number of blooming forbs and blooming forb richness overall.

Habitat Type	Shrubs within browsing reach of ungulates (< 2 m in height)		Shrubs above browsing reach of ungulates (> 2 m in height)		Scat Encounters
	Shrubs with fruit	Shrubs without fruit	Shrubs with fruit	Shrubs without fruit	Average number of pellet groups
Grazing Lawns	n = 3 1%	n = 278 99%	n = 3 25%	n = 9 75%	5
Ungulate Refugia	n = 3 1%	n = 269 99%	n = 37 51%	n = 35 49%	2

Appendix Table 0.25: Summary table of the fruit producing shrubs above and below the maximum browsing height of elk (2 m) in grazing lawns and ungulate refugia plots. In each cell, the total number of shrubs (n) is reported to the left and the percentage (%) value of shrubs with or without fruit out of all shrubs of the same height class (> 2 m or < 2 m) is reported on the right. Scat encounters are reported by the average number of pellet groups recorded across all lawns and refugia plots, respectively.

Site	Woody plants protected by spruce		Woody plants not protected by spruce	
	With browsing evidence	Without browsing evidence	With browsing evidence	Without browsing evidence
USFS Lawn	n = 7 (41%)	n = 10 (59%)	n = 62 (89%)	n = 8 (11%)

Appendix Table 0.26: Summary table of the browsing rates of plants protected and not protected by Sitka spruce in the USFS grazing lawn. Woody plants consist of all shrubs and saplings present within the lawn, excluding spruce. The total number of plants (n) and the percentage (%) value of shrubs and saplings with and without browsing evidence is reported within each cell pertaining to spruce protection.